Teradata Visual Explain

User Guide

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Preface

Purpose

This book contains information on Teradata® Visual Explain, which is a Teradata® Tools and Utilities product. Teradata Tools and Utilities is a group of products designed to work with Teradata Database.

Teradata Visual Explain adds another dimension to the EXPLAIN modifier by visually depicting the execution plan of complex SQL statements in a simplified manner. It presents a graphical view of the statement broken down into discrete steps that show the flow of data during execution.

Audience

This book is intended for use by:

- System and application programmers
- System administrators
- Database Administrators and relational database developers
- System operators and other database specialists

Supported Releases

This book supports the following releases:

- Teradata Database 16.00
- Teradata Tools and Utilities 16.00
- Teradata Visual Explain 16.00

  Note: See "Help Menu Commands and Standard Toolbar" on page 35 to verify the Teradata Visual Explain version number.

To locate detailed supported-release information:

1. Go to http://www.info.teradata.com/.
2. Under Online Publications, click General Search.
3. Type *3119* in the Publication Product ID box.
4. Under Sort By, select Date.
5. Click Search.

6 Open the version of the *Teradata Tools and Utilities ##.# Supported Platforms and Product Versions* spreadsheet associated with this release.

The spreadsheet includes supported Teradata Database versions, platforms, and product release numbers.

Prerequisites

The following prerequisite knowledge is required for this product:

- Relational database management systems
- Teradata SQL
- Basic concepts and facilities of Teradata Database
- Connectivity software, such as ODBC or CLIv2
- The Microsoft Windows operating system

Changes to This Book

The following changes were made to this book in support of the current release. Changes in the chapters are marked with change bars. For a complete list of changes to the product, see the *Teradata Tools and Utilities Release Definition* associated with this release.

Date and Release	Description
November 2016 16.00	Chapter 5: Updated path to script files in "Creating Restricted Views." Removed prerequisite for viewing DBQL plans and steps to download XML Services in “Prerequisites” section in Preface.

Related Documentation

Additional information that supports this product and Teradata Tools and Utilities is available at the web sites listed in the table that follows.

Type of Information	Description	Source
Release overview Late information	The Release Definition provides the following information: • Overview of all the products in the release • Information received too late to be included in the manuals • Operating systems and Teradata Database versions that are certified to work with each product • Version numbers of each product and the documentation for each product • Information about available training and support center	1 Go to http://www.info.teradata.com/. 2 Under **Downloadable Publications**, click **General Search**. 3 Type *2029* in the **Publication Product ID** box. 4 Click **Search**. 5 Select the appropriate Release Definition from the search results.

Type of Information	Description	Source
Additional product information	Use the Teradata Information Products web site to view or download specific manuals that supply related or additional information to this manual.	1 Go to http://www.info.teradata.com/. 2 Under the **Downloadable Publications** subcategory, **Browse by Category**, click **Data Warehousing**. 3 Do one of the following: • For a list of Teradata Tools and Utilities documents, click **Teradata Tools and Utilities**, and then select an item under **Releases** or **Products**. • Select a link to any of the data warehousing publications categories listed. Specific books related to Teradata Visual Explain are as follows: • *Data Dictionary* (B035-1092) • *Database Administration* (B035-1093) • *ODBC Driver for Teradata User Guide* (B035-2509) • *SQL Data Definition Language* (B035-1144) • *SQL Data Manipulation Language* (B035-1146) • *SQL Request and Transaction Processing* (B035-1142) • *Teradata Call-Level Interface Version 2 Reference for Network-Attached Systems* (B035-2417) • *Teradata Index Wizard User Guide* (B035-2506) • *Teradata SQL Assistant for Microsoft Windows User Guide* (B035-2430) • *Teradata System Emulation Tool User Guide* (B035-2492) • *Teradata SQL Assistant/Web Edition User Guide* (B035-2505) • *Teradata Statistics Wizard User Guide* (B035-2503) • *Teradata Tools and Utilities for Microsoft Windows Installation Guide* (B035-2407) • *Teradata Visual Explain User Guide* (B035-2504)

Type of Information	Description	Source
CD-ROM images	Access a link to a downloadable CD-ROM image of all customer documentation for this release. Customers are authorized to create CD-ROMs for their use from this image.	1 Go to http://www.info.teradata.com/. 2 Under the Downloadable Publications subcategory, Browse by Category, click Data Warehousing. 3 Click Documentation CD Images. 4 Follow the ordering instructions.
Ordering information for manuals	Use the Teradata Information Products web site to order printed versions of manuals.	1 Go to http://www.info.teradata.com/. 2 Under Print & CD Publications, click How to Order. 3 Follow the ordering instructions.
General information about Teradata	The Teradata home page provides links to numerous sources of information about Teradata. Links include: • Executive reports, case studies of customer experiences with Teradata, and thought leadership • Technical information, solutions, and expert advice • Press releases, mentions and media resources	Teradata.com

Product Safety Information

This document may contain information addressing product safety practices related to data or property damage, identified by the word Notice. A notice indicates a situation which, if not avoided, could result in damage to property, such as equipment or data, but not related to personal injury.

Example:

Notice: Improper use of the Reconfiguration utility can result in data loss.

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CHAPTER 1

Introduction

This chapter is an overview of Teradata Visual Explain (VE) and the prerequisites for launching this tool.

About Teradata Visual Explain

Teradata Visual Explain adds another dimension to the EXPLAIN modifier by depicting the execution plans of complex SQL statements visually and simply. The graphical view of the statement is displayed as discrete steps showing the flow of data during execution.

By making the comparison of optimized queries easier, Teradata Visual Explain helps application developers, Database Administrators, and database support personnel fine-tune SQL statements to ensure Teradata Database accesses data in the most effective manner.

How Teradata Visual Explain Can Help You

Teradata Visual Explain makes query plan analysis easier. Plan steps can be captured and represented graphically, and two or more plans can be compared. Also, the performance implications of data skew and bad or missing statistics can be identified. These features make it easier to understand why Teradata Optimizer chooses a particular plan for a given SQL query. All of the information required for query plan analysis such as database object definitions, data demographics, and cost and cardinality estimates is available through the interface.

Teradata Visual Explain also is used to capture query plans in an emulated database environment. This function is helpful for comparing query plans for different configurations or row counts to see the impact of system expansion or table growth for a particular query.

How Teradata Visual Explain Works

Teradata Visual Explain reads the execution plan, which has been stored in the Query Capture Database (QCD), and converts the plan into a series of icons. View the execution plan either by entering a QCD using the Query Capture Facility (QCF), or by logging into the query plan using DBQL query logging.

Viewing an execution plan using QCF includes the INSERT EXPLAIN and INSERT EXPLAIN in XML commands (when connected to Teradata Database 13.10 and higher).

Viewing an execution plan *without* using QCF includes the EXPLAIN IN XML command, which generates the plan from an XML document (direct output from the database). Note that plan information is not stored on the database.

See "Entering a Query and Viewing Its Execution Plan" on page 78 and "View the Execution Plan" on page 62.

Prerequisites

To run Teradata Visual Explain:

- Establish the appropriate Teradata Visual Explain rights on Teradata Database.

  For more information, see "Using the Administrative Functions" on page 88.
- Increase the DBS Control Record MaxParseTreeSegs performance value to a higher limit to analyze complex queries.

  For example, increase the value to 1500 (or 3000 for 64-bit systems). MaxParseTreeSegs define the number of 64-KB tree memory segments the parser allocates when parsing a Teradata SQL statement. The valid range of values for MaxParseTreeSegs is 12 to 3000 (6000 for 64-bit systems).

  For information on setting DBS Control fields to help ensure that Teradata Database operates at peak performance based on applications and processing needs, refer to *Utilities* B035-1102.

CHAPTER 2

Interface

The Teradata Visual Explain user interface is designed to make viewing and analyzing query execution plans simple. The following topics describe the interface:

- Windows
- Menu Bar and Toolbars
- Display Icons
- Tool Tips and Pop-Up Menus
- Connectors
- Dialog Boxes and Shortcut Menus

Windows

The Teradata Visual Explain interface has the following windows:

- Teradata Visual Explain Main Window

  Used to perform most functions.
- Summary Information Window

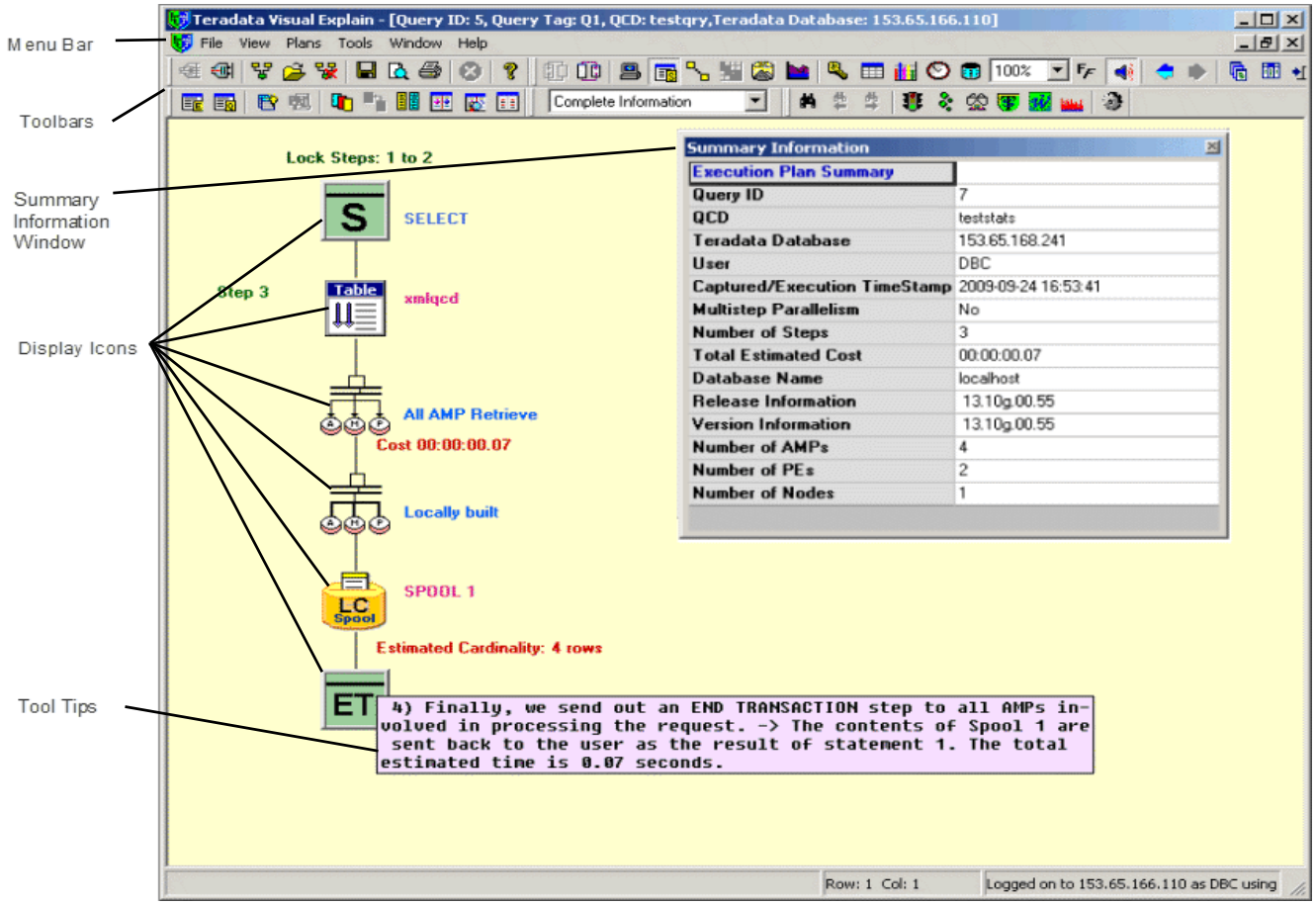
  Used to see summarized plan information from the main window.
- Teradata Visual Explain Bulk-Compare Window

  Used when working with bulk-compare reports.

Teradata Visual Explain Main Window

Figure 1 is a representative view of the Teradata Visual Explain main window, which opens by default when starting Teradata Visual Explain. Information that appears in the window is dependent on the task being performed.

Figure 1: Teradata VE Main Window



The basic features of the interface are:

- The Menu Bar and Toolbars used to access commands
- Display Icons that depict database objects (for example, tables, spools), operations, and statement types
- Tool Tips and Pop-Up Menus, shown as overlaying boxes in the window, that provide summary or detailed information about each icon
- Connectors that show the relationship between the displayed icons
- A Summary Information Window that provides information related to the displayed execution plan

Teradata Visual Explain also features shortcut menus, resizable columns in dialog boxes, and online help to make it easier to work with the application and on-screen information.

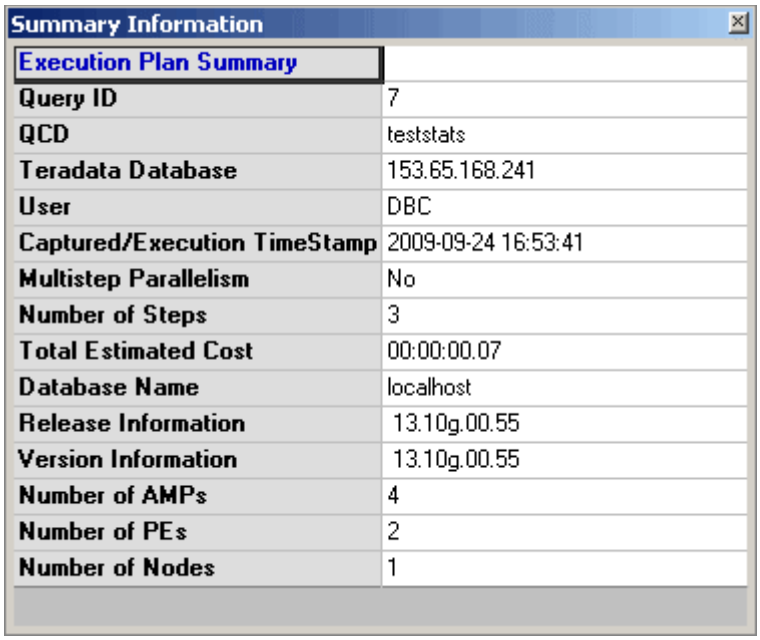
For more information about:

- Shortcut menus, see “Shortcut Menus” on page 51
- Resizing columns, see “Dialog Boxes” on page 51
- Online help, see “Online Help” on page 52

Summary Information Window

The Summary Information window (Figure 2) presents summarized information related to the displayed execution plan. The window opens in addition to the main window when viewing a plan.

Figure 2: Summary Information Window



Summary Information	
Execution Plan Summary	
Query ID	7
QCD	teststats
Teradata Database	153.65.168.241
User	DBC
Captured/Execution TimeStamp	2009-09-24 16:53:41
Multistep Parallelism	No
Number of Steps	3
Total Estimated Cost	00:00:00.07
Database Name	localhost
Release Information	13.10g.00.55
Version Information	13.10g.00.55
Number of AMPs	4
Number of PEs	2
Number of Nodes	1

Table 1 describes the information displayed in the Summary Information window. When comparing plans, the information in the Summary Information window changes according what is displayed in the main window.

Table 1: Summary Information Window Description

Summary Information	Description
Execution Plan Details	
Query ID	Query ID number.
QCD	QCD name.
Teradata Database	Teradata Database name.
User	User name.
Captured/Execution TimeStamp	Date and time the execution plan was captured and executed in TIMESTAMP (0) format.
Multistep Parallelism	Indicates whether parallelism is invoked.
Number of Steps	Number of steps required to execute the plan.
Total Estimated Cost	Estimated cost of executing the query. Cost is displayed in hours, minutes, seconds, and milliseconds (*HH:MM:SS.ss* format).

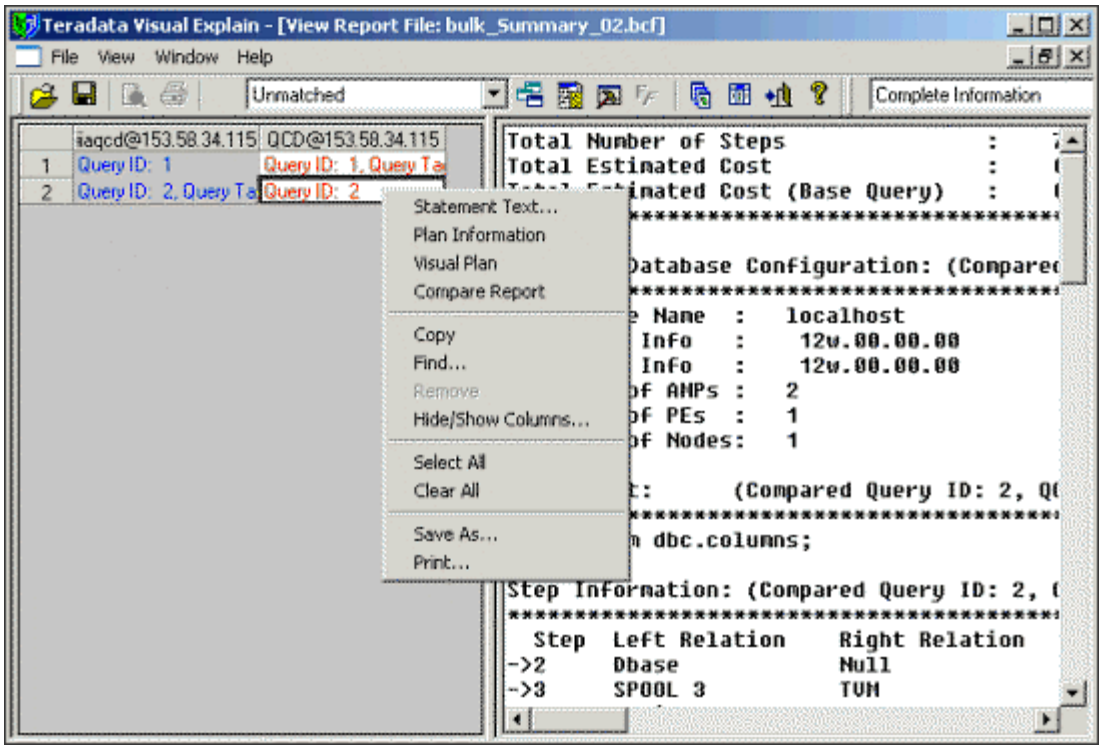
Table 1: Summary Information Window Description (continued)

Summary Information	Description
Join Types	Types of joins used in the execution of the query. Note: This category appears only when the execution plan uses joins.
Join (all types)	Number of joins in the plan. Note: This category appears only when the execution plan uses joins.
Teradata Database Configuration Note: The categories listed change according to the cost parameter settings. For more information, see "Setting or Resetting Cost Parameters" on page 146.	
Database Name	Name of the database the user was connected to at the time the execution plan was captured.
Release Information	When the plan is captured, the release level of the software loaded on Teradata Database. For example: • V2R.*nn.nn.nn.nn* for releases earlier than 12.00.00 • *nn.nn.nn.nn* for release 12.00.00 and later This attribute comes from the DBC.DBCInfoTbl system table.
Version Information	The version of the Teradata Database software used when the plan was captured. For example, 05.01.02.03 represents the: • Major release number, 05 • Minor release number, 01 • Maintenance release number, 02 • E-fix or patch release number, 03 These numbers come from the DBC.DBCInfoTbl system table.
Number of AMPs	Number of AMPs on the system where the plan was captured.
Number of PEs	Number of PEs on the system where the plan was captured.
Number of Nodes	Number of nodes on the system where the plan was captured.

Teradata Visual Explain Bulk-Compare Window

Figure 3, The Teradata Visual Explain bulk-compare window is only available when a bulk compare has been performed. Information that appears in the window is dependent on the task being performed.

Figure 3: Teradata VE Bulk-Compare Window



The features of the bulk-compare window are:

- The View Menu and Toolbar for the Bulk-Compare Report Viewer used to work with the report
- Shortcut Menus, which provide quick access to commands

Menu Bar and Toolbars

The menu bar and toolbars are located at the top of the Teradata Visual Explain windows. The menu bar displays menus that contain the commands used to operate the application. The toolbars contain buttons used as shortcuts to the more commonly used menu commands.

The interface displays the following menus and toolbars, depending on the window in use and other user-controlled settings:

- File Menu Commands and Standard Toolbar
- View Menu Commands and Toolbar
- View Menu and Toolbar for the Bulk-Compare Report Viewer
- Plans Menu Commands and Toolbar

- Tools Menu Commands and Toolbar
- Navigate Menu Commands and Toolbar
- Window Menu Commands and View Toolbar
- Help Menu Commands and Standard Toolbar

Note: A modified View menu appears in the menu bar when using the bulk-compare report viewer. For more information, see "View Menu and Toolbar for the Bulk-Compare Report Viewer" on page 30. The Window menu only appears when a plan is open.

File Menu Commands and Standard Toolbar

Use the File menu or Standard toolbar, described in Table 2, to perform the basic functions of working with plans. For example, establish a connection to a Teradata Database, disconnect from a Teradata Database, and load or delete execution plans. Use the menu to save and print plans as well as exit the application. Use (only on the toolbar) to abort the current transaction and roll back changes.

Table 2: File Menu Commands and Standard Toolbar Buttons

Command	Button	Description
Connect		Connect to a Teradata Database.
Disconnect		Disconnect from a Teradata Database.
Open Plan from Database		Load an execution plan from a QCD database.
Open Plan from File		Load an execution plan from a previously saved file.
Delete Execution Plans		Delete execution plans from a QCD database.
Save As		Save one or more execution plans to one or more files. Note: This command is only available when a plan is open.
Print		Print the details of the execution plan. Note: This command is only available when a plan is open.
Print Preview		Preview the plan details to be printed. Note: This command is only available when a plan is open.
Exit	None	Exit the application.
None		Abort the current transaction and roll back any changes.

Note: The Standard toolbar includes one toolbar button (not described in Table 2) for the most common Help menu command. See "Help Menu Commands and Standard Toolbar" on page 35 for a description of this button.

View Menu Commands and Toolbar

Use the View menu and toolbar, described in Table 3, to work with the information in the Teradata Visual Explain window. View database configurations, connectors, indexes, object

definitions, statistics, and summary information related to the displayed execution plan. Commands on this menu also set the display font, set the zoom level, and provide a view pad (an easy way to navigate large plans).

Table 3: View Menu Commands, Submenus, and Toolbar Buttons

Command	Submenu	Button	Description
Toolbars			
	Standard	None	Toggle the **Standard** toolbar on or off. For more information, see "File Menu Commands and Standard Toolbar" on page 28.
	View	None	Toggle the **View** toolbar on or off. For more information, see "View Menu Commands and Toolbar" on page 28.
	Plans	None	Toggle the **Plans** toolbar on or off. For more information, see "Plans Menu Commands and Toolbar" on page 33.
	Tools	None	Toggle the **Tools** toolbar on or off. For more information, see "Tools Menu Commands and Toolbar" on page 33.
Status Bar	None	None	Toggle the status bar on or off.
Spool Space Utilization	None		View the current and maximum spool space of the current user.
Compressed	None		View the data-flow-oriented layout of the execution plan. **Note:** The command is only available when a plan is open.
Expanded	None		View the control-flow-oriented layout of the execution plan. **Note:** This command is only available when a plan is open.
Teradata Database Configuration	None		View the configuration of the database the execution plan was captured on. **Note:** This command is only available when a plan is open.
Summary Information	None		Toggle the **Summary Information** window open or closed. The window displays information about the execution plan. **Note:** This command is only available when a plan is open.
Show Connectors	None		Display additional connectors (made up of dashed lines) depicting spool reuse.
View Pad	None		Show an overall view of plans that span multiple horizontal and vertical pages. **Note:** This command is only available when a plan is open.
Show Demographics	None		View the demographics of the tables used in the plan. **Note:** This command is only available when a plan is open.

Table 3: View Menu Commands, Submenus, and Toolbar Buttons (continued)

Command	Submenu	Button	Description
Object Use Count	None	None	View a report of all objects used in the displayed plan. Note: This command is only available when a plan is open.
Current vs. Captured Note: This command and related submenus are only available when a plan is open.			
	Indexes		View the indexes on objects used in the execution plan.
	Object Definitions		View the definitions of the database objects used in the execution plan.
	Statistics		View statistics for the tables used in the execution plan.
Actual vs. Estimated Note: This command and related submenus are only available when a plan is open.			
	Cardinality		View the actual and estimated cardinality of each spool in the execution plan.
	Cost		View the actual and estimated cost of each step in the execution plan.
Set Font	None		Display the Font dialog box to change the display font.
Zoom	None	100%	Set the size of the display.
None	None		Toggle audible notification on or off.
None	None		Go to the previous spool reference.
None	None		Go to the next spool reference.

Note: The View toolbar includes toolbar buttons (not described in Table 3) for some of the more commonly used menu commands. See "Window Menu Commands and View Toolbar" on page 35 for a description of these buttons.

View Menu and Toolbar for the Bulk-Compare Report Viewer

Use the modified View menu and toolbar, described in Table 4, to:

- Access reports easily
- Set viewing preferences for the toolbar, status bar, and summary information
- Set variance thresholds and font preferences in the bulk-compare report viewer

The modified View menu and toolbar appear when using the bulk-compare report viewer.

Table 4: View Menu Commands, Toolbar Buttons, and Lists for the Bulk-Compare Report Viewer

Command	Button or List	Description
Toolbar	None	Toggle the toolbar on or off.
Status Bar	None	Toggle the status bar on or off.
Summary Information		View summary information for the execution plan.
Filter By	All Unmatched Matched All	Specify the filter-by criteria. Select a criterion from the **Filter By** list. For more information, see step 3 on page 129.
Compare Attribute	Complete Information Complete Information Teradata Database Configura Step Information Estimated Cost Operation Operands Residual Condition	Specify the comparison attribute. Select an attribute from the **Select Item** list. For more information, see step 4 on page 129. For information on reports, see "To display a report type" on page 186. **Note:** The Compare Attribute combo box is visible only for the Bulk Compare Report window.
Variance Threshold		Display the **Set Variance Threshold** dialog box to set a threshold (percent variance) for the estimated cost and row size. When the percent difference at any step is above the limit set, the differences are highlighted in the reports.
Multiple Windows		Control the number of windows displayed: multiple windows versus a single window. For more information, see "Viewing the Bulk Compare Results" on page 128.
Set Font		Open the **Font** dialog box to change the display font only in the bulk-compare report viewer. **Note:** Place the cursor in the report viewer, then click **View>Set Font**.

Note: The modified toolbar includes toolbar buttons (not described in Table 4) for some of the more commonly used commands on other menus. See "File Menu Commands and Standard Toolbar" on page 28, "Window Menu Commands and View Toolbar" on page 35, and "Help Menu Commands and Standard Toolbar" on page 35 for a description of these buttons.

View Menu and Toolbar for the Textual Compare Report

Use the modified **View** menu and toolbar, described in Table 5, to:

- Split the report view vertically or horizontally
- See the base query information
- See the compare query information
- See the base and compare query information

The modified **View** menu and toolbar appear when plans are compared.

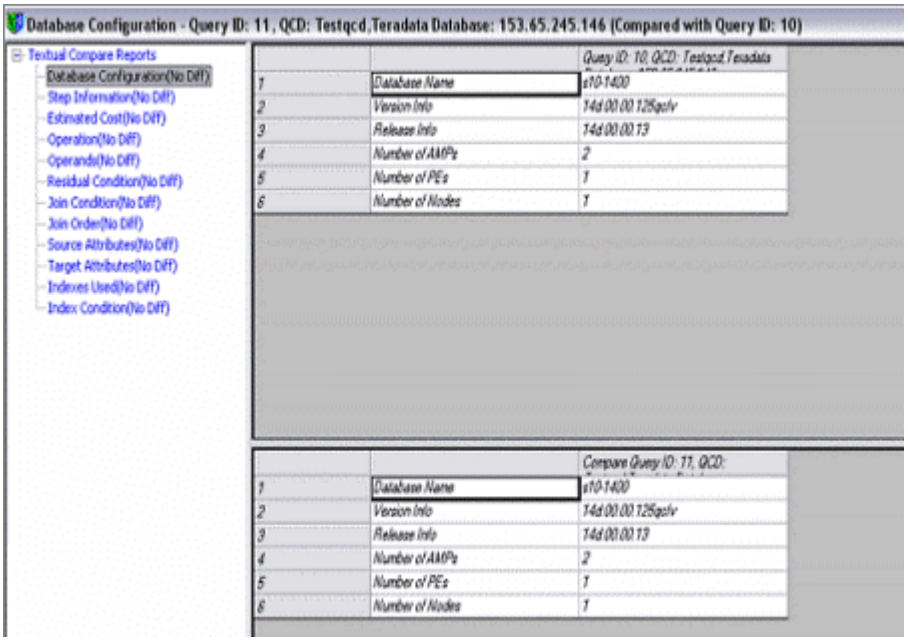
Table 5: View Menu Commands and Toolbar Buttons for Textual Compare Reports

Command	Button	Description
Split Window Vertically		Split the report view vertically. Note: This command is only available when the window is split horizontally. Click the button to toggle between a vertical and a horizontal view.
Split Window Horizontally		Split the report view horizontally. Note: This command is only available when the window is split vertically. Click the button to toggle between a vertical and a horizontal view.
Base Query Report	None	Show the base query information.
Compare Query Report	None	Show the compare query information.
Show Base/Compare Report	None	Show the base and query compare information.

Note: The modified toolbar includes toolbar buttons (not described in Table 5) for some of the more commonly used commands on other menus. See "File Menu Commands and Standard Toolbar" on page 28, "View Menu and Toolbar for the Bulk-Compare Report Viewer" on page 30, "Window Menu Commands and View Toolbar" on page 35, and "Help Menu Commands and Standard Toolbar" on page 35 for a description of these buttons.

The Textual Compare Plans screen is shown below in Figure 4.

Figure 4: Textual Compare Plans



Plans Menu Commands and Toolbar

Use the Plans menu and toolbar, described in Table 6, to compare execution plans, either textually by showing the differences in Explain text, or visually by using icons to describe the execution plan.

Table 6: Plans Menu Commands, Submenus, and Toolbar Buttons

Command	Submenu	Button	Description
Explain Text	None		View the Explain text for the current execution plan.
Statement Text	None		View the statement text for the current execution plan.
Index Analysis	None		Perform an analysis on a group of currently opened online execution plans.
Monitor SQL	None		Submit the query text to the database that belongs to the active plan, and view the execution of the query. Note: Valid for SELECT statements only. Monitor SQL command available only if connected using CLIv2.
Compare	None		Compare execution plans visually.
Compare Steps	None		Compare the steps of execution plans.
Compare Report	None		Compare execution plans textually.
Bulk Compare	Bulk Compare		Compare multiple execution plans.
Bulk Compare	View Report		View the results of the Bulk Compare.
Explain Text Differences	None		View the differences in Explain text.

Tools Menu Commands and Toolbar

Use the Tools menu, described in Table 7, to launch the Control Center, which is a powerful tool for working with QCDs and QCF-related objects, creating and administering users, and specifying the connectivity type. The Tools menu and toolbar serve many functions. When an execution plan is open, use the menu or toolbar to find differences in compared plans and to go to specific plan steps. Also use the Tools menu or toolbar to execute SQL commands, capture execution plans and insert them into the QCD, and set other program options.

Table 7: Tools Menu Commands and Toolbar Buttons

Command	Button	Description
None		Open the Find Difference dialog box to find a specific difference in the current compare.
Go To Step		Open the Go To Step dialog box to locate a specific step in the current plan.
Find Next		Find the next difference in the current compare.
Find Previous		Find the previous difference in the current compare.
Execute SQL		Submit SQL to a Teradata Database.

Table 7: Tools Menu Commands and Toolbar Buttons (continued)

Command	Button	Description
View Execution Plan(s)		Capture and visualize an execution plan for a query.
Teradata System Emulation Tool		Start Teradata System Emulation Tool. Note: Only available if the application is installed on the client machine.
Teradata Index Wizard		Start Teradata Index Wizard. Note: Only available if the application is installed on the client machine.
Teradata SQL Assistant		Start Teradata SQL Assistant. Note: Only available if the application is installed on the client machine.
Teradata Statistics Wizard		Start Teradata Statistics Wizard. Note: Only available if the application is installed on the client machine.
Control Center		Open the Control Center to: • Delete execution plans, databases, and QCF-related objects. • Create, upgrade, and clean up QCDs. • Export and import QCDs to and from a file. • Create users, and grant and revoke privileges on QCDs. • Specify the connectivity type (CLIv2 or ODBC). • Define data sources.
Define Data Source	None	Administer ODBC data sources.
Options	None	Set the connectivity, default path, and other program options.

Navigate Menu Commands and Toolbar

Use the Navigate menu and toolbar, described in Table 8, to find differences in Explain text. The Navigate menu only appears when viewing Explain-text differences.

Table 8: Navigate Menu Commands and Toolbar Buttons

Command	Button	Description
First Difference		Find the first difference in the Explain text.
Previous Difference		Find the previous difference in the Explain text.
Current Difference		Find the current difference in the Explain text.
Next Difference		Find the next difference in the Explain text.
Last Difference		Find the last difference in the Explain text.

Note: The Navigate toolbar includes toolbar buttons (not described in Table 8) for some of the more commonly used commands on other menus. See "Menu Bar and Toolbars" on page 27 for a description of these buttons.

Window Menu Commands and View Toolbar

Use the Window menu and buttons on the View toolbar, described in Table 9, to close plans and report windows and to arrange the reports in the Teradata Visual Explain window so they are easier to see.

Table 9: Window Menu Commands and View Toolbar Buttons

Command	Button	Description
Close		Close the active document.
Close All		Close all open documents.
Close Report Windows	None	Close all open reports.
Cascade		Display all open windows so their title bars are visible.
Tile Horizontal	None	Spread all open windows horizontally across the display.
Tile Vertical		Spread all open windows vertically across the display.

Note: The View toolbar includes buttons for the most commonly used Window menu commands. See "View Menu Commands and Toolbar" on page 28 for a description of the other buttons on the View toolbar.

Help Menu Commands and Standard Toolbar

Use the Help menu and Standard toolbar, described in Table 10, to access Teradata Visual Explain Online Help. Also use the Help menu to identify Teradata Visual Explain version and copyright information.

Table 10: Help Menu Commands

Command	Button	Description
Help Topics		Display a list of Help topics.
About Visual Explain	None	Display version and copyright information. See Table 11 on page 36 for more information.

Note: The Standard toolbar includes buttons for Help menu commands. See "File Menu Commands and Standard Toolbar" on page 28 for a description of the other buttons on the Standard toolbar.

From the Help menu, open the About Teradata Visual Explain dialog box to display the version and copyright information for the Teradata Visual Explain software in use.

To open the About dialog box

- Click Help > About Teradata Visual Explain.

Table 11 describes the About dialog box.

Table 11: About Dialog Box Description

Element Name	Description
Version	Version number of the software installed.
Date Created	Date the installed version of software was created.
System	Name of the system to which the user was last connected. The connectivity type is indicated as a suffix to the system name. For example, -ODBC or -CLI. This field is blank when not connected to a Teradata Database.
Connection Type	When connected to a Teradata Database, always displays Logged on. This information only appears when connected to a Teradata Database.
Character Set	Name of the session character set. This information only appears when connected to a Teradata Database.
User	Name of the user connected to Teradata Database. This information only appears when connected to a Teradata Database.

Toolbar Access

Toolbars contain buttons used to perform the functions of the more commonly used menu commands. The following toolbars can be hidden or displayed, as needed, to manage screen space:

- Standard
- View
- Plans
- Tools

For more information on the Teradata Visual Explain toolbars, see "Menu Bar and Toolbars" on page 27.

To hide a toolbar

1 From the menu bar, click View>Toolbars.

2 From the names with a check mark, select the name of the toolbar to hide.

The check mark next to the toolbar name is cleared, and the toolbar is no longer displayed in the window.

To display a toolbar

1 From the menu bar, click View>Toolbars.

2 From the names without a check mark, select the name of the toolbar to display.

A check mark is placed next to the name of the toolbar, and the toolbar appears in the window.

Display Icons

Icons are used in Teradata Visual Explain to visually depict the execution plan. The various icons provide information on:

- The type of statement executed
- The type of join used
- The eligibility of a table or spool for synchronized scanning
- Whether the table or spool is cached
- The method of retrieval (for example, single-AMP retrieval, multi-AMP retrieval or all-AMP retrieval)
- The distribution of rows (for example, whether the rows were locally built, duplicated on all AMPs or redistributed on all AMPs)
- End transaction information

For icon descriptions, see:

- Statement Icons
- Retrieval-Method Icons
- Data-Redistribution-Method Icons
- Table and Spool Icons
- Derived-Table Icons
- Index Icons
- Aggregate-Step Icons
- Join-Method Icons
- Other Operation Icons
- Error Icon

Note: Tool Tips and Pop-Up Menus provide additional information about each icon.

Statement Icons

Table 12 describes the SQL statement icons.

Table 12: Statement Icons

Icon	Statement	Description
	Abort	Terminate the current transaction.
	Alter Trigger	Allow users to enable or disable a trigger, and to change its creation timestamp.

Table 12: Statement Icons (continued)

Icon	Statement	Description
EP	Call Procedure	Execute a stored procedure.
CM	Create/Replace Macro	Insert or replace a macro in the database.
CP	Create/Replace Procedure	Insert or replace a procedure in the database.
CG	Create/Replace Trigger	Create new triggers or change the definition of a trigger.
AT	Alter Table	Alter a specified table.
CI	Create Index	Create an index to table.
CT	Create Table	Alter a specified table.
D	Delete	Delete rows from a table.
DI	Drop Index	Drops an index on a specified table.
DT	Drop Table	Drops a specified table.
ET	End Transaction	Indicate that the processing is complete and any locks on the data can be removed.
EM	Execute Macro	Execute a specified macro.
I	Insert	Add rows to a table.
M	Merge into	Merge row or rows into a table.

Table 12: Statement Icons (continued)

Icon	Statement	Description
END MLOAD	End Mload	Signifies the end of a Multiload task and initiates task processing.
BEGIN MLOAD	Begin Mload	Signifies the beginning of a Multiload import task.
ML	Mload Job	Invokes a Multiload job.
CHECK WL	Set Check Workload	Set check workload during Mload.
CHKPT Load	Checkpoint Loading	Threshold limits that can be assigned for a specific load for Fastload, Multiload, or TPT.
BL	Begin Loading	Initiates the loading process.
QB	Query Band	Sets a query band.
RB	Roll Back	Terminate the current transaction, and reset the database to the pre-transaction state.
S	Select	Retrieve specified fields from a table.
U	Update	Modify data in one or more rows of a table.

Retrieval-Method Icons

Table 13 describes the retrieval-method icons.

Table 13: Retrieval-Method Icons

Icon	Method	Circumstances When Used
	All-AMP Retrieve	The data resides on more than two AMPs.
	Multi AMP Retrieve	Teradata Database uses various hashing combinations.

Table 13: Retrieval-Method Icons (continued)

Icon	Method	Circumstances When Used
	Single AMP Retrieve	Teradata Database can use the row hash to determine which AMP contains the requested row.
	Two AMP Retrieve	Teradata Database uses various hashing combinations.

Data-Redistribution-Method Icons

Table 14 describes the icons that depict the data-redistribution methods.

Table 14: Data-Redistribution-Method Icons

Icon	Method	Circumstances When Used
	Duplicated On All AMPs	The resultant rows are duplicated on all AMPs.
	Redistributed On All AMPs	The resultant rows are redistributed on all AMPs.
	Locally Built On All AMPs	The resultant rows are locally built on all AMPs.

Table and Spool Icons

Table 15 describes the table and spool icons.

Table 15: Table and Spool Icons

Icon	Object	Definition
	Table	A table involved in the operation.
	Table eligible for synchronized scanning (not cached)	A table involved in the operation that is eligible for synchronized scanning.
	Spool	A spool, which is where intermediate query results are stored.

Table 15: Table and Spool Icons (continued)

Icon	Object	Definition
	Low Confidence Spool	A target spool, where the number of rows is estimated with a low confidence.
	High Confidence Spool	A target spool, where the number of rows is estimated with a high confidence.
	Index Join Confidence Spool	A target spool, where the number of rows is estimated with an index join confidence.
	No Confidence Spool	A target spool, where number of rows is estimated with no confidence, which may indicate lack of collected statistics or certain query constraints.
	High Confidence IN-List Spool	A spool whose rows are made up from the values of an IN condition. Used to maximize the performance of a special class of star join queries. The confidence is always high because the cardinality is known.

Derived-Table Icons

Teradata Visual Explain uses derived-table icons. A derived table is a spool in the visual plan.

Table 16 describes the derived-table icons.

Table 16: Derived-Table Icons

Icon	Object	Definition
	Derived table	Information from a table, which is stored in the spool.
	Derived table with high confidence	A derived table with high confidence level.
	Derived table with low confidence	A derived table with low confidence level.

Table 16: Derived-Table Icons (continued)

Icon	Object	Definition
	Derived table index join confidence	A derived table index with join confidence. The label "JC" indicates that the number of rows is estimated with index join confidence.
	Derived table with no confidence	A derived table with no confidence level.

Index Icons

Table 17 describes the index-type icons.

Table 17: Index Icons

Icon	Index Type
	Unique Primary Index
	Unique Secondary Index
	Non-Unique Primary Index
	Non-Unique Secondary Index

Aggregate-Step Icons

Table 18 describes the aggregate-step icons.

Table 18: Aggregate-Step Icons

Icon	Function	Description
	Sampling	An aggregate step that performs the sampling of data.
	Statistics	An aggregate step that operates on grouped rows and computes a new function value for the row given the function specification. This is similar to aggregates in that it operates on grouped rows and a filtering QUALIFY clause analogous to the HAVING clause. It is different from aggregates because it returns a value for each participating row (unless filtered out), and it allows other descriptive columns that are not in the group-by list to be specified in the SELECT list.
	Sum	An aggregate operator that returns a column value. The value is the arithmetic sum for a specified expression for a group. An expression is a constant or column expression for which the sum value is computed and that cannot contain any ordered analytic or aggregate functions.

Join-Method Icons

Table 19 describes the join-method icons.

Table 19: Join-Method Icons

Icon	Join Step	Description
	Correlated Exclusion Merge join	A correlated version of the Exclusion Merge Join. A subquery is correlated when it references columns of outer tables in the enclosing (outer) query.
	Correlated Exclusion Product join	A correlated version of the Exclusion Product Join. A subquery is correlated when it references columns of outer tables in the enclosing (outer) query.
	Correlated Inclusion Merge join	A correlated version of the Inclusion Merge Join. A subquery is correlated when it references columns of outer tables in the enclosing (outer) query.
	Correlated Inclusion Product join	A correlated version of Inclusion Product Join. A subquery is correlated when it references columns of outer tables in the enclosing (outer) query.
	Dynamic Hash join	An equality join between a small table and a large table on nonprimary index columns without placing the large table into a spool file. For Dynamic Hash join to be used, the left table must be small enough to fit in a single partition.

Table 19: Join-Method Icons (continued)

Icon	Join Step	Description
	Exclusion Hash Join	Exclusion Hash Join returns the outer row when there is no matching inner row. The cost of Exclusion Hash Join is similar to Inner Hash Join with the exception that Exclusion Hash Join returns only the outer row when there is no matching inner row.
	Exclusion Merge join	A merge join where only the rows that do not satisfy (are NOT IN) any condition specified in the request are joined.
	Exclusion Product join	A Product Join where only the rows that do not satisfy (are NOT IN) any condition specified in the request are joined.
	Exists join	A join that returns all the left table rows that satisfy a condition, if the right table is non-empty.
	Full Outer Merge join	A join in which full outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the tables that were not returned in the result of the inner join, extending these rows with null values.
	Full Outer Product join	A join in which full outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the tables that were not returned in the result of the inner join, extending these rows with null values.
	Hash join	A hash join is applicable only to equijoins. This join eliminates the need to sort the tables being joined before performing the join operation.
	Hash Star join	A join in which two or more small relations are joined to large a relation.
	Inclusion Hash Join	Inclusion Hash Join is similar to Inner Hash Join with the exception of the cost of building a row and the probe cost. With Inclusion Hash Join, when there is a match, only the outer row is returned where the Inner Hash Join returns the join of the inner and outer row.
	Inclusion Merge join	A join in which only the rows that satisfy (are IN) any condition specified in the request are joined.
	Inclusion Product join	A join in which only the rows that satisfy (are IN) any condition specified in the request are joined.

Table 19: Join-Method Icons (continued)

Icon	Join Step	Description
	Inner Merge join	A join that retrieves rows from two tables, then puts the rows onto a common AMP, based on the row hash of the columns involved.
	Inner Product join	A product join that compares every qualifying row from one table to every qualifying row from another table. The number of comparisons needed is the product of the number of qualifying rows in the two tables.
	Intersect join	A join in which the INTERSECT ALL clause is used and where the rows returned are common to both tables, including the duplicate values.
	Left Outer Merge join	A join in which left-outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the *left* join table that were not returned in the result of the inner join, extending these rows with null values.
	Left Outer Product join	A join in which left outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the *left* join table that were not returned in the result of the inner join, extending these rows with null values.
	Minus/Except join	A join that occurs when the MINUS ALL clause is used. Returns the rows that are present only in the left and not in the right table, including duplicate values.
	Nested join	A join in which a WHERE condition specifies a constant value for a unique index of one table. The conditions can also match some column of that single row to the primary or secondary index of the second table.
	Not Exists Join join	A join that returns all the left table rows that do not satisfy a condition, if the right table has no rows.
	Right Outer Merge join	A join in which right outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the *right* join table that were not returned in the result of the inner join, extending these rows with null values.
	Right Outer Product join	A join in which right outer joins of two or more tables perform an inner join of those tables according to a join condition. In addition, rows are returned from the *right* join table that were not returned in the result of the inner join, extending these rows with null values.
	Row ID join	A join in which the condition specified in the query must match a column in the first table to a NUSI or USI of the second table. A subset of the NUSI or USI values is qualified, and a nested join between the two tables retrieves the row IDs from the second table.

Other Operation Icons

Table 20 describes the Teradata Visual Explain icons used to depict other operations.

Table 20: Other Operation Icons

Icon	Operation	Function
	Merge-Delete	Merge the data, then delete it from the table.
	Merge	Perform a single-row merge.
	Multiple-row merge	Perform a multiple-row merge.
	Update	Update data in a table.
	Merge Update	Update merged data in a table.
	Bitmap	Insert bitmap images into a table.
	Insert	Insert rows into a table.
	Delete	Delete specified rows from a table.
	Abort	Terminate the current transaction.
	Materialized temporary trace table step	Realize temporary tables (actual creation and data insertion).
	Sort step	Sort the rows in a target relation and eliminates duplicate rows.

Table 20: Other Operation Icons (continued)

Icon	Operation	Function
Spoil	Spoil step	Spoil the dictionary cache.
Flush	Flush Database step	Flush the DISKSPACE and AMPUSAGE caches.
Begin Loading	Begin Loading	Initiate loading process.
CIX	Create Index Subtable	Create Index Subtable on table.
MTH	Modify Table Header	Modify the Table Header.
MVN	Modify Table Version	Modify Table Version in the data dictionary.
CTRts	Insert Access Rights	Insert Access Rights in the data dictionary.
CTH	Create Table Header	Create a Table Header.
DTH	Drop Table Header	Drop the Table Header.
ELog	End Logging	End logging on the table.
Set Query Band	Set Query Band	Identifies the type and source of a query to determine prioritization of queries.
CSU	Save updated stats	Save the updated Statistics for columns of a table.

Table 20: Other Operation Icons (continued)

Icon	Operation	Function
CSS	Collect Stats	Collect Stats for the given Table or Columns.
End Mload	End Mload	Signifies the end of the task script and initiates task processing by the Teradata Database.
Begin Mload	Begin Mload	Initiates or restarts a Multiload import task.
Mload	Mload Job	Invokes Multiload
CHECK WL	Check Workload Step	Check Workload during Mload.
Chkpt Loading	Checkpoint Loading	Checkpoints are entries posted to a restart log table at regular intervals of a data transfer operation.
i	Configuration Information	Describes the Configuration Information.
LIN	Fast Load Insert	INSERT is a Teradata SQL statement that inserts data records into the rows of the Teradata FastLoad table.
EDM	Edit DML step for MLOAD	Edit DML step for MLOAD controls the processing of an MLOAD insert, update, or delete step.

Error Icon

When there is an error in QCD data, Teradata VE displays the warning ?.

Tool Tips and Pop-Up Menus

Tool tips provide additional information about plan icons displayed in the Teradata Visual Explain window. When a visual plan is displayed in the main window, a pop-up menu can also be displayed to access additional information about the plan icons.

To display a tool tip

- Position the mouse pointer over the icon.

  A tool tip appears.

To view general icon-description information

1. From the **Teradata Visual Explain** window, right-click the icon.

   A pop-up menu appears.
2. Select **Icon Description**.

   An icon-information dialog box appears and displays information about the icon selected.

To view detailed icon-description information

1. From the **Teradata Visual Explain** window, right-click the icon.

   A pop-up menu appears.
2. Select **Detail Information**.

   A dialog box opens and displays information about the particular icon selected.

Table 21 describes tool tip and pop-up information.

Table 21: Icon Tool Tips and Pop-up Information

Icon Type	Tool Tip Information	Pop-Up Information
Table	Table Name Cardinality Type Property Index Fields Partitioned Primary Index Alias Name View Name Partition Count	Icon Description Detail Information Object Definition Indexes Statistics Table Demographics Recommendations Object Use Count
Spool	Spool Name Confidence Level Sorted Cached Estimated Cardinality Compressed Columns Spool Size	Icon Description Detail Information Spool Reference Show Connector

Table 21: Icon Tool Tips and Pop-up Information (continued)

Icon Type	Tool Tip Information	Pop-Up Information
IN-List Spool	The spool name Estimated Cardinality	Icon Description Detail Information
Operation	Operation name Type of join used, in case of a join condition Type of row redistribution Join column in case of a join condition Number of columns involved in the join condition Estimated CPU time required to execute the step Estimated I/O time required to execute the step Estimated network time required to execute the step Estimated heuristic time required to execute the step Estimated time required to execute the step	Icon Description Detail Information
Statement	Statement text	Icon Description Detail Information
Step Number	Step text	Step text Compare Steps: • Base Step (makes the selected step the base for comparison in the Select Steps for Compare window) • Compare Step (signifies that the selected step is the one to compare to the base step) • Remove Sel (removes any previous base step or compare step selections)
Any blank area of the screen	Not applicable	Explain Text Statement Text Teradata Database Configuration Object Definitions Indexes Statistics Show Connectors Summary Information View Pad Index Analysis Object Use Count

Connectors

Teradata Visual Explain uses two types of connectors to show the relationship between the database objects. These connectors turn the individual icons of the visual plan into an easy-to-understand graphical flow chart. Table 22 describes the connector types.

Table 22: Connector Descriptions

Connector Type	Description
Solid line	The relationships between database objects displayed in the visual plan (for example, a table, a spool, and the result set). These connectors are always on.
Dashed line	Spool reuse in the visual plan. To view these connectors, click View>Show Connectors.

Clicking on a dashed-line connector changes the line to a solid line. Right-click on a connector to go to the source or go to a target.

Use the menu options to navigate between the steps where the spool is reused.

For additional information, see "Locating Spool References" on page 142.

Dialog Boxes and Shortcut Menus

Dialog boxes and shortcut menus make it easier to work with and modify plans. For more information about dialog boxes and shortcut menus, see:

- Dialog Boxes
- Shortcut Menus

Dialog Boxes

Use dialog boxes to establish application settings or to modify a plan and see the results of the modifications in order to optimize the plan.

To maximize screen space, the columns of most dialog boxes can be resized by dragging the column border in either direction until the intended size is reached. The size and state of the columns remain even after restarting Teradata Visual Explain.

Shortcut Menus

Use shortcut menus to access routine functions that appear in most dialog boxes. The options available on a shortcut menu are specific to the active window.

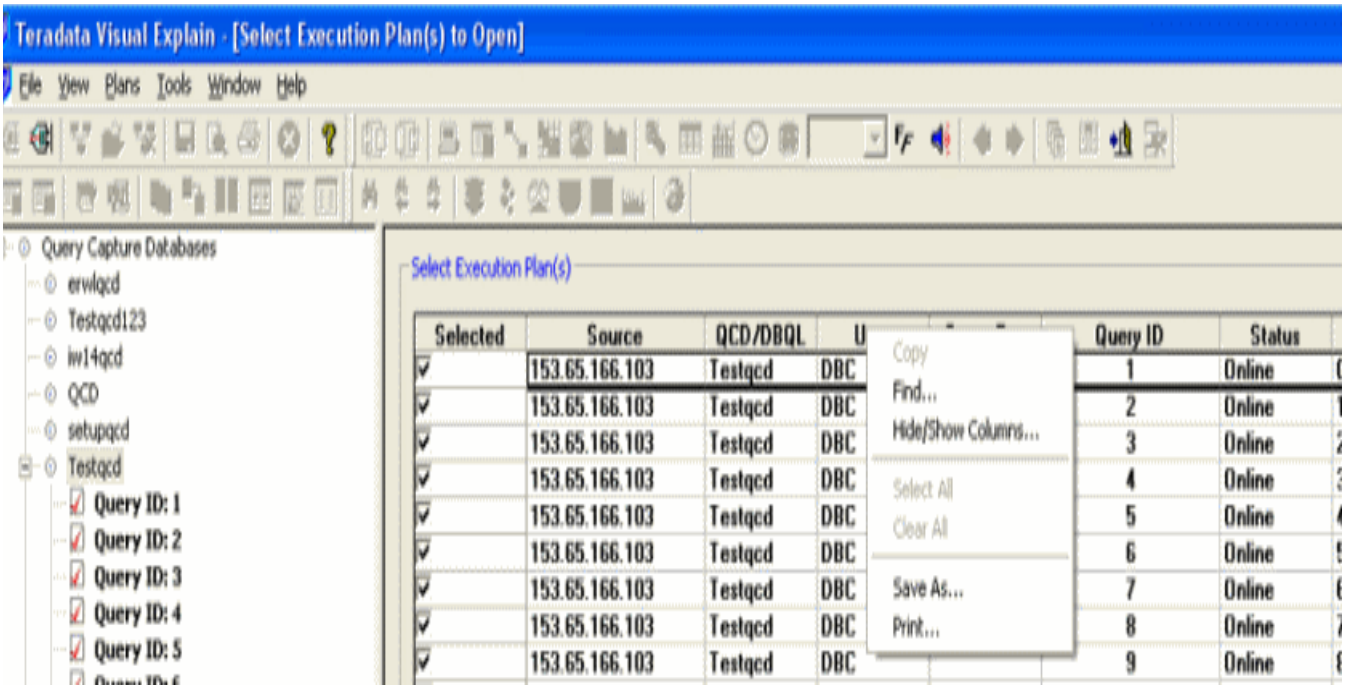
Available functions include:

- Copy (copy selected data to the Clipboard)

- Find (find items on a spreadsheet)
- Hide/Show (hide or show specific columns or all columns in a spreadsheet)
- Select All (select all rows in a spreadsheet)
- Clear All (clear all rows in a spreadsheet)
- Save As (save tables as text files [*.txt*] or Excel files [*.xls*])
- Print (print spreadsheets)

For example, right-click on a row heading or pane to access a shortcut menu (see Figure 5). If Hide/Show Columns is selected, the Column List dialog box appears. Use the Column List dialog box to choose which columns to hide or show or to restore the default column display.

Figure 5: Shortcut Menu Example



Online Help

Teradata Visual Explain has online help that can answer questions about how to use its features. Help opens in a browser window. Microsoft Internet Explorer, version 6.0 or later, is the recommended browser for viewing Teradata Visual Explain Online Help.

To open online help

- From the toolbar, click ?.

  From the online help window, navigate to or search for other topics.

To open context-sensitive help

- From most windows and dialog boxes, press `F1` to see context-sensitive help for the active window or dialog box.

Occasionally when `F1` does not open help, press `Shift+F1`.

CHAPTER 3

Getting Started Using Teradata Visual Explain

The following topics explain some basic functions of Teradata Visual Explain. Use these procedures to become more familiar with the way Teradata Visual Explain works:

- Specifying the Type of Connectivity
- Setting Up a Query Capture Database
- Upgrade or Revert QCD Version
- Displaying an Execution Plan
  - Viewing Explain Text
  - Viewing Statement Text
  - Adjusting Window Height and Width
- Entering a Query and Viewing Its Execution Plan
- Disconnecting and Closing Teradata Visual Explain

Other topics that might be useful when beginning to use Teradata Visual Explain include:

- Setting the Default Directory
- Setting Display Options
- Setting General Options
- View Low Level Error Log

Specifying the Type of Connectivity

The type of connectivity used to log on to Teradata Database can be specified before logging on or changed after logging on to Teradata Database. The available connectivity types are ODBC or CLIv2.

Note: A connection to Teradata Database is required in order to use most features in Teradata VE.

To specify the type of connectivity when not connected

1 From the menu bar, click Tools > Options.

   The Options dialog box appears, and the Connectivity tab is selected.

2 Select ODBC or CLIv2.

3 Click Apply.

4 Click OK to close the Options dialog box.

For information on how to log on using ODBC, see "Defining a Data Source and Connecting through ODBC" on page 56. To log on using CLIv2, see "Connecting through CLIv2" on page 57.

To change the connectivity type once connected

To change connectivity type after connecting to a Teradata Database, follow this procedure.

Note: If the connectivity type is changed while an execution plan is open, a message appears warning that information about the plan will be lost and providing the option to continue.

1 Click Tools > Options.

   The Options dialog box appears.

2 Click the Teradata Database Connectivity tab.

3 Click Specify Connection type (CLIv2/ODBC).

   The Teradata Database Connectivity dialog box appears.

4 Select ODBC or CLIv2.

5 Click OK.

   A message appears providing the option to terminate the connection.

6 Click Yes.

   The connection is terminated. Clicking No retains the current connection and returns to the Teradata Database Connectivity dialog box.

For information on how to log on using ODBC, see "Defining a Data Source and Connecting through ODBC" on page 56. To log on using CLIv2, see "Connecting through CLIv2" on page 57.

Defining a Data Source and Connecting through ODBC

Teradata Visual Explain can be connected to a Teradata Database using ODBC. Define a data source before connecting through ODBC.

To define a data source

1 Do one of the following:

- From the menu bar, click Tools > Define Data Source.
- From the menu bar:

  i Click Tools > Control Center.

  ii Click the Teradata Database Connectivity tab.

  iii Click Define Data Source.

The ODBC Data Source Administrator dialog box appears.

2 Fill in the boxes in the ODBC Data Source Administrator dialog box.

Refer to *ODBC Driver for Teradata User Guide* (B035-2509).

To connect using ODBC

1 Click .

The Select a Data Source dialog box appears.

2 Select a data source.

3 Click OK.

The Teradata Database Connect dialog box appears.

4 Fill in the boxes in the Teradata Database Connect dialog box.

5 Click OK.

Refer to *ODBC Driver for Teradata User Guide* (B035-2509).

Connecting through CLIv2

Teradata Visual Explain can be connected to a Teradata Database using CLIv2.

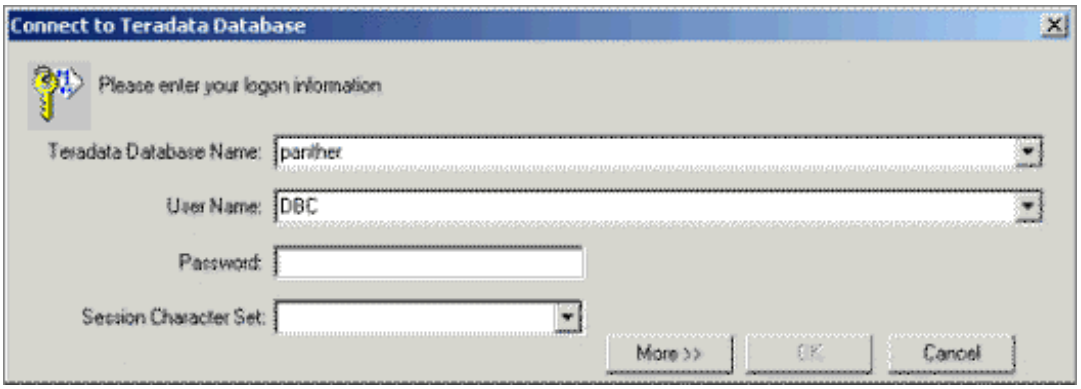
To connect using CLIv2

Set the connectivity type to CLIv2 before performing this procedure. See "Specifying the Type of Connectivity" on page 55.

1 Click .

The Connect to Teradata Database dialog box appears.

Figure 6: Connect to Teradata Database



2 Enter the logon information in the dialog box.

Box	Description
Teradata Database Name	Select a Teradata Database from the list or type a valid Teradata Database name. The list contains recently connected databases. The IP address of the server must be stored in the hosts file to use a Host name.
User Name	After selecting or entering the database name, the name of the user who last connected to the database appears in the User Name box. Use the name that appears, or type a different user name.
Password	Enter the password for the user name entered in the User Name box.
Session Character Set	[Optional] Select the session character set from the list. Choose one of the following: • ASCII • KANJISJIS_0S • UTF-16 • UTF-8
More	Click to view Advanced Options.
Authentication Mechanism	Select from the list of mechanisms that automatically appear in this box. Each Teradata site implements their own security measures. If a list of mechanisms does not appear and it is not possible to log on to Teradata Database, consult the Database Administrator or the person responsible for Teradata Database security.
Authentication Parameter	The authentication parameter is a password required by the selected mechanism. When typing in this box, asterisks appear to conceal the password. Contact the Database Administrator for a valid password.
Default Database	[Optional] Select the default database for the session.
Account String	[Optional] Specify the logon account string.
Less	Click to hide Advanced Options.

Setting Up a Query Capture Database

To capture and visualize new query execution plans using Teradata Visual Explain, a QCD must be set up.

To set up a new QCD

If a QCD was created for an earlier Teradata Database release, the QCD must be upgraded before being used with the current release. For instructions, see "Upgrade or Revert QCD Version" on page 60. To set up a new QCD, perform this procedure.

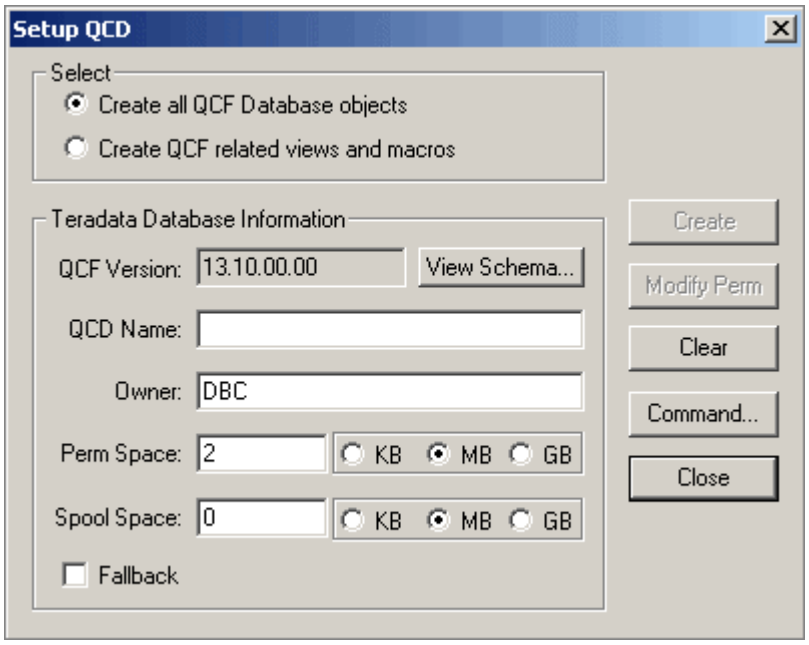
1 From the menu bar, click **Tools > Control Center**.

   The **Control Center** dialog box appears, and the **Manage QCD** tab is selected.

2 Click **Setup QCD**.

   The **Setup QCD** dialog box appears.

   Figure 7: Setup New QCD



3 Click **Create all QCF Database objects**.

4 In the **QCD Name** box, enter a name for the QCD.

5 In the **Owner** box, enter an owner name.

   If the box is left blank, the owner defaults to the name of the logged-on user.

6 The **Perm Space** box specifies the amount of permanent space for a QCD.

   a Enter a value greater than or equal to the default value. The default value is an estimate of space for a QCD.

   b Click one of the following units of space: **KB**, **MB**, or **GB**.

7 In the Spool Space box:

   a Enter a value.

   b Click **KB**, **MB**, or **GB**.

   The default is 0 MB.

8 [Optional] Select the **Fallback** check box to use fallback protection.

9 [Optional] Click **View Schema** to view the schema of the tables and macros created in the new QCD.

The **View Schema** dialog box appears and displays the DDL and macro definitions of the QCD. What is displayed depends on the version of QCF that is running.

10 [Optional] Click **Clear** to clear all boxes and restore the default options.

11 [Optional] Click **Command** to open the **Execute SQL** window.

For more information, see "Executing an SQL Query" on page 148.

12 Click **Create** to set up the QCD.

Note: Click **Close** to stop the set up of the QCD (before set up is complete) and return to the main window.

To modify the permanent space of a QCD

Use this procedure to modify the permanent space of a QCD.

1 Perform steps step 1 through step 3 of the procedure "To set up a new QCD" on page 58.

2 In the **QCD Name** box, enter the name of the QCD to be modified.

3 In the **Perm Space** box:

a Enter a value.

b Click **KB**, **MB**, or **GB**.

4 Click **Modify Perm** to modify the perm space of the existing QCD.

Upgrade or Revert QCD Version

Upgrade any QCD created prior to the latest Teradata Database release before using the QCD with the current release. With some limitations, it is also possible to revert to an earlier version.

Note: The following limitations apply when upgrading an existing QCD or reverting to an earlier version:

- If the existing QCD is version 03.00.00, upgrade to 03.01.00, and then upgrade to version 12.00.00, then upgrade to version 13.00.00 before upgrading to version 13.10.00.
- If a QCD contains Unicode® data and has been upgraded to version 12.00.00 or was created using version 12.00.00, the QCD *cannot* be reverted to the previous version.
- If a version 12.00.00 QCD contains only non-Unicode® data, it must be reverted to version 03.01.00 before being reverted to version 03.00.00.

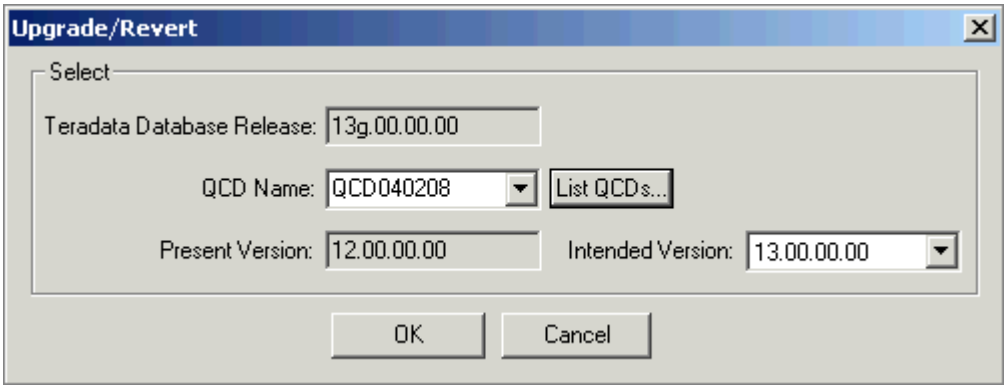
To upgrade an existing QCD or revert to an earlier QCD version

1 From the menu bar, click **Tools > Control Center**.

2 From the **Manage QCD** tab, click **Upgrade/Revert QCF Version**.

The **Upgrade/Revert** dialog box appears and displays the current Teradata Database version number in the **Teradata Database Release** box.

Figure 8: Upgrade/Revert Existing QCD



3 In the **QCD Name** box, do one of the following:

- Type the name of the QCD.
- Click **List QCDs** to select a QCD from the **QCD Name** list.

The **Present Version** box displays the selected QCF version.

4 [Optional] Click the **Intended Version** arrow to select a version to upgrade or revert to.

Note: The **Intended Version** list displays the available options (the next and previous version) unless QCF 12.00.00.00 or 03.01.00 is used.

5 Click **OK** to begin the conversion.

Displaying an Execution Plan

Use Teradata Visual Explain to present even the most complex textual Explain statement visually as a flow diagram of easily recognized icons. The plan must first be inserted into a QCD using Teradata Visual Explain or other applications, such as BTEQWIN or Teradata SQL Assistant.

Use Teradata VE to Insert an Execution Plan into a QCD

Use this procedure to insert a plan into a QCD using Teradata Visual Explain.

To insert an execution plan into a QCD

Connect to Teradata Database before inserting an execution plan.

1 From the menu bar, click **Tools > Execute SQL**.

   The **Execute SQL** window opens.

2 Do one of the following to execute a query containing the INSERT EXPLAIN statement:
   - To select a query to be executed from a file, click **File > Open Query**.
   - Enter the text for the query in the **Query** window.

3 Click **File > Execute Query**.

   The results of the query appear in the **Answer Set** window.

4 From the menu bar, click **File > Close** to close the **Execute SQL** window.

View the Execution Plan

View the execution plan graphically or textually from a QCD database or from the Database Query Log (DBQL) when:

- Loading plans associated with a workload
- Loading a specific plan
- Loading a range of plans
- Loading plans from a different server
- Loading one or more offline plans
- Loading one or more DBQL plans

  Note: DBQL XML plans larger than 31K cannot be viewed using this procedure when connected to Teradata Database 13.00. This limitation is removed for Teradata Database 13.10 and higher.

Use the Explain graphics or text to check for query performance issues.

Note: After a plan is loaded using the **Select Execution Plan(s) to Open** dialog box, right-click on a column heading under **Available Execution Plans** to access the shortcut menu. For more information, see "Shortcut Menus" on page 51. Also, the columns can be resized. For more information, see "Dialog Boxes" on page 51.

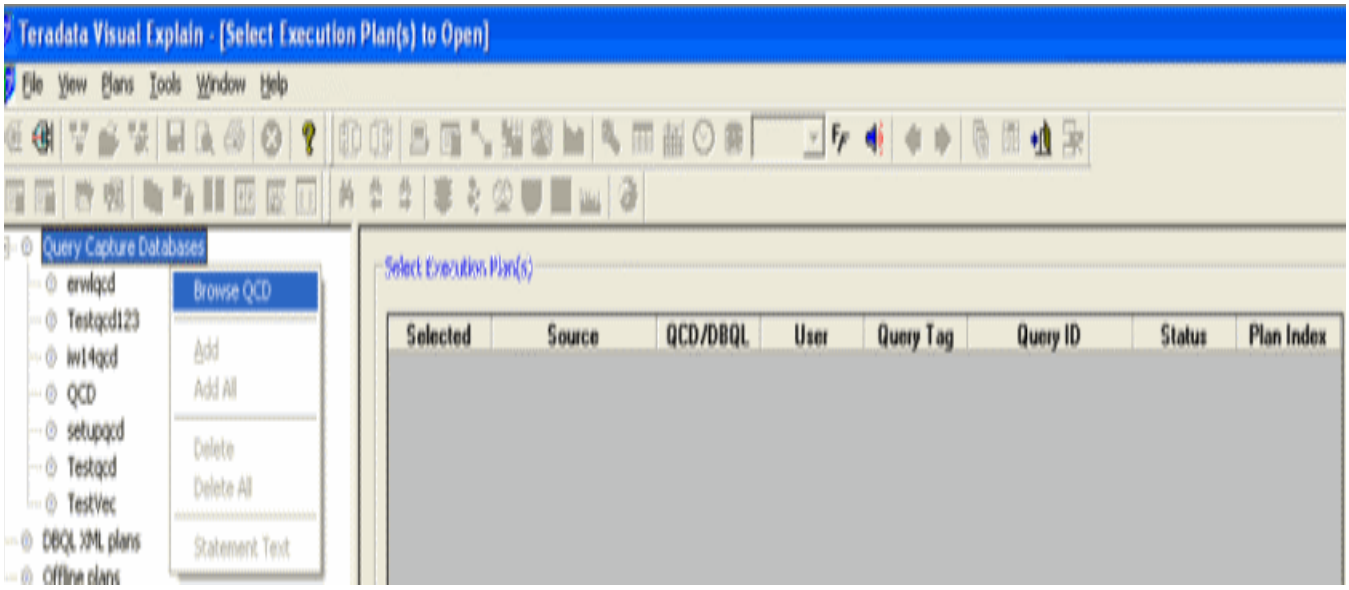
To view the execution plan graphically or textually from a QCD database (workload)

After inserting an execution plan into a QCD using INSERT EXPLAIN, view the Explain results graphically or textually.

1 Click .

The **Select Execution Plan(s) to Open View** window appears.

Figure 9: Select Execution Plan(s) to Open View (Workload)

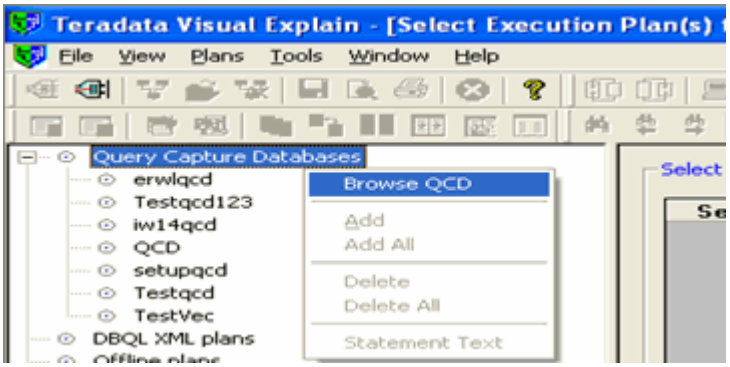


2 Use the following steps to select a QCD:

a In the **Tree View**, right-click on the **Query Capture Databases Node** to display a shortcut menu.

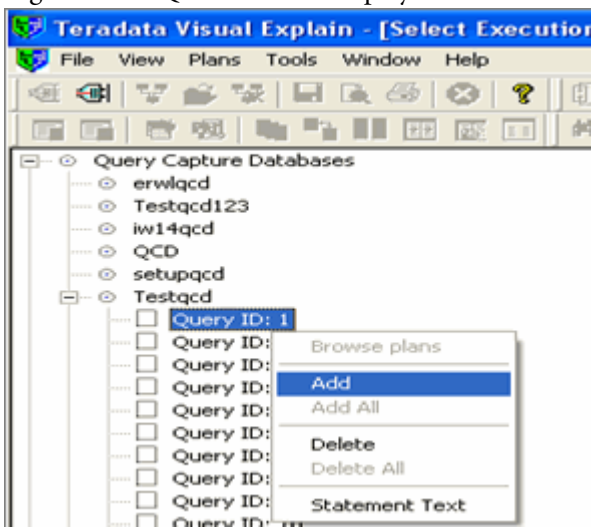
b In the shortcut menu, select **Browse QCD** to list all Available QCD names under **Query Capture Databases Node**. See figure below.

Figure 10: Browse QCD



3 Select a QCD from the list.

a Right-click a QCD name to display a menu.



b Select the **Browse Plans** menu item to display the **Browse QCD Plans** dialog box.

Figure 11: Browse QCD Plans

c Use one of the following options:

- Enter the workload name in the **Workload** box.
- Click the arrow button in the **Workload** box to select a workload from a list of available workloads.
- Click the **XML Plans** checkbox to view XML plans available in the selected QCD.

Note: This option is available only for Teradata Database version 13.10 and later.

4 Click the **OK** button to retrieve the specified plans.

5 Select the checkbox next to one or more plans in the **Available Execution Plans** list.

Note: Click **Add All** to all available plans to the **Selected Execution Plans** list, and skip step 6.

6 [Optional] To see complete statement text, select a QueryID in the **Available** list, right-click and select the **Statement Text** menu item for a specific QueryID. For more information on the **Statement Text** dialog box, see "Viewing Statement Text" on page 77.

Click **Open** to display the plans. The plans appear in the main window. for more information, see "Summary Information Window" on page 25.

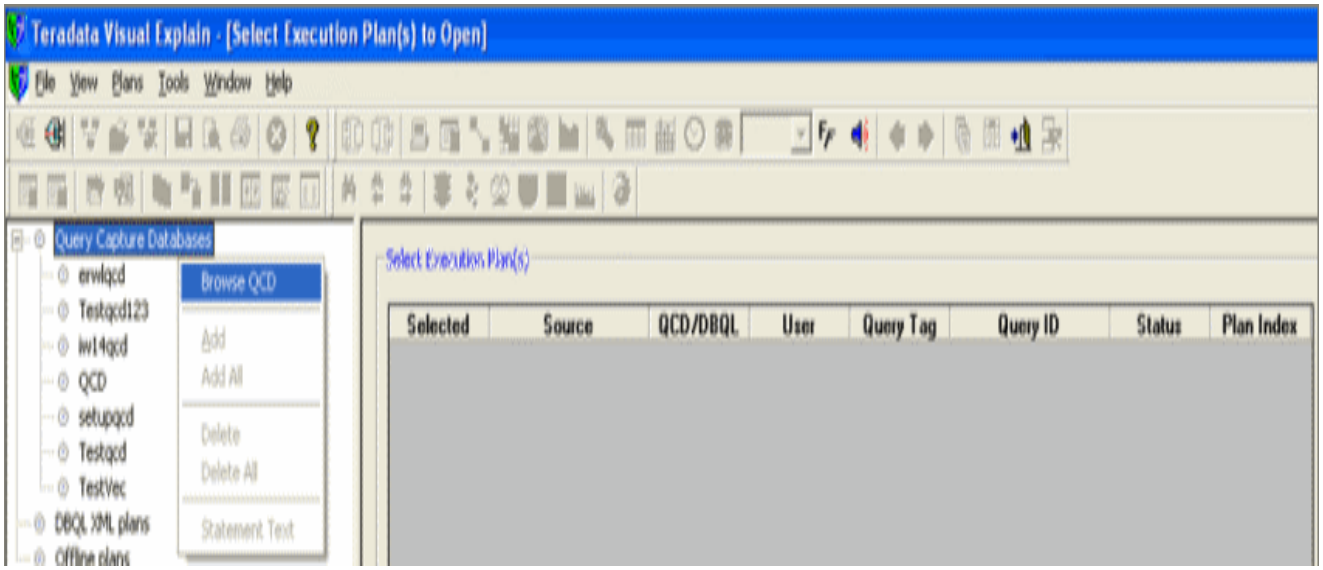
To view the execution plan graphically or textually from a QCD database (specific plan)

After inserting an execution plan into a QCD using INSERT EXPLAIN, view the Explain results graphically or textually.

1 Click .

The Select Execution Plan(s) to Open View window appears.

Figure 12: Select Execution Plan(s) to Open View (Specific Plans)

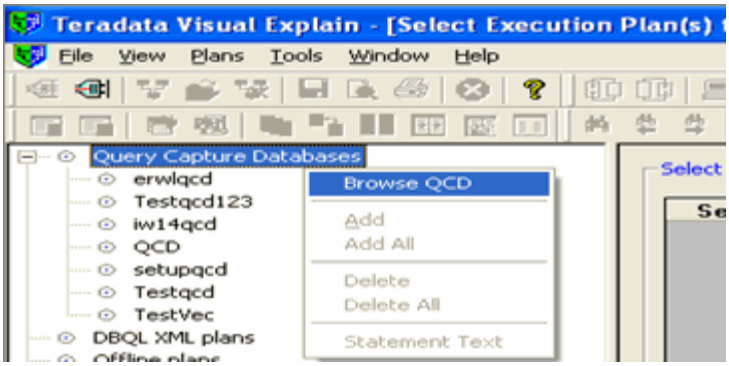


2 Use the following steps to select a QCD:

a In the Tree View, right-click on the Query Capture Databases Node to display a shortcut menu.

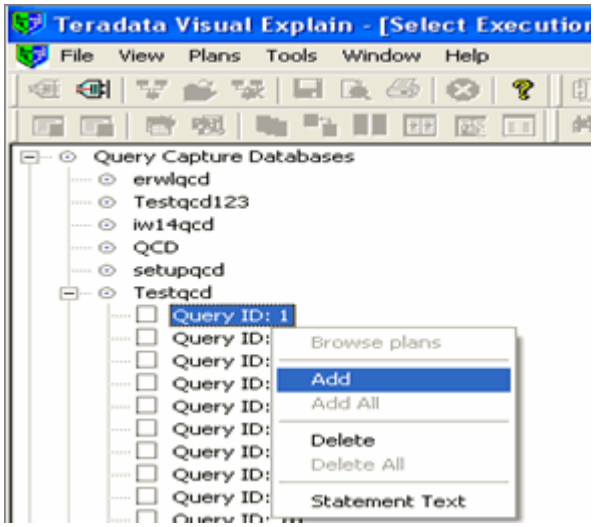
b In the shortcut menu, select Browse QCD to list all Available QCD names under Query Capture Databases Node. See Figure 13 below.

Figure 13: Browse QCD



3 Select a QCD from the list.

a Right-click a QCD name to display a shortcut menu.



b Select the **Browse Plans** menu item to display the **Browse QCD Plans** dialog box.

Figure 14: Browse QCD Plans

c Use one of the following options:

- Enter a query tag name in the **Query Tag** box.
- Enter the query ID number in the **Query ID** box.

Note: In the **Query Tag** box, the symbols % and _ (underscore) are wildcards that can be used independently or in combination. The % symbol represents any string of zero or more arbitrary characters. For example, query% matches query, query1, query12, and so on. The _ symbol represents one arbitrary character. Therefore, any single character is acceptable in the position in which the _ appears.

4 Select **XML Plans** to view the available XML plans in the selected QCD.

Note: This option is available only for Teradata Database 13.10 and later.

5 Click the **OK** button to retrieve the specified plans.

6 Select the checkbox next to one or more plans in the **Available Execution Plans** list.

Note: Click **Add All** to add all available plans to the Selected Execution Plans list and skip step 7.

7 [Optional] To see complete statement text, select a QueryID in the Available List, right-click to select the **Statement Text** menu item for a specific QueryID. For more information on the **Statement Text** dialog box, see "Viewing Statement Text" on page 77.

8 Click **Open** to display the plans. The plans appear in the main window. for more information, see "Summary Information Window" on page 25.

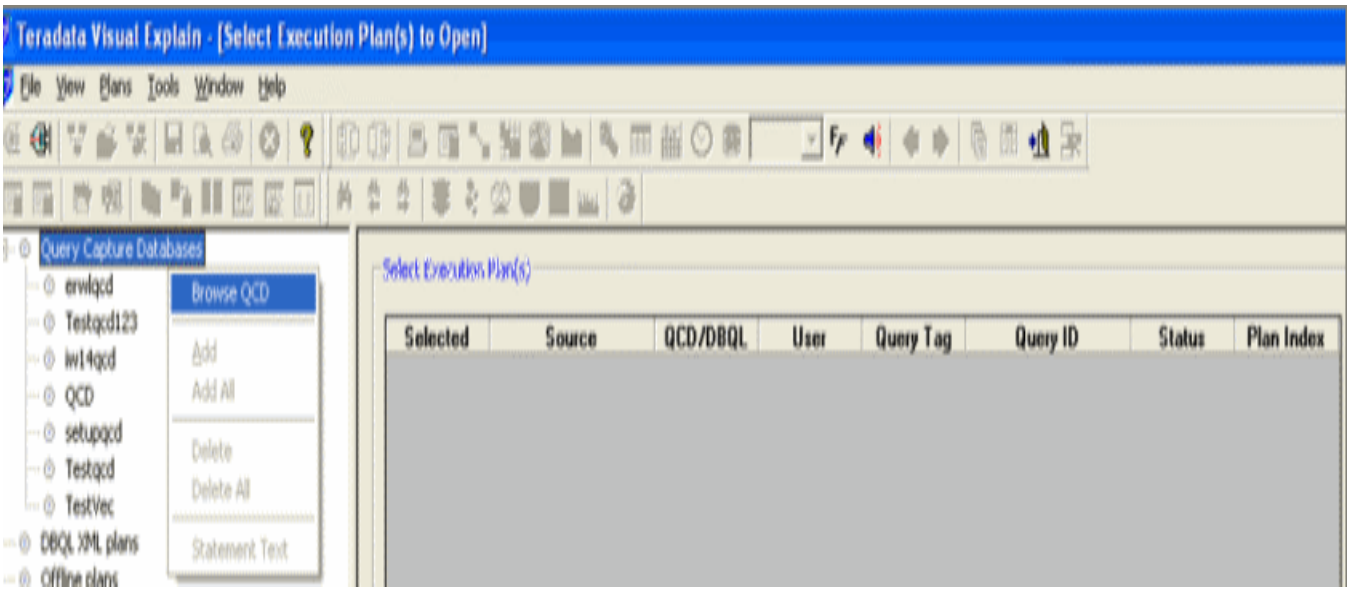
To view the execution plan graphically or textually from a QCD database (range of plans)

After inserting an execution plan into a QCD using INSERT EXPLAIN, view the Explain results graphically or textually.

1 Click .

The **Select Execution Plan(s) to Open View (Range of Plans)** window appears.

Figure 15: Select Execution Plan(s) to Open View (Range of Plans)



2 Use the following steps to select a QCD:

- In the **Tree View**, right-click on the **Query Capture Databases** node to display a shortcut menu.
- In the shortcut menu, select **Browse QCD** to list all Available QCD names under **Query Capture Databases Node**.

3 Select a QCD from the list. See Figure 13 on page 65.

a Right-click a QCD name to display a shortcut menu.

b Select the Browse Plans menu item to display the Browse QCD Plans dialog box.

c In the Query ID box, enter a range such as, 1-100, or 1,3,6-10.

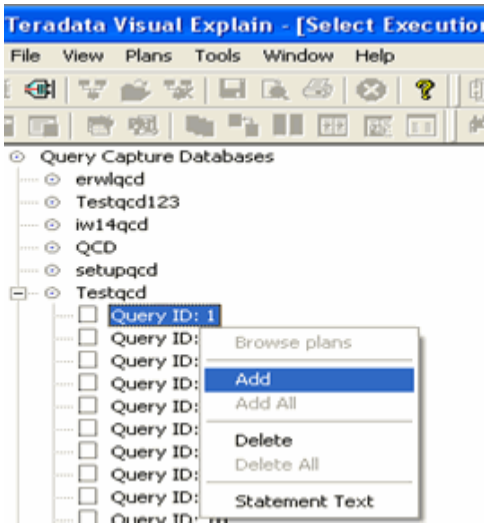


Figure 16: Browse QCD Plans

4 Select XML Plans to view available XML plans in the selected QCD.

Note: This option is available only for Teradata Database version 13.10 and later.

5 Click the OK button to retrieve the specified plans.

6 Select the check box next to one or more plans in the Available Execution Plans list.

Note: Click Add All to add all available plans to the Selected Execution Plans list and skip step 7.

7 [Optional] To see complete statement text, select a QueryID in the Available list, right-click to select the Statement Text menu item for a specific QueryID. For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

8 Click Open to display the plans.

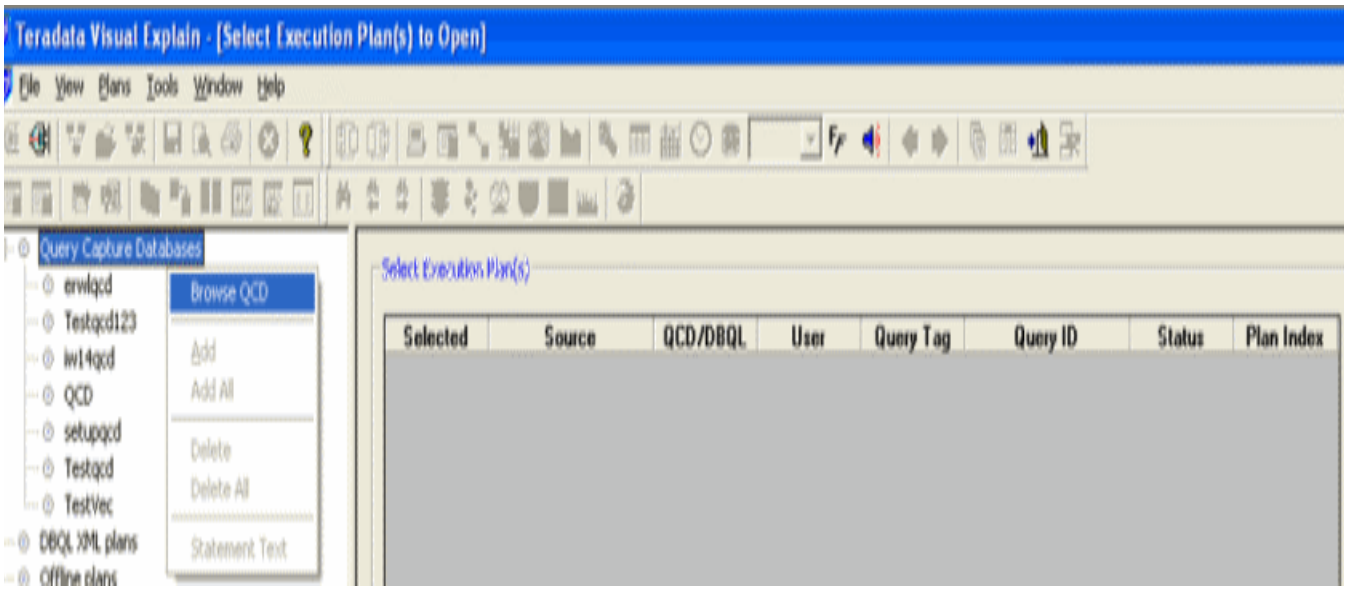
To view the execution plan graphically or textually from a QCD database (different server)

After inserting an execution plan into a QCD using INSERT EXPLAIN, view the Explain results graphically or textually.

1 Click .

The Select Execution Plan(s) to Open View (Different Server) window appears.

Figure 17: Select Execution Plan(s) to Open View (Different Server)



2 Use the following steps to select a QCD:

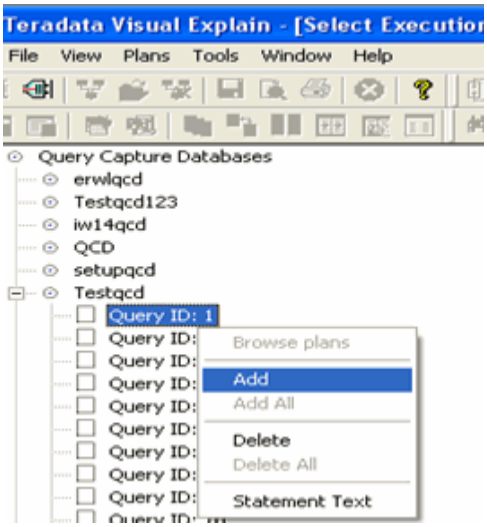
- In the Tree View, right-click on the Query Capture Databases node to display a shortcut menu.
- In the shortcut menu, select Browse QCD to list all Available QCD names under Query Capture Databases Node.

3 Load plans from a different server:

a Click the Connect button in the toolbar of the Open Plan window.

b Enter logon information.

c In the shortcut menu, select Browse QCD to list all Available QCD names under Query Capture Databases Node. In the shortcut menu, select Browse QCD to list all Available QCD names under Query Capture Databases Node.



4 Under the Query Capture Databases Node, Select a specific QCD. Right-click on the QCD to display the shortcut menu. Select the Browse Plans menu item to display the Browse QCD Plans dialog box. Select XML Plans to view the available XML plans in the selected QCD.

Note: This option is available only for Teradata Database version 13.10 and later.

Figure 18: Browse QCD Plans

5 Click the OK button to retrieve the specified plans.

6 Select the check box next to one or more plans in the Available Execution Plans list.

Note: Click Add All to add all available plans to the Selected Execution Plans list and skip step 7.

7 [Optional] To see complete statement text, select a QueryID in the Available list, right-click to select the Statement Text menu item for a specific QueryID. For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

8 Click Open to display the plans.

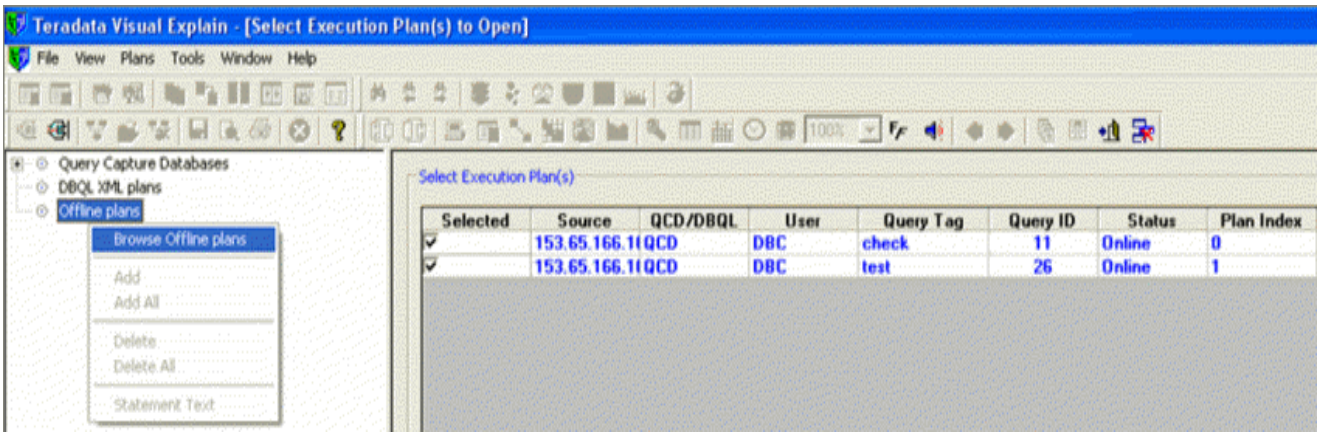
To view the execution plan graphically or textually from a QCD database (offline plans)

After inserting an execution plan into a QCD using INSERT EXPLAIN, view the Explain results graphically or textually.

1 Click .

The Select Execution Plan(s) to Open View window appears.

Figure 19: Select Execution Plan(s) to Open View (Offline Plans)



2 Load one or more offline plans:

a Right-click the **Offline Plan** menu item and select **Browse Offline Plans** to display the **Open** dialog box.

b From the **Open** dialog box, choose one or more plans.

When selecting a file, the description of the plan appears in the **Description** box. If the file contains more than one plan, the **Description** box indicates how many plans the file contains.

c Click **Open**.

3 Right-click on the **Plan Node** and select the **Add** menu item to add the single plan to the **Selected Execution Plans** list.

4 [Optional] To see complete statement text, select a QueryID in the Available list, right-click to select the **Statement Text** menu item for a specific QueryID. For more information on the **Statement Text** dialog box, see "Viewing Statement Text" on page 77.

5 Click **Open** to display the plans.

The plans appear in the main window. For more information, see "Summary Information Window" on page 25

To view the execution plan graphically or textually from the DBQL

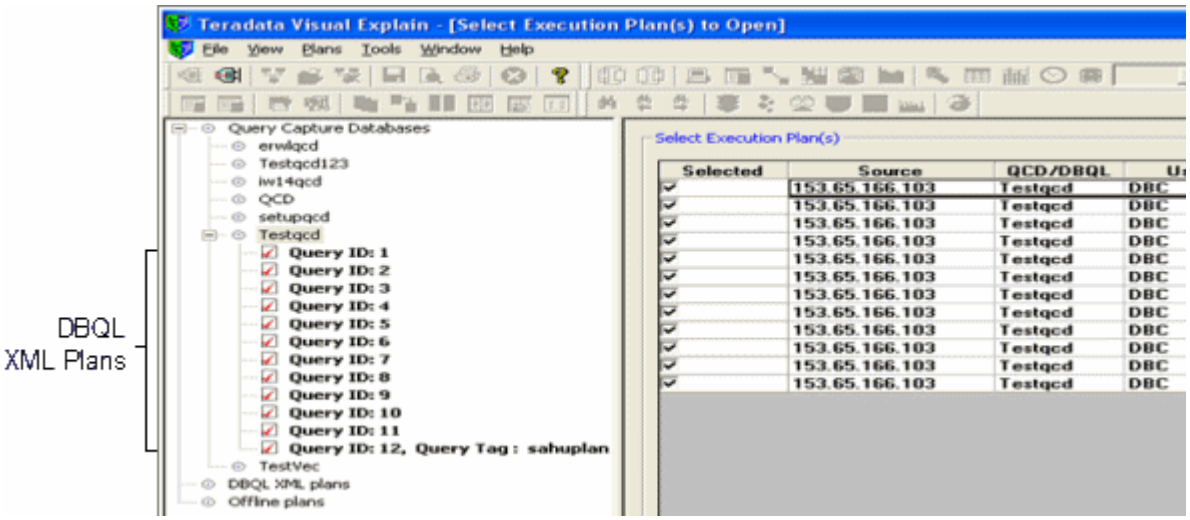
View the Explain results graphically or textually after the XML execution plan has been logged into the DBQL.

Note: DBQL XML plans larger than 31K cannot be viewed using this procedure when connected to Teradata Database 13.00. This limitation is removed for Teradata Database 13.10 and higher.

1 Click .

The **Select Execution Plan(s) to Open View** (DBQL) window appears.

Figure 20: Select Execution Plan(s) to Open View (DBQL)



2 In the tree view, under **Available Plans**, right-click on the **DBQL XML Plans** node and select the **Browse DBQL Plans** menu item.

The **Select DBQL Criteria** dialog box opens.

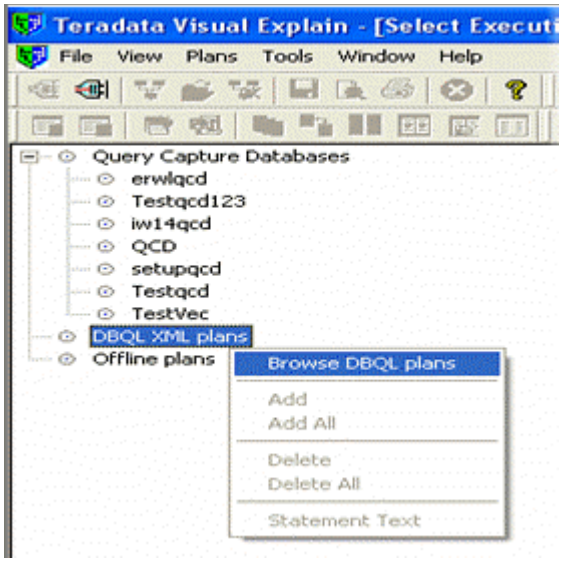
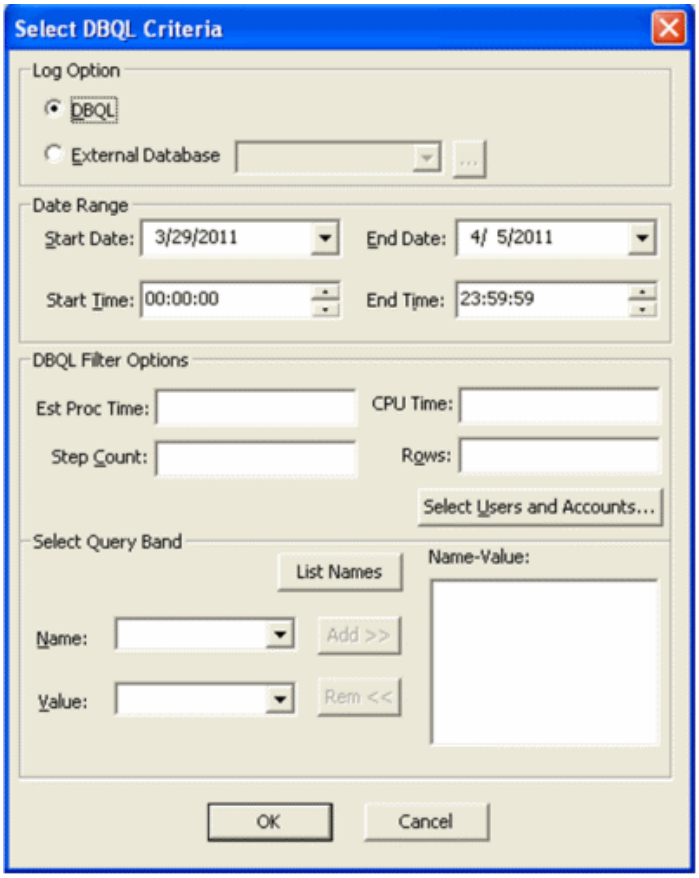


Figure 21: Select DBQL Criteria



3 Under **Log Option**, select one of the following:

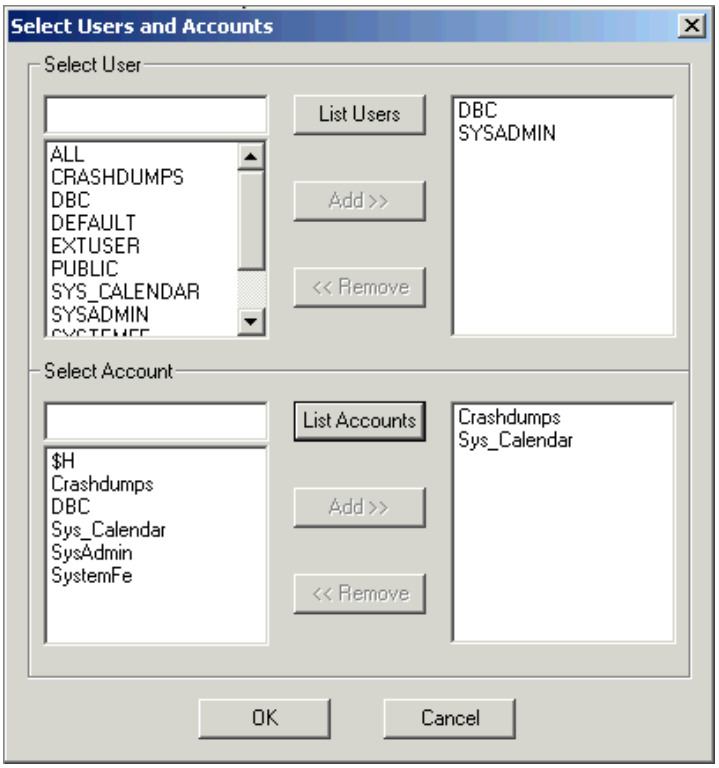
- DBQL (default) – to browse XML queries from the DBQL.
- External Database – to browse XML queries from an external database.

4 [Optional] Under DBQL Filter Options, enter filter options:

- Est Proc Time – is the minimum estimated processing time for the queries received from the DBQL.
- CPU Time – is the minimum processing time for the queries received from the DBQL.

Descriptions of DBQL criteria selection are given in the following table.

Group	Subgroup	Box or Button	Description
Select Criteria	Select Date	Start Date	The start date, in the format *MM/DD/YYYY*, to use when searching for queries on or after the date. If no start date is specified, all statements from the DBQL are loaded.
		End Date	The end date, in the format *MM/DD/YYYY*, to use when searching for queries on or before the date. If a start date is specified but no end date is specified, all statements from the DBQL beginning at the start date are loaded.
	Select Time	Start Time	A start time in the format *HH:MM:SS* to use in searching for queries on or after the start time. If no start time is specified, the default start time of 00:00:00 is used.
		End Time	An end time in the format *HH:MM:SS* to use in searching for queries on or before the end time. If no start time is specified, the default end time of 13:59:59 is used.
	None	Step Count	The minimum number of steps for the queries received from the DBQL.
	None	Result Rows	The minimum number of rows returned for the queries retrieved from the DBQL.

Group	Subgroup	Box or Button	Description
	None	Select Users and Accounts	Use this button to open the Select Users and Accounts dialog box and browse queries submitted by a set of users or accounts. From the Select Users and Accounts dialog box, browse a list of users and accounts, and select one or more users or one or more accounts to include in the search. Selected user or account names are displayed on the right-hand side of the dialog box. 
Select QueryBand	None	Name	The name of a Query Band Name-Value pair. See "SET QUERY_BAND" in *SQL Data Definition Language* B035-1144 for more information.
		Value	The value of a Query Band Name-Value pair.
		List Names	Use to list all the Query Band Names in the Name-Value list.
		Add	Use to add the selected Query Band Name-Value pair to the Name-Value list.
		Rem	Use to remove the selected Query Band Name-Value pair from the Name-Value list.
		Name-Value	The combination of the Name and Value entries (Name-Value pair).
		Filter	Use to filter out some of the Query Band Name-Value pairs.
	Estimated Time Proc		The minimum estimated processing time for the queries received from the DBQL.

Group	Subgroup	Box or Button	Description
	CPU Time		The minimum processing time for the queries received from the DBQL.

5 Click OK to close the Select DBQL Criteria window.

6 In the Select Execution Plan(s) to Open View (DBQL Plans) window, right-click on the QueryID and select the Add menu-item checkbox in the Available Execution Plans list.

Note: Right-click a specific DBQL and select Add All to add all available plans to the Selected Execution Plans list.

7 [Optional] To see a complete statement text, right-click Statement Text and select a statement and click. For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

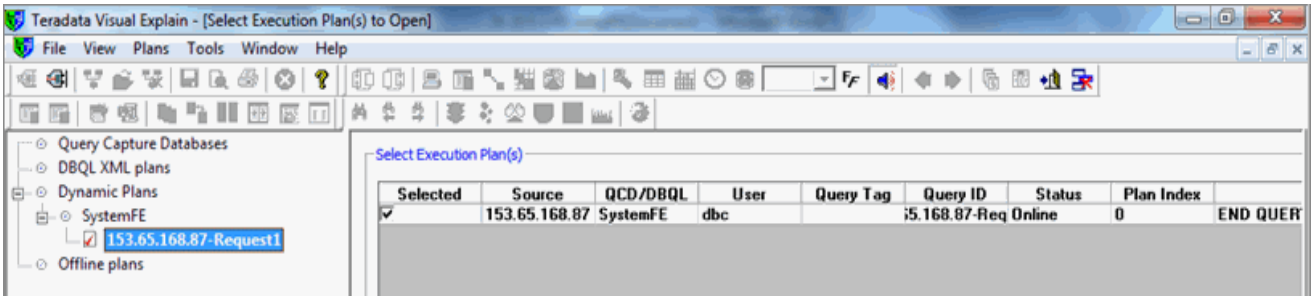
8 Click Open to display the plans.

9 Click OK to close the Select Criteria dialog box.

To view the Dynamic execution plans graphically or textually

After Capturing a Dynamic (IPE) plan into SystemFE.Opt_XMLPlan_Table, view the Explain results graphically or textually. Select Dynamic Plans from the following screen using the steps below the screen.

Figure 22: Select Execution Plans to View Dynamic Plans



1 In the Tree View, right-click on the Dynamic Plans node to display a shortcut menu.

2 In the shortcut menu, select Browse Dynamic Plans to list SystemFE node and right-click SystemFE shortcut menu to display all Available Dynamic Plan Names under Dynamic Plans Database Node.

3 Right-click on the Plan Node and select the Add menu item to add the single plan to the Selected Execution Plans list.

4 [Optional] To see complete statement text, select a Request Name in the Available list, right-click to select the Statement Text menu item for a specific Request. For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

5 Click Open to display the plans.

Other Options

Table 23 lists ways to view additional plan information about the execution plan using the menu bar.

Table 23: View Additional Plan Information

Task	Instructions	For more information
View statement text	Click **Plans > Statement Text.**	"Viewing Statement Text" on page 77
View Explain text	Click **Plans > Explain Text.**	"Viewing Explain Text" on page 76
View step text	1 Position the cursor over the step number. 2 Right-click the mouse button. 3 Click **Step Text.**	Not applicable
View summary information	Click **View > Summary Information.**	"Summary Information Window" on page 25
Navigate larger plans	Click **View > View Pad.**	"Navigating Execution Plans" on page 141
View the data-flow layout of the execution plan	Click **View > Compressed.**	Not applicable
View the control-flow layout of the execution plan	Click **View > Expanded.**	Not applicable

To view a previously saved execution plan, see "To load and view a saved execution plan" on page 140.

Viewing Explain Text

After an execution plan displays, view the Explain text for the execution plan.

To better understand and interpret the Explain text of an execution plan, see *SQL Data Manipulation Language* B035-1146 and refer to the section "EXPLAIN Modifier."

To view Explain text

- Click .

  The Explain text appears in a separate window to the left of the plans.

  Note: Click to toggle between opening and closing the window that displays the Explain text.

Viewing Statement Text

To view statement text for an execution plan from the main window

- Click .

  The statement text appears in a separate window at the bottom of the plan.

  Note: Click to toggle between opening and closing the window that displays the statement text.

To view statement text during procedures

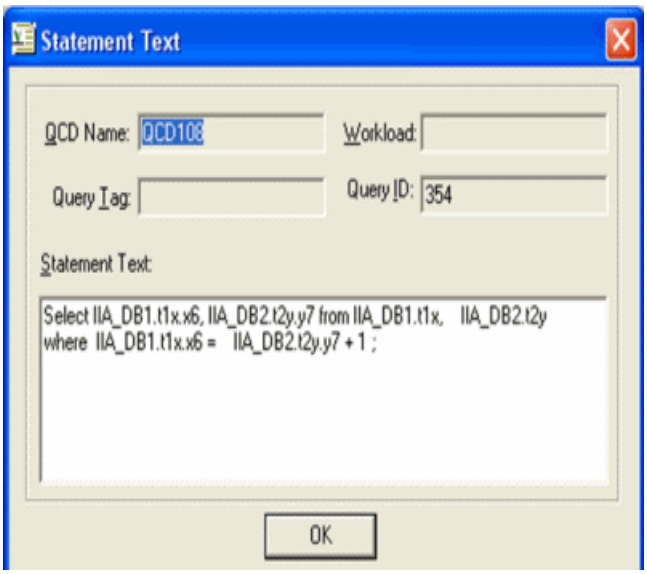
Complete statement text can be viewed while deleting execution plans, loading and viewing execution plans, textually comparing execution plans, or visually comparing execution plans.

1 Select a statement.

2 Click .

  The Statement Text dialog box appears.

  Figure 23: Statement Text



Adjusting Window Height and Width

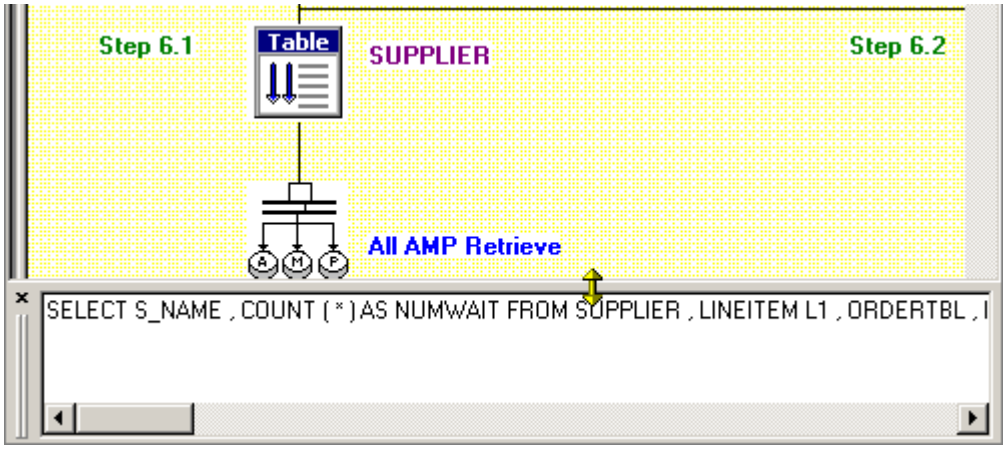
In most windows, the window height and width can be adjusted.

To adjust the window height

1 Place the mouse pointer to the line directly above the window.

  When correctly positioned, the mouse pointer changes to a two-headed vertical arrow.

Figure 24: Adjust Window Height



2 Drag the arrow up or down to size the window, releasing the mouse button when the intended size is reached.

To adjust the window width

1 Place the mouse pointer on the line directly to the right of the window.

When correctly positioned, the mouse pointer changes to a two-headed horizontal arrow.

2 Drag the arrow left or right to size the window and then release the mouse button.

Entering a Query and Viewing Its Execution Plan

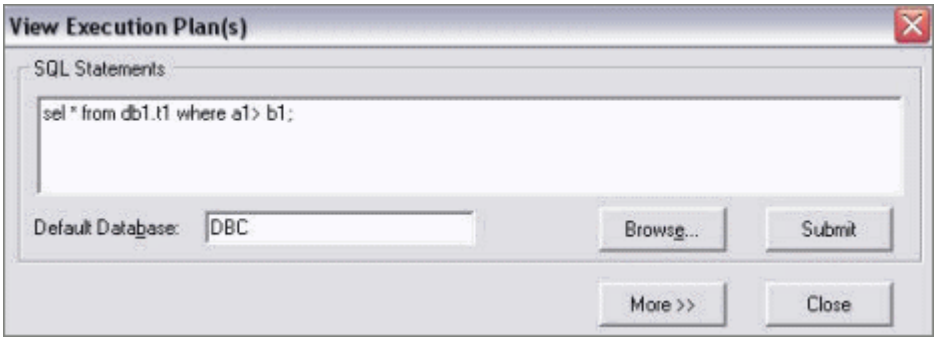
As stated, a query can be run using the INSERT EXPLAIN statement, then located, and the results displayed. Alternatively, Teradata Visual Explain can be used to enter the statement text and immediately view the execution plan for that query.

To enter a query and view its execution plan

1 Click **Tools > View Execution Plan(s)**.

The **View Execution Plan(s)** dialog box appears.

Figure 25: View Execution Plan(s) Dialog Box

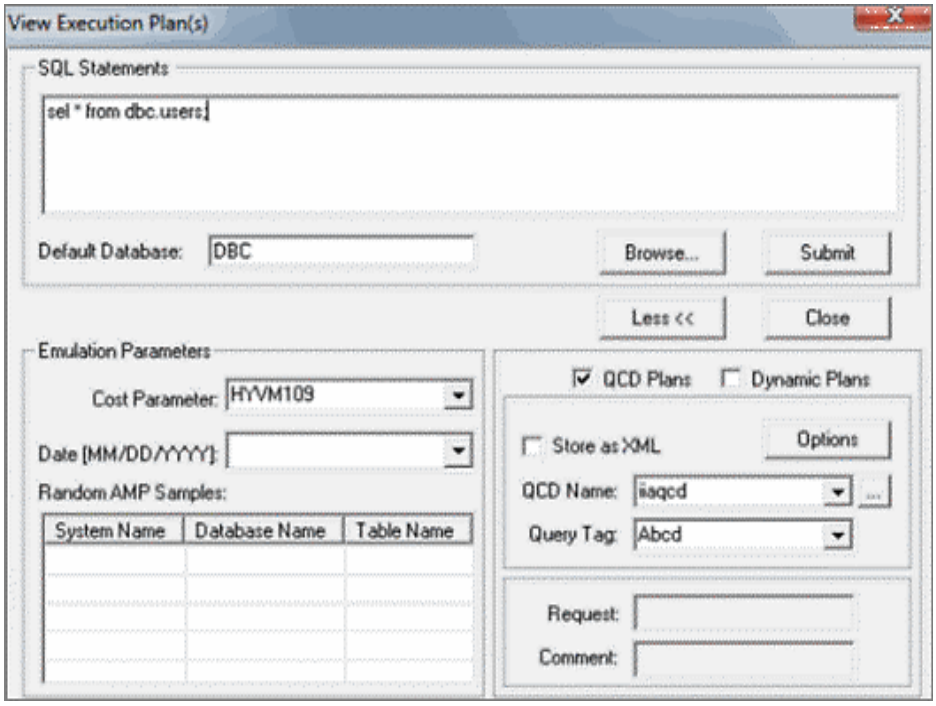


2 Use one of the following options:

- The default option is **Explain in XML**, which generates an execution plan but does not store plan information in the database.
- To insert execution plan information into the specified QCD, click the **More** button and select the **QCD Plans** checkbox.
- To capture the execution plan in XML form in QCD (i.e to submit INSERT EXPLAIN in XML), select the **Store as XML** checkbox.
- To capture dynamic (IPE) plan information, click the **More** button and select the Dynamic Plans checkbox. Enter a Request Name and optional Comment in the respective edit boxes.

  Note: This option is enabled only if the **Capture in QCD** checkbox is selected.

Figure 26: View Execution Plan - Expanded



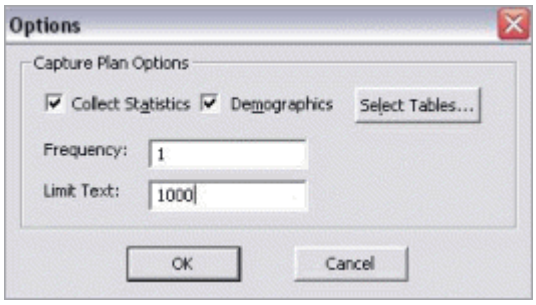
3 [Optional] To place a limit on the size of the query and DDL text captured in the QCD:

a Click the **Options** button to open the **Options** window.

b Enter **Limit Text** in the **Limit Text** checkbox and click **OK**.

If no value is entered, 0 is the default.

Figure 27: Options window



4 [Optional] Use the following steps to specify the number of times an SQL statement is typically performed within its identified workload:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Enter a value in the **Frequency** box in the **Options** window and click **OK**.

This value is used to weight the respective benefits of each column analyzed for inclusion in the index recommendation computed by Teradata Index Wizard. Any positive integer up to 4 B is valid. If a value is not specified, frequency defaults to 1.

5 In the **Default Database** box, enter the name of the default database.

6 In the **QCD Name** box, enter the name of the QCD database.

Note: This field is enabled only when **Capture in QCD** is selected.

7 In the **Query Tag** box, enter a name used to identify the query.

8 Do one of the following:

- To enter statement text, enter it in the **Query** window.
- To open a file containing statement text:
  - i Click **Browse**.
  - ii Locate the *.sql* or *.txt* file.

9 [Optional] To collect statistics on tables involved in the plan:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Select the **Collect Statistics** checkbox in the **Options** window.

Statistics are collected on columns that are index candidates and for all columns and indexes referenced explicitly with values in the SQL statement when the **Collect Statistics** checkbox is selected. A candidate index column is defined as a column for which the value or join-range access frequencies stored in the **Field** table of the specified QCD are greater than 0.

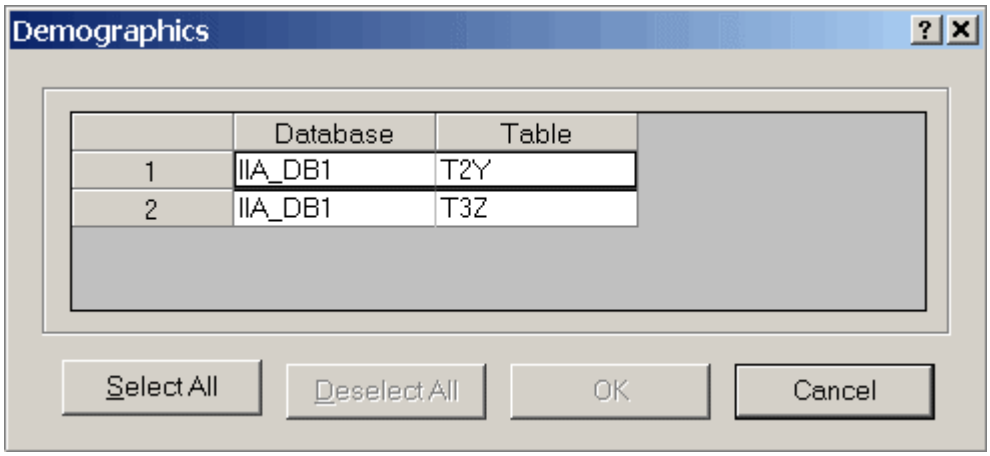
10 [Optional] To collect demographics on some tables involved in the plan:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Select the **Demographics** checkbox.

d Click **Select Tables**.



11 Select the tables.

12 Click OK.

13 To specify the set of tables for which data demographics and statistics details are either included in or excluded from analysis, click Select Tables.

14 To emulate a target environment at various levels or scopes.

Note: Only the DBC or SystemFE user can set costs at the IFP and SYSTEM levels. An error occurs when these conditions are not met.

For more information, see "Setting or Resetting Random AMP Samples" on page 145 or "Setting or Resetting Cost Parameters" on page 146.

15 Click Submit to create the execution plan.

Disconnecting and Closing Teradata Visual Explain

When finished using Teradata Visual Explain, disconnect from the Teradata Database using the Disconnect feature. Closing Teradata Visual Explain also disconnects the session from a Teradata Database.

Disconnecting from a Teradata Database

To disconnect from a Teradata Database

- Click .

To connect to another database, see "Defining a Data Source and Connecting through ODBC" on page 56. To close Teradata Visual Explain, see "Closing Teradata Visual Explain" on page 82.

Closing Teradata Visual Explain

When finished using Teradata Visual Explain, close the application and return to the Windows desktop.

To close Teradata Visual Explain

1 Click File > Exit.

If connected to a database, a confirmation message appears.

2 Click Yes to close Teradata Visual Explain and return to the Windows desktop.

Setting the Default Directory

Set the default directory where Teradata Visual Explain saves files such as *.bcf, .bteq, .dat, .ddl, .dem, .log, .mcf, .opt, .qcd, .ras, .rpt, .sql, .tmp, .txt, .undo, .vec*.

To set the default directory

1. From the menu bar, choose **Tools > Options**.
2. Click the **Path** tab.

   The **Path** box indicates the current default directory.
3. [Optional] To change the default directory, click **Browse**.

   The **Browse for Folder** dialog box appears.
4. Choose a new folder as the default directory
5. Click **OK**.

   The new directory appears in the **Path** box.
6. Click **OK** to close the **Options** dialog box.

Setting Display Options

Change the display options to change the number of:

- Characters and lines displayed in Tool Tips
- Characters displayed for table names in the visual plan
- Entries saved in the registry

To set display options

1. Click **Tools > Options**.
2. Click the **Display** tab.
3. Fill in the boxes.

Box	Description
Tip text limit: Characters	Maximum number of characters displayed in tool tips
Tip text limit: Lines	Maximum number of lines displayed in tool tips
Table name length	Maximum number of characters displayed for table names in the visual plan
Keep History	Maximum number of entries that can be saved in the registry

Setting General Options

A number of general options can be set in Teradata Visual Explain.

To set general options

1 Click **Tools > Options.**
2 Click the **General** tab.
3 Complete the dialog box.

Check Box	Description
Provide audible notification	If selected, the system emits a tone: • When closing a plan window • When finding steps in the plan • While loading queries in the **Select Execution Plan(s)** dialog box
Use X views (DBC and QCD)	If selected, X views are used to access Data Dictionary and QCD data. X views limit access to those plans owned (Data Dictionary and QCD data captured) by the user currently logged-on. See Chapter 5: "Restricted DBC Views" for information on restricted views.
Use X views (QCD only)	If selected, X views are used to access QCD data. X views limit access to those plans (QCD data only) that are owned (or were captured) by the logged-on user. This option can be used, along with the predefined user categories of the Grant/Revoke Privileges on QCD feature, to control plan access. For more information, see "Changing User Rights on the QCD" on page 102. See Chapter 5: "Restricted DBC Views" for information on restricted views.
Use multiple windows for demographics display	See "Chapter 9 Data Demographics" on page 167.
Open separate window for each table	See "Chapter 9 Data Demographics" on page 167.
Use SQL Assistant instead of Execute SQL Window	If selected, invokes Teradata SQL Assistant as the interface for executing an SQL query. See "Executing an SQL Query" on page 148.

View Low Level Error Log

Teradata Visual Explain logs low level information useful for GSC in the *VECOMP.LOG* file located in following directory.

Drive:\Documents and Settings\user\My Documents\Teradata\DBM Logs

CHAPTER 4

Privileges

Specific privileges to Teradata Database are required in order to use Teradata Visual Explain. The following topics explain the access requirements:

- Using the Basic Functions
- Using the Administrative Functions

Using the Basic Functions

The following are required to use the basic functions of Teradata Visual Explain:

- Access to a Teradata Database system with a valid Teradata Database user ID and password. The user must log on to a Teradata Database system after starting the application and before loading the queries.
- The SELECT access right on the QCD (used by Teradata Visual Explain to display the execution plan) is required for QCDs of version 03.00.00 and later. Also Exec on QCD macros is required.

The following categories of users are defined on a QCD to enhance its security:

- Normal user
- Power user
- Administrator

Normal users load, view, and delete only their own plans or workloads.

Power users load and view plans or workloads inserted by any user. Power users delete only their own plans or workloads.

Administrators load, view, and delete any plan created by any user. Administrators drop and delete QCD tables. By default, the QCD creator has administrator privileges.

For more information about the Query Capture Facility, refer to *SQL Request and Transaction Processing* B035-1142.

- Insert and Update on QCD tables is required to insert the plans.
- To execute Monitor SQL feature, user should have MONRESOURCE privileges on the DBS.
- To view actual vs. estimated cost and actual vs. estimated cardinalities, user should have SELECT on DBQL tables.

- To get the current information, user should have necessary privileges to execute the SHOW QUALIFIED SQL statement.

Using the Administrative Functions

Table 24 describes the privileges required to use the administrative features available in the Control Center area of Teradata Visual Explain.

Table 24: Privileges for Control Center Features

Control Center Feature	Privileges Needed
Set up a new QCD	CREATE DATABASE CREATE TABLE CREATE MACRO on Teradata Database
The Upgrade/Revert feature of Manage QCD	CREATE TABLE CREATE MACRO DROP TABLE DROP MACRO UPDATE on the QCD
The Delete Execution Plans option in the Cleanup feature of Manage QCD	EXECUTE MACRO DELETE on the QCD
The Delete Database Objects option in the Cleanup feature of Manage QCD	DROP TABLE DROP MACRO DROP VIEW on the QCD
The Delete Entire Database option in the Cleanup feature of Manage QCD	DROP TABLE DROP MACRO DROP VIEW DROP DATABASE on the Teradata Database
The Create User feature in Security	CREATE USER on Teradata Database
The Change Privileges feature in Security	User must have the right being granted plus the GRANT privilege for that right on Teradata Database.
The Export QCD feature of Data Exchange	EXECUTE SELECT on the QCD
The Import QCD feature of Data Exchange	SELECT INSERT UPDATE on the QCD

CHAPTER 5

Restricted DBC Views

Restricted DBC database views can be created to limit access to information. Restricted DBC views are identified by the suffix *X* and called X views.

Teradata Visual Explain uses restricted views to limit access to those plans owned (Data Dictionary and QCD data) by the users currently logged on.

Beginning with Teradata Database 12.00.00, each system view in the DBC database includes a view that supports Unicode®. All Unicode® system views are identified by a suffix, *V* or *VX*, when a restricted Unicode® view is used.

Note: To learn more about X views, V views, and VX views, see *Data Dictionary* B035-1092 and *Database Administration* B035-1093.

Creating Restricted Views

Additional restricted views required by Teradata Visual Explain must be created in Teradata Database. Teradata Visual Explain creates restricted views from a script file. There are five versions of the script file:

- *DBCXViews62.bteq*; used on Teradata Database V2R6.2, V2R6.1, or V2R6.0
- *DBCVXViews120.bteq*; used on Teradata Database 12.00.00
- *DBCVXViews123.bteq*; used on Teradata Database 13.00.00 and 13.10.00
- *DBCVXViews140.bteq*; used on Teradata Database 14.0
- *DBCVXViews141.bteq*; used on Teradata Database 14.10 and higher

The script files are located in the Teradata installation folder. For example:
C:\Program Files(x86)\Teradata\Client\16.00\bin\DBCVXViews141.bteq

To create restricted views

1. Open Teradata Visual Explain.
2. Select Tools > Options from the menu.

   The Options dialog box appears with the General tab selected.
3. Select Use X views (DBC and QCD).
4. A confirmation message appears. Do one of the following:
   - Click Yes to create restricted views.

- Click No to proceed using non-X views, and then click OK.

CHAPTER 6

Administering the QCD

An important aspect of using Teradata Visual Explain is the administration of the QCD. The following topics explain the QCD:

- Managing the QCD
- Setting Up a New QCD
- Upgrading a QCD
- Cleaning Up a QCD
- Security
- Creating a User
- Exporting QCD Data to a File
- Changing User Rights on the QCD
- Importing QCD Data from a File

Note: The administrative functions require additional Teradata Database permissions. For more information, see "Using the Administrative Functions" on page 88.

Managing the QCD

Use the Control Center to set up a QCD, upgrade or revert the QCF version, or clean up a QCD.

To manage the QCD

1 Click Tools > Control Center.

The Control Center dialog box appears, and the Manage QCD tab is selected.

2 Choose one of the following options:

- Setup QCD (see "Setting Up a New QCD" on page 92 for more information)
- Upgrade/Revert QCF Version (see "Upgrading a QCD" on page 93 for more information)
- Cleanup QCD (see "Cleaning Up a QCD" on page 95 for more information)

Setting Up a New QCD

The QCF provides a method to capture and store the steps from any query plan in a set of predefined relational tables called the QCD.

Note: A QCD created prior to the latest Teradata Database release must be upgraded for use with the current release. For instructions, see "Upgrading a QCD" on page 93.

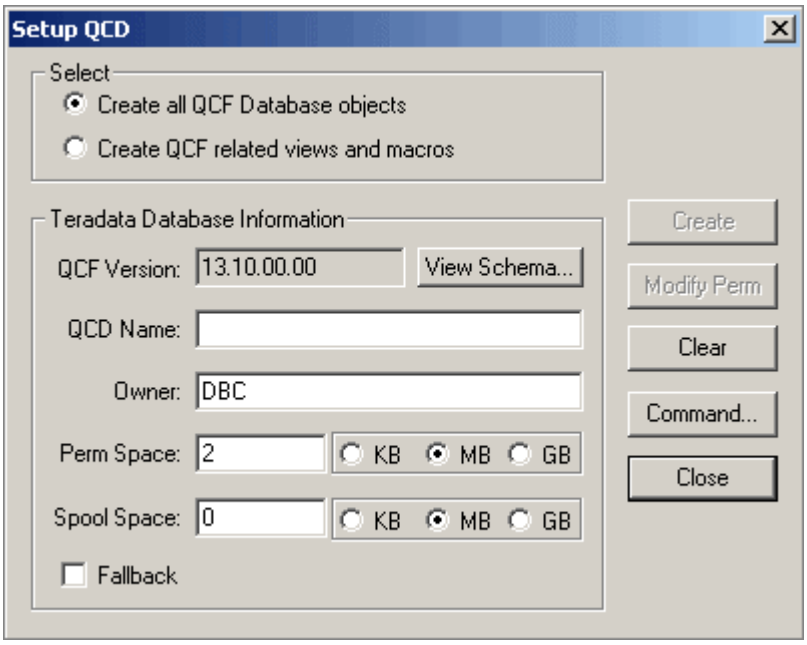
To create a new QCD

1 Click Tools > Control Center.

The Control Center dialog box appears, and the Manage QCD tab is selected.

2 Click Setup QCD.

The Setup QCD dialog box appears.



3 Click Create all QCF Database objects.

4 In the QCD Name box, enter a name for the QCD.

5 In the Owner box, enter an owner name.

If this box is left blank, the owner defaults to the name of the logged-on user.

6 In the Perm Space box, define the permanent space:

a Enter a value.

b Click one of the following: KB, MB, or GB.

7 In the Spool Space box, define the spool space:

a Enter a value.

b Click one of the following: KB, MB, or GB.

The default is 0.

8 [Optional] Select Fallback to use fallback protection.

9 Click View Schema to view the schema of the tables and macros created in the new QCD.

The View Schema dialog box opens and displays the DDL and macro definitions of the QCD. What is displayed depends on the version of QCF that is running.

10 Click Modify Perm to modify the permanent space of the existing QCD.

11 Click Create to set up the QCD.

12 Click Clear to clear all boxes to their default options.

13 Click Command to open the Execute SQL window.

For more information, see "Executing an SQL Query" on page 148.

14 Click Close to stop the set up of the QCD and return to the main window.

For more information on the QCF, refer to *SQL Request and Transaction Processing* B035-1142.

To create the macros and views in a QCD, see "Creating QCF-Related Views and Macros in a QCD" on page 93.

Creating QCF-Related Views and Macros in a QCD

To create views and macros in a QCD

1 Click Tools > Control Center.

The Control Center dialog box appears, and the Manage QCD tab is selected.

2 Click Setup QCD.

The Setup QCD dialog box appears.

3 Click Create QCF related views and macros.

4 In the QCD Name box, enter a name for the QCD.

5 Click View Schema to view the schema of the tables and macros created in the new QCD.

The View Schema dialog box appears and displays the QCD Data Definition Language (DDL) and macro definitions. What is displayed depends on the version of QCF that is running.

6 Click Create to create the QCD views and macros.

Upgrading a QCD

A QCD created prior to the latest Teradata Database release must be upgraded before it can be used.

Note: The following limitations apply when upgrading an existing QCD or reverting to an earlier version:

- If the existing QCD is version 03.00.00, upgrade to 03.01.00, then upgrade to version 12.00.00, then upgrade to version 13.00.00.
- If a QCD contains Unicode® data and has been upgraded to version 12.00.00 or was created using version 12.00.00, the QCD *cannot* be reverted to the previous version.
- If a version 12.00.00 QCD contains only non-Unicode® data, it must be reverted to version 03.01.00 before being reverted to version 03.00.00.

To upgrade a QCD

Before beginning, note the following:

- A QCD that is currently version 02.00.00 must first be upgraded to version 03.00.00, then version 03.01.00, then version 12.00.00, and then version 13.00.00.
- While upgrading QCDs from QCF version 02.00.00 to QCF version 03.00.00, the data migration from the statistics table to the Index Table might fail if the index field names have a comma character.
- There is no reversion possible for QCF 03.00.00.

1. Click Tools > Control Center.

   The Control Center dialog box appears, and the Manage QCD tab is selected by default.
2. Click Upgrade/Revert QCF Version.

   The Upgrade/Revert dialog box appears.
3. In the QCD Name box, do one of the following:
   - Enter the name of the QCD.
   - Select the name of a QCD:
     i. Click List QCDs.
     ii. Select a QCD from the QCD Name list.

   The selected QCF version appears in the Present Version box.
4. In the Intended Version box, select the version to change to.
5. Click OK to begin the conversion.

Note: An upgrade can be aborted or canceled, provided the upgrade process has not been completed. Depending on the point at which an upgrade is aborted, a prompt appears and displays an option to rollback the upgrade process. At this point, the upgrade can be rolled back or canceled. Rolling back the upgrade process undoes all upgrade operations; that is, the QCD is restored to its pre-upgrade state. Canceling the upgrade leaves the upgrade process unfinished. The partially upgraded QCD is not a valid QCD. To complete the upgrade, the QCD name must be re-entered in the QCD Name box and the process restarted from the step where the rollback was canceled.

Cleaning Up a QCD

Clean up a QCD to get rid of unwanted execution plans or database objects residing in the QCD.

To clean up a QCD

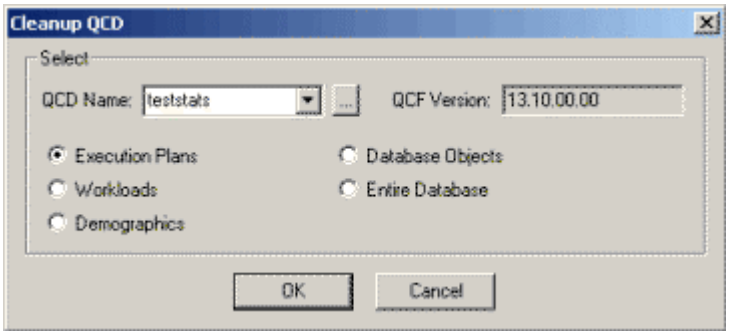
Connect to Teradata Database before cleaning up a QCD.

1 Click **Tools > Control Center**.

The **Control Center** dialog box appears, and the **Manage QCD** tab is selected.

2 Click **Cleanup QCD**.

The **Cleanup QCD** dialog box appears.



3 In the **QCD Name** box, do one of the following:

- Select a QCD to clean up.
- Click ... for a list of QCDs that are available on the server.

4 From the **Cleanup QCD** dialog box, click one of the following:

- **Execution Plans**
- **Database Objects**
- **Workloads**
- **Entire Database**
- **Demographics**

5 Go to the procedure that corresponds, respectively, to the option selected in step 4:

- "Deleting Execution Plans from a QCD" on page 95 (execution plans)
- "Deleting QCD-related Database Objects" on page 97 (database objects)
- "Deleting Workloads" on page 97 (workloads)
- "Deleting the QCD" on page 99 (entire database)
- "Deleting Demographics" on page 100 (demographics)

Deleting Execution Plans from a QCD

A QCD must be cleaned up before deleting associated execution plans.

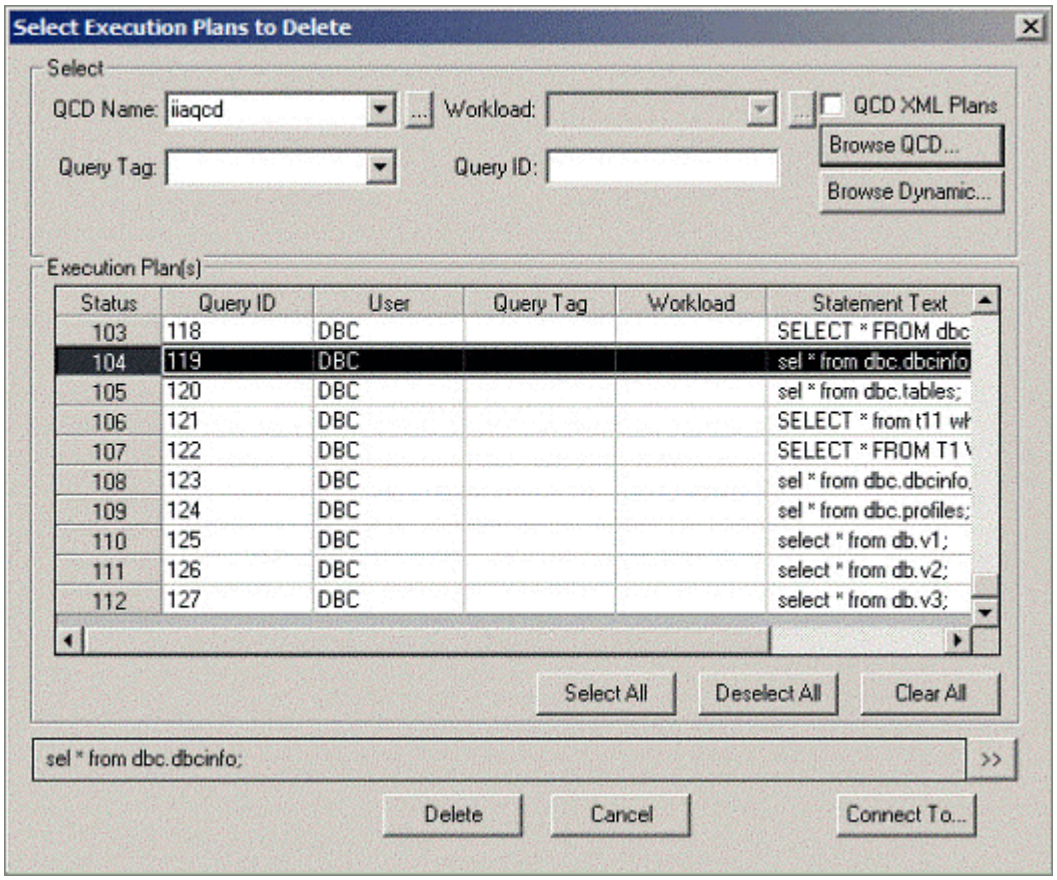
To delete execution plans from a QCD

Before deleting execution plans, go to "Cleaning Up a QCD" on page 95 and perform step 1 through step 3 of the procedure.

1 From the **Cleanup QCD** dialog box, click **Execution Plans**.

2 Click **OK**.

The **Select Execution Plans to Delete** dialog box appears.



3 [Optional] To delete **QCD XML plans**, select the "QCD XML Plans" check box.

XML plans are listed in the **Execution Plans** list for the selected QCD. Plans are listed based on the range given in the **Query ID** field.

4 [Optional] To delete **DynamicXML** plans, select the **Browse Dynamic** button. Dynamic XML plans are listed in the Execution Plans list.

5 Select the plans to delete.

Note: Execution plans with a gray background are referenced in one or more workloads and are not deleted.

6 [Optional] To see complete statement text:

a Select a statement.

b Click .

For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

7 Click Delete.

A message appears indicating the status of the cleanup process.

Note: To stop the delete action immediately, click Abort. A message appears indicating the delete action has been canceled.

8 Click Cancel to close the Select Execution Plans to Delete dialog box and return to the Cleanup QCD dialog box.

Deleting QCD-related Database Objects

A QCD must be cleaned up before deleting QCD-related database objects.

To delete QCD-related database objects

Before deleting QCD-related database objects, go to "Cleaning Up a QCD" on page 95 and perform step 1 through step 3 of the procedure.

1 From the Cleanup QCD dialog box, click Database Objects.

2 Click OK.

A warning message appears.

3 Click Yes to continue.

A message appears indicating the status of the cleanup process.

Note: To stop the delete action immediately, click Abort. A message appears indicating the delete action has been canceled.

4 Click Cancel to close the dialog box and return to the Cleanup QCD dialog box.

Deleting Workloads

There are several options when deleting workloads. More than one workload can be selected for deletion. Workload information can be deleted without deleting the associated execution plans. One or more selected recommendations can be deleted.

Note: Execution plans referenced in one or more workloads cannot be deleted.

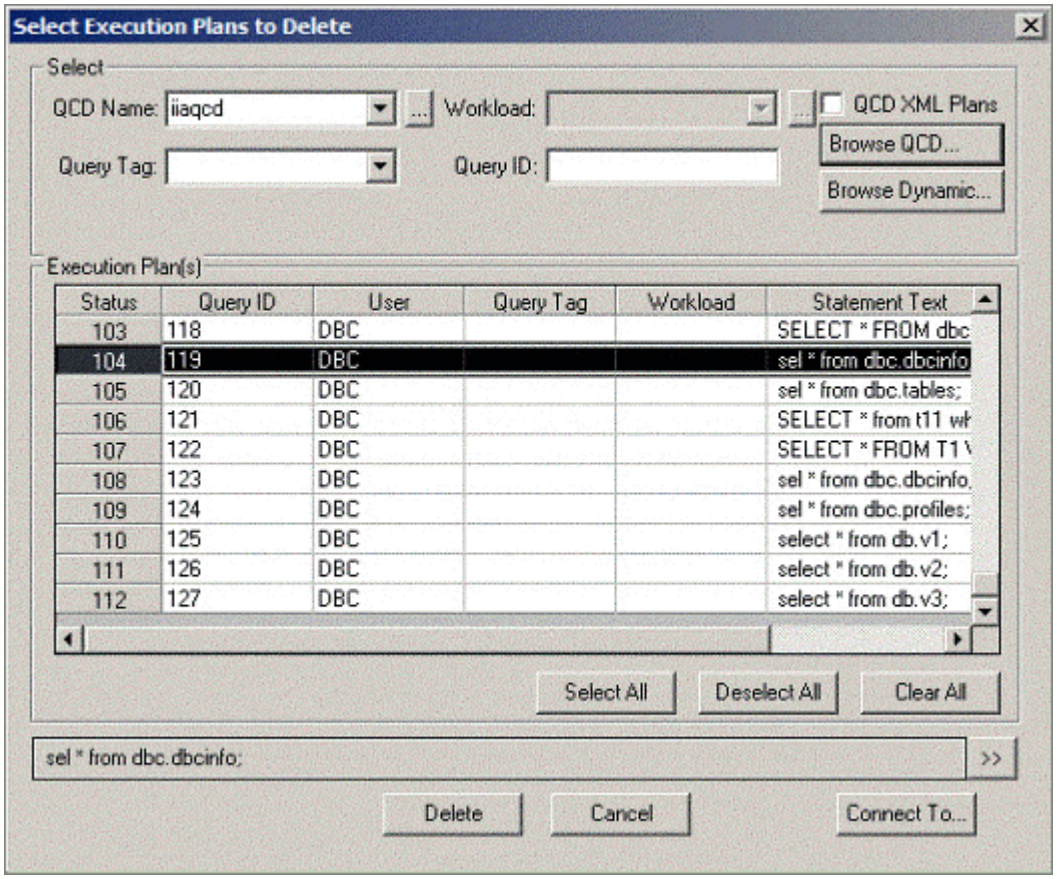
To delete one or more workloads

Before deleting a workload, go to "Cleaning Up a QCD" on page 95 and perform step 1 through step 3 of the procedure.

1 From the Cleanup QCD dialog box, click Workloads.

2 Click OK.

The **Select Workload Plans to Delete** dialog box appears.



3 In the **Workload** box, do one of the following:

- Click the **Workload** arrow to select a workload to delete.
- Select a workload to delete from a list of all available workloads:

  i Click ⋯.

  ii Select a workload.

4 Click **Browse QCD**.

The queries related to the workload selected in step 3 appear under **Execution Plan(s)**.

5 [Optional] To delete the execution plans associated with the workload in addition to the workload itself, select the **Delete Plans** check box.

6 [Optional] If recommendations exist on a particular workload, click **View Recommendations** to display the **View Recommendations** dialog box and additional information.

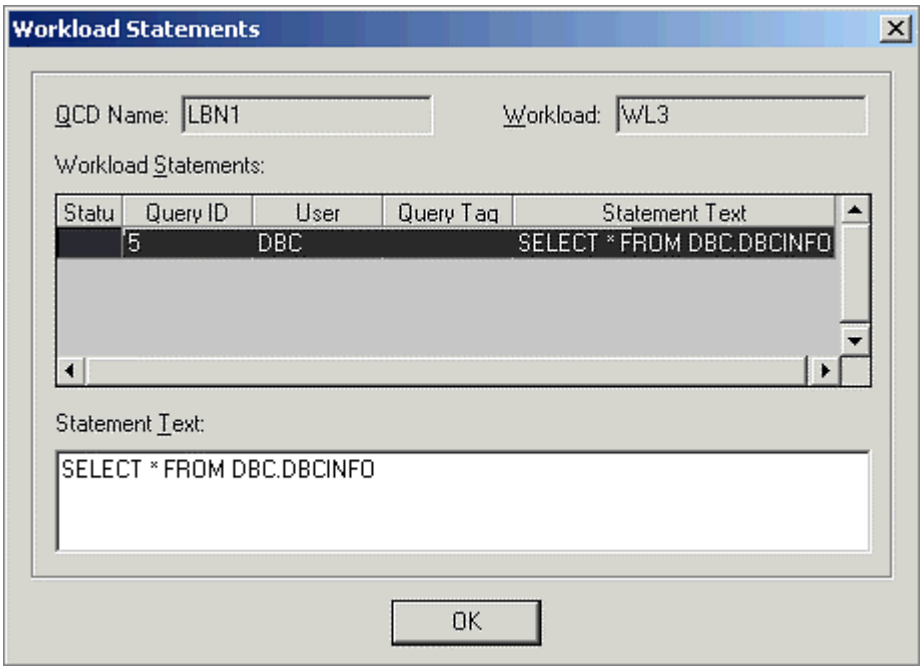
For more information about this dialog box, see step 6 on page 108.

7 [Optional] To see complete statement text:

a Select a statement.

b Click ».

The **Workload Statements** dialog box appears.



c Select a statement to view.

The **Statement Text** box displays the full text.

d Click **OK** to close the **Workload Statement** dialog box.

8 Click **Delete**.

A message appears indicating the status of the cleanup process.

Note: To stop the delete action immediately, click **Abort**. A message appears indicating the delete action has been canceled.

9 Click **Cancel** to close the **Select Workload to Delete** dialog box and return to the **Cleanup QCD** dialog box.

Deleting the QCD

When deleting the QCD, all QCD-related and user-defined objects also are deleted.

To delete the QCD

Before deleting the QCD, go to "Cleaning Up a QCD" on page 95 and perform step 1 through step 3 of the procedure.

1 From the **Cleanup QCD** dialog box, select **Entire Database**.

2 Click **OK**.

A warning message appears.

3 Click Yes to continue.

A message appears indicating the status of the cleanup process.

Note: To stop the delete action immediately, click Abort. A message appears indicating the delete action has been canceled.

4 Click Cancel to close the dialog box and return to the Cleanup QCD dialog box.

Deleting Demographics

Use the COLLECT DEMOGRAPHICS statement to delete the demographics collected in the QCD.

To delete demographics

Before deleting demographics, go to "Cleaning Up a QCD" on page 95 and perform step 1 through step 3 of the procedure.

1 From the Cleanup QCD dialog box, click Demographics.

2 Click OK.

The Delete window appears and displays the demographics that can be removed from the QCD.

3 Select one or more tables from the list.

4 Click Delete.

Security

Use the Teradata Visual Explain security features when creating a new user or changing user privileges on a QCD.

To secure the QCD

1 Click Tools > Control Center.

The Control Center dialog box appears.

2 Click the Security tab.

3 Choose one of the following:

- Create a New User (see "Creating a User" on page 101 for more information)
- Grant/Revoke Privileges on QCD (see "Changing User Rights on the QCD" on page 102 for more information)

Creating a User

Use this procedure to create a user on Teradata Database.

To create a user

Connect to Teradata Database before creating a user.

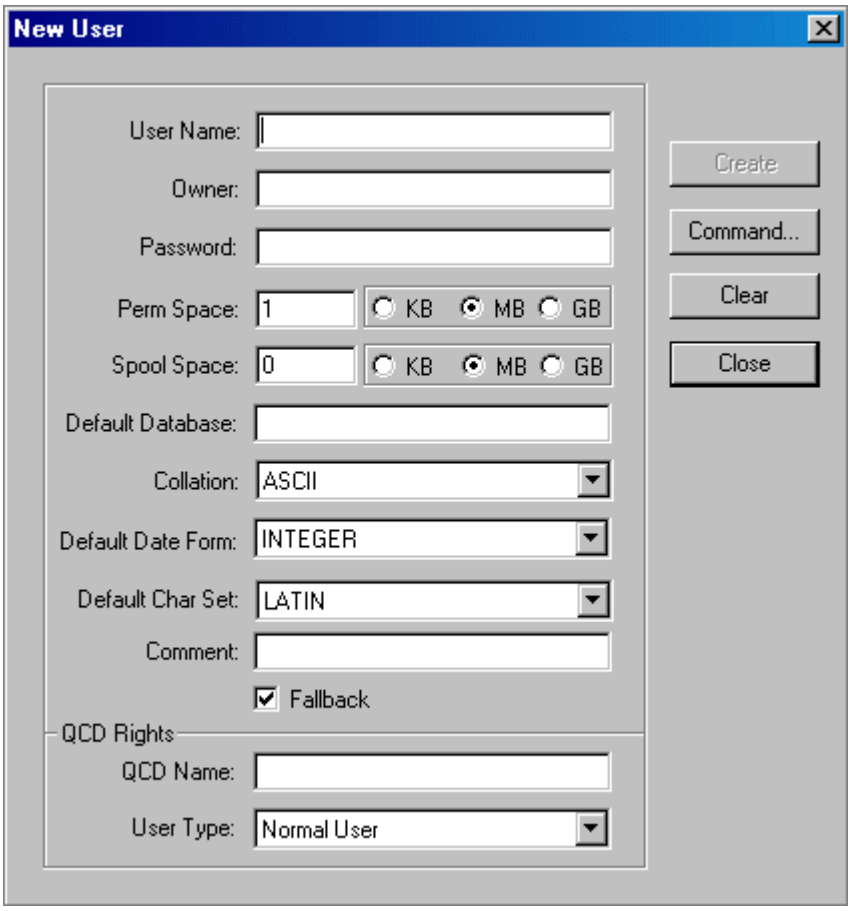
1 Click **Tools > Control Center**.

The **Control Center** dialog box appears.

2 Click the **Security** tab.

3 Click **Create a New User**.

The **New User** dialog box appears.



4 In the **User Name** box, enter the name of the user.

5 In the **Owner** box, enter the owner of the user.

6 In the **Password** box, enter the password for the user.

7 In the **Perm Space** box, define the permanent space:

a Enter a value.

b Click one of the following: KB, MB, or GB.

8 In the Spool Space box, define the spool space:

a Enter a value.

b Select one of the following: KB, MB, or GB.

The default is 0.

9 In the Default Database box, do one of the following:

- Enter the user name entered in step 4.
- Enter an alternate name for the default database.

10 In the Collation box, select a collation type.

11 In the Default Data Form box, select a default data form.

12 In the Default Character Set box, choose a default character set.

13 [Optional] In the Comment box, enter a comment up to 255 characters in length.

14 [Optional] Select Fallback to use fallback protection.

15 In the QCD Name box, enter the name of the QCD in which the user is being given rights.

16 In the User Type box, select a user type.

User Type	Associated Rights
Normal User	Load, view, or delete user's own plans or workloads only. Note: The Use X-views for QCD information check box must be selected in the Options dialog box. (See "Setting General Options" on page 84 for more information.)
Power User	Load and view plans or workloads inserted by any user. Delete user's own plans or workloads only.
Administrator	Load, view, or delete any plan created by any user. Drop or delete QCD tables. Note: By default, the QCD creator has Administrator privileges.

17 Click Command to execute an SQL statement.

For more information, see "Executing an SQL Query" on page 148.

18 Click Create to create the user.

Note: Appropriate rights on Teradata Database are required to create a user.

Changing User Rights on the QCD

Use the Control Center to grant or revoke specific user rights on the QCD.

To grant or revoke user rights

Connect to Teradata Database before changing user rights.

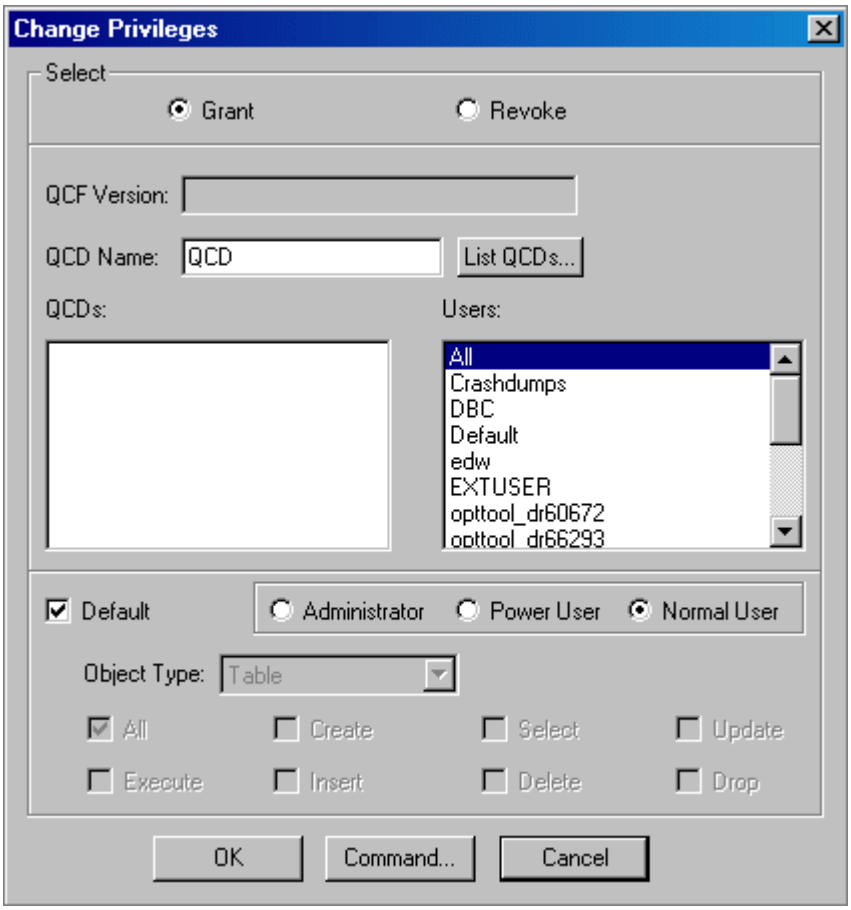
1 From the menu bar, click **Tools > Control Center**.

The **Control Center** dialog box appears.

2 Click the **Security** tab

3 Click **Grant/Revoke privileges on QCD**.

The **Change Privileges** dialog box appears.



4 Select one of the following:

- **Grant**, to give the user additional rights.
- **Revoke**, to take away user rights.

5 In the **QCD Name** box, do one of the following:

- Enter the name of the QCD.
- Click **List QCDs** to display all QCDs on the current server.

  The QCDs are listed under **QCDs**.

6 Under **Users**, select one or more users.

Note: To select more than one user, press `Ctrl` while making selections.

7 Do one of the following:

- To define the user category for the users listed for the specified QCD:
  - i Select the Default check box.
  - ii Select a user category: Administrator, Power User, or Normal User.

  Note: By default, the QCD creator has Administrator rights. For a definition of the user categories, see "Creating a User" on page 101.
- Grant or revoke all privileges for selected users:
  - i Select the All check box.
  - ii In the Object Type box, select table, view or macro to grant or revoke privileges for tables, views, or macros.
- Grant or revoke specific privileges for selected users:
  - i Clear the All check box.
  - ii Select the check boxes for the appropriate individual privileges.

8 Click OK to change the privileges.

9 [Optional] Click Command, to execute an SQL statement.

For more information, see "Executing an SQL Query" on page 148.

10 Click Cancel to stop changing privileges and return to the Control Center dialog box.

Exporting QCD Data to a File

Using the Control Center dialog box, export specific execution plans, workloads, or the entire contents of the QCD to a text file.

To export QCD data to a file

Connect to Teradata Database before exporting QCD data.

1 Click Tools > Control Center.

The Control Center dialog box appears.

2 Click the Data Exchange tab.

3 Click Exports QCD to a File.

The Export QCD dialog box appears.

4 In the QCD Name box, specify the name of the QCD.

5 Select one of the following options:
   - To export execution plans, see "Exporting Execution Plans" on page 105.
   - To export workloads, see "Exporting Workloads" on page 107.
   - To export QCD data, see "Exporting QCD Data" on page 110.

Exporting Execution Plans

The export operation for execution plans stores the information in an internal Teradata Visual Explain format (*.mcf* file), as opposed to a text file containing SQL statements that could be executed using BTEQ.

To export execution plans

Before exporting execution plans, go to "Exporting QCD Data to a File" on page 104 and perform step 1 through step 4 of the procedure.

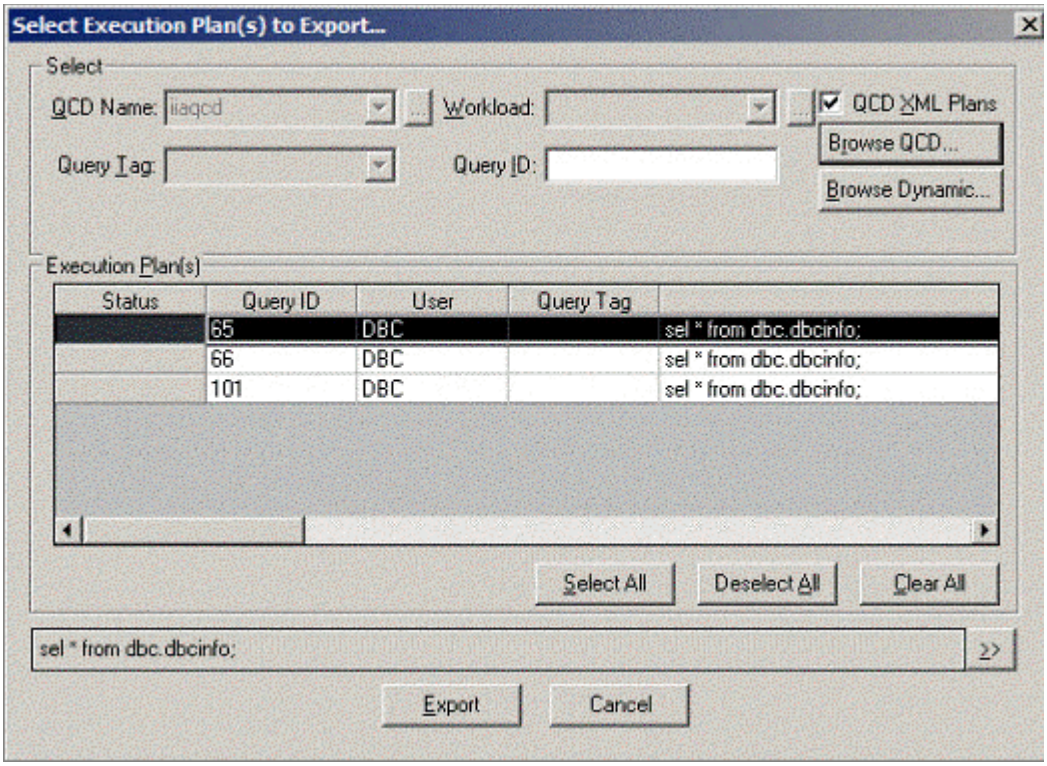
1 In the Export QCD dialog box, ensure that Execution Plans is selected (default setting).
2 Click Export.

   The Save As dialog box appears.
3 To save the QCD information:
   a Enter a name for the file in the File name box.
   b Click Save.

   Note: If an existing *.mcf* file is selected, a message appears indicating the file already exists. Click Yes to replace the existing file or click No to return to the Save As dialog box and enter a new file name.

The **Select Execution Plan(s) to Export** dialog box appears.

Figure 28: Select Execution Plan(s) to Export



4 Click **Browse QCD**.

The execution plans load.

5 [Optional] To see complete statement text information:

a Select a statement.

b Click .

For more information on the **Statement Text** dialog box, see "Viewing Statement Text" on page 77.

Note: To browse XML plans, select the QCD XML Plans check box and click the Browse button.

6 Select the plans to export.

7 Click **Export**.

A progress message appears. When the export is complete, the **Control Center** dialog box appears.

Note: To stop the export action immediately, click **Abort**. A message appears indicating the export action has been canceled.

Exporting Workloads

The export operation for workloads stores the information in an internal Teradata Visual Explain format (*.mcf* file).

To export workloads

Before exporting execution plans, go to "Exporting QCD Data to a File" on page 104 and perform step 1 through step 4 of the procedure.

1 In the Export QCD dialog box, select Workloads.

2 Click Export.

The Save As dialog box appears.

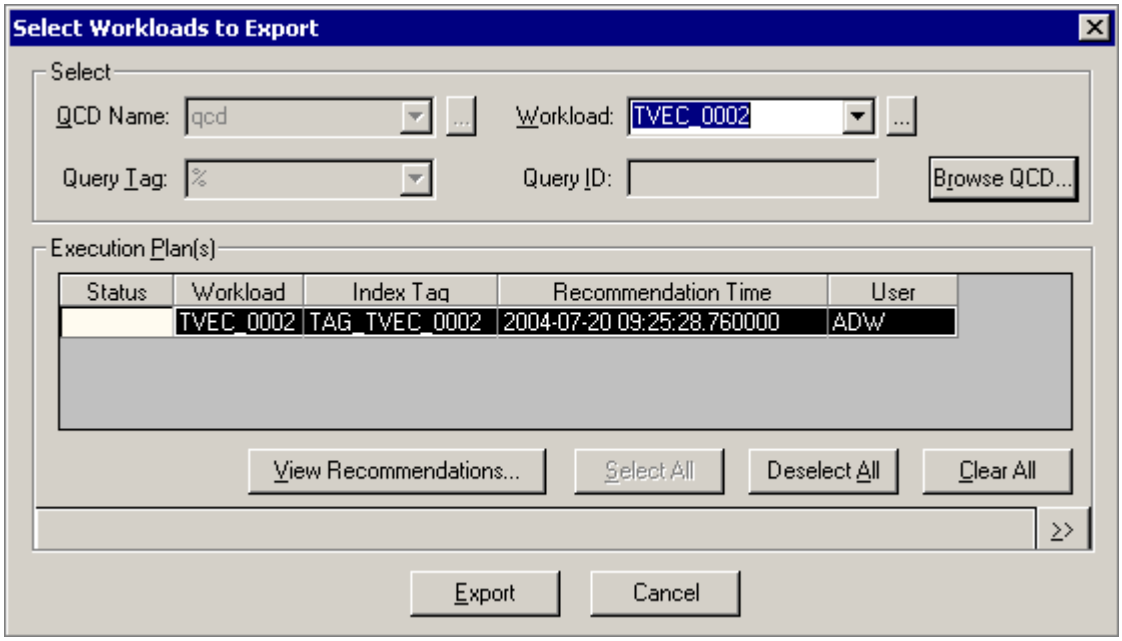
3 To save the QCD information:

a Enter a name for the file in the File name box.

b Click Save.

Note: If an existing *.mcf* file is selected, a message appears indicating the file already exists. Click Yes to replace the existing file or click No to return to the Save As dialog box and enter a new file name.

The Select Workloads to Export dialog box appears.



4 Click Browse QCD.

The workloads are inserted.

5 [Optional] To view the workload statements and the complete statement text of each query:

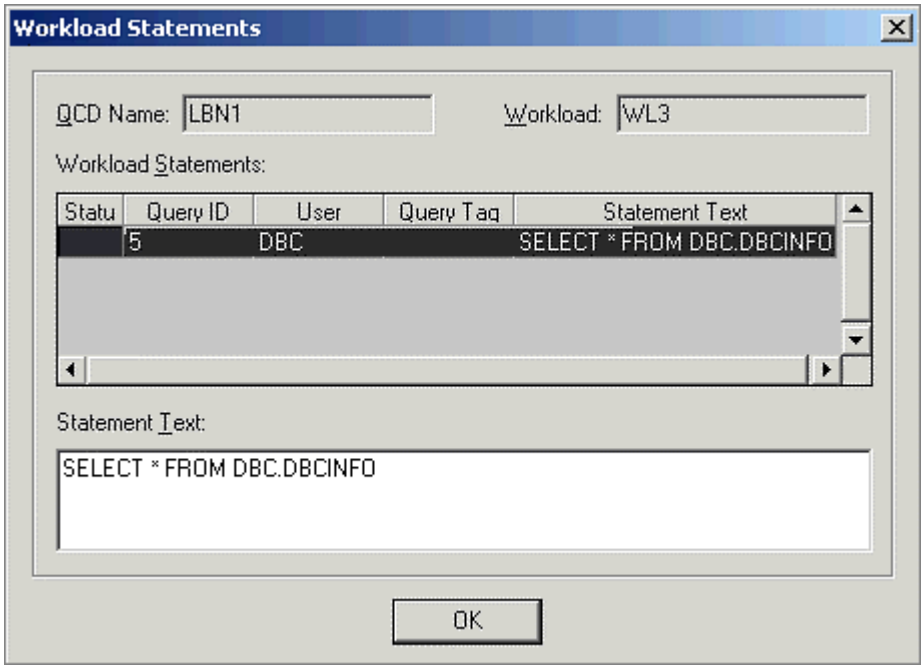
a Click .

The **Workload Statements** dialog box appears.

b Click the statement to view.

The full statement text appears in the **Statement Text** box.

c Click **OK** to close the **Workload Statements** dialog box.

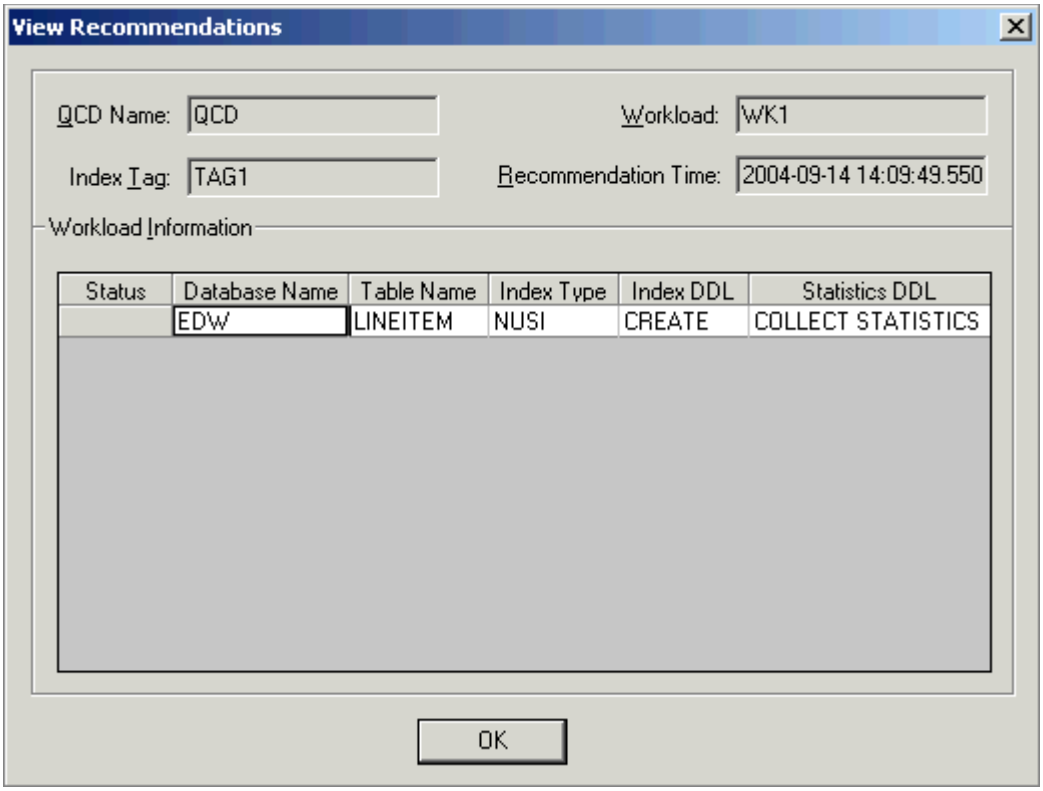


6 [Optional] To view recommendations when recommendations exist on a particular workload:

a Select the workload.

b Click **View Recommendations**.

The **View Recommendations** dialog box appears and displays details of the index recommendation.



Column	Description
Database Name	Name of the database the table belongs to.
Table Name	Name of the table on which the index recommendations were made.
Index Type	Type of the index recommended. Possible values are USI, VOSI, NUSI and HOSI. For more information, refer to the Index Recommendations section in *SQL Request and Transaction Processing* B035-1142.
Index DDL	Index recommendation DDL statement text used to create/drop the index.
Statistics DDL	Statistics recommendation DDL statement text used to create/drop the statistics.

7 Select the workloads to export.

8 Click **Export**.

A progress message appears. When the export is complete, the **Control Center** dialog box appears.

Note: To stop the export action immediately, click **Abort**. A confirmation message appears.

Exporting QCD Data

The export operation for database objects stores the information in an internal Teradata Visual Explain format, as opposed to text files containing SQL statements that could be executed using BTEQ.

To export QCD data

Before exporting execution plans, go to "Exporting QCD Data to a File" on page 104 and perform step 1 through step 4 of the procedure.

1 In the Export QCD dialog box, select QCD Data.

2 Click Export.

   The Save As dialog box appears.

3 To save the QCD information:

   a Enter a name for the file in the File name box.

   b Click Save.

   Note: If an existing *.mcf* file is selected, a message appears indicating the file already exists. Click Yes to replace the existing file or click No to return to the Save As dialog box and enter a new file name.

   A progress message appears. When the export is complete, the Control Center dialog box appears.

   Note: To stop the export action immediately, click Abort. A confirmation message appears.

Importing QCD Data from a File

QCD data can be imported from a file into a QCD.

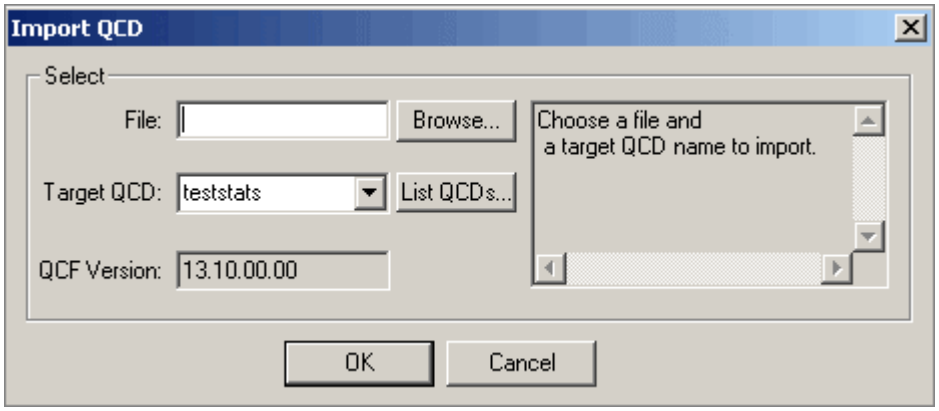
To import QCD data from a file

1 Click Tools > Control Center.

2 Click the Data Exchange tab.

3 Click Imports QCD from a File.

The Import QCD dialog box appears.



4 In the File box, do one of the following:
   - Enter the name of the file containing the plans to import
   - Click Browse to locate the file.

   Instructions on how to complete this process are in the read-only text box located on the right side of the Import QCD dialog box.
5 In the Target QCD box, do one of the following:
   - Enter name of the QCD to import the plans to.
   - Click List QCDs to see a list of all available QCDs.
6 Click OK to begin the import.

SQL Statement Failed Dialog Box

When importing QCD information, errors, such as insufficient privileges or permanent space, might occur. Use the SQL Statement Failed dialog box to retry the failed statement.

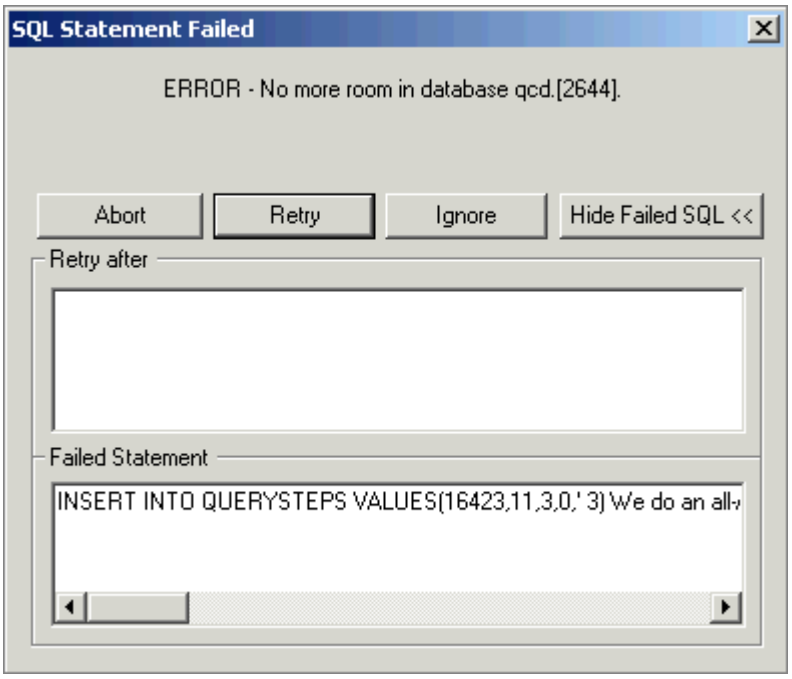
See "Chapter 4 Privileges" for more information on privileges.

To correct a failed statement

The SQL Statement Failed dialog box must be visible to perform this procedure.

1 In the SQL Statement Failed dialog box, do one of the following:
   - Enter another SQL statement.

     For example, enter a statement to grant the necessary privileges or increase the permanent space, and click Retry.
   - Click Abort to automatically abort the import operation.
   - Click Retry to retry the failed statement if the error has been corrected by other means.
   - Click Ignore to ignore the error and proceed with the import of the remaining objects.
2 Click Hide Failed SQL to hide the SQL Statement Failed dialog box.


SQL Statement Failed
ERROR - No more room in database qcd.[2644].
Abort
Retry
Ignore
Hide Failed SQL <<
Retry after
Failed Statement
INSERT INTO QUERYSTEPS VALUES(16423,11,3,0,' 3) We do an all-

CHAPTER 7

Comparing, Printing, and Saving Execution Plans

Manage execution plans using the compare, print, and save execution plans functions. See the following topics:

- Visual Comparison
- Textual Comparison
- Comparing Steps in Execution Plans
- Displaying Differences in Explain Text
- Finding Differences in Compared Execution Plans
- Comparing Multiple Plans Using Bulk Compare
- Printing
- Saving an Execution Plan

Visual Comparison

Use Teradata Visual Explain to visually compare one or more execution plans to a designated base plan. Red arrows indicate differences in the depicted plan.

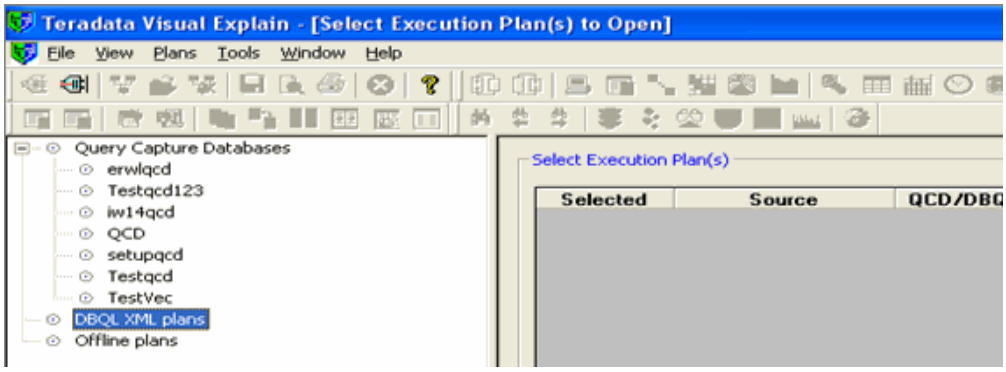
To compare execution plans visually

1 Click .

The **Select Execution Plans for Visual Compare** dialog box appears. The following selections are available:

- Plans from a database
- Open plans
- Offline plans
- All plans

Figure 29: Select Execution Plans for Visual Compare



2 [Optional] Use one of the following options to select plans from a database:

a In the tree view, right-click on the **Query Capture Databases** node to display the shortcut menu. Select the **Browse QCD** menu item to display the Available QCD names in the tree view.

b To select a plan associated with a workload, use one of the following options in the **Workload** box:

- Right-click on the **Query Capture Databases** node to display the shortcut menu. Select the **Browse QCD** menu item to display the Available QCD names in the tree view.
- Select a specific QCD from the tree view. Right-click the QCD to display the shortcut menu. select the **Browse Plans** menu item to display the **Browse QCD Plans** dialog box.
- Enter a workload name.
- Click for a list of available workloads.
- To view XML plans available in the selected QCD, select the **QCD XML Plans** check box.

c To select a specific plan, use one of the following options:

- In the tree view, right-click on the **Query Capture Databases** node to display the shortcut menu. Select the **Browse QCD** menu item to display the Available QCD names in the tree view.

- Select a specific QCD from the tree view. Right-click the QCD to display the shortcut menu. select the Browse Plans menu item to display the Browse QCD Plans dialog box.
- Enter a query-tag name in the Query Tag box.

  Note: In this field, the % and _ symbols can be used as wildcard characters and can be used in any combination. The % symbol represents any string of zero or more arbitrary characters. For example, query% would match query, query1, query12, and so on. The _ symbol represents one arbitrary character; therefore, any single character is acceptable in the position in which _ appears.

  Type a query ID number in the Query ID box.

  To select a range of plans, type a range in the Query ID box.

  For example, type *1-100* or *1,3,6-10*.

d To select plans from a different server:

i Close the Browse QCD Plans dialog box.

ii Click the Disconnect button in the Status Bar of the Open Plan Tree window.

iii Click the Connect button in the Status Bar of the Open Plan Tree window.

iv Type logon information.

v Right-click on the Query Capture Databases node to display the shortcut menu. Select the Browse QCD menu item to display the available QCD names.

vi Select a specific QCD from the tree view. Right-click the QCD to display the shortcut menu. select the Browse Plans menu item to display the Browse QCD Plans dialog box.

3 [Optional] To select plans that are open:

a Click Browse Open Plan(s). The plans are listed under Select Execution Plans.

b Click Open to display the plans.

4 [Optional] To select offline plans:

a In the tree view, click the Offline Plan node and select Browse Offline Plans.

b From the Open dialog box, select one or more plans.

- The description of the selected plan appears under Description.
- If the file contains more than one plan, the Description box identifies the number of plans the file contains.

c Click Open to display the plans in the Offline Plan node.

d Click Add to add the selected plans to the Selected Execution Plans list.

5 [Optional] To load one or more DBQL plans:

a In the tree view, right-click on the DBQL XML Plan node to open the shortcut menu.

b Select the DBQL Plans menu item to display the Select DBQL Criteria dialog box.

c Under Selection, use one of the following options:

- Select DBQL (default) to browse XML queries from the DBQL.
- Select External Database to browse XML queries from an external database.

d [Optional] Set DBQL filter options:

i DBQL filter options are present in the Select DBQL Criteria dialog box.

ii Click OK.

e The filtered DBQL plans are populated below the DBQL XML Plan node in the tree view.right-click on the

f In the Available Execution Plans, right-click on the QueryID to open the shortcut menu and select the Add menu-item checkbox in the Available Execution Plans list.

g To select a base plan for the comparison, double-click on a row under Selected Execution Plan.

6 [Optional] Use the following steps to see the complete statement text:

a Select a QueryID from QCD, DBQL, or a file name from the Offline Plans in the tree view.

b Right-click on the selected item to open the shortcut menu.

c Select the Statement Text menu item to open the Statement Text dialog box.

d Select a statement in the grid.

e Click the icon.

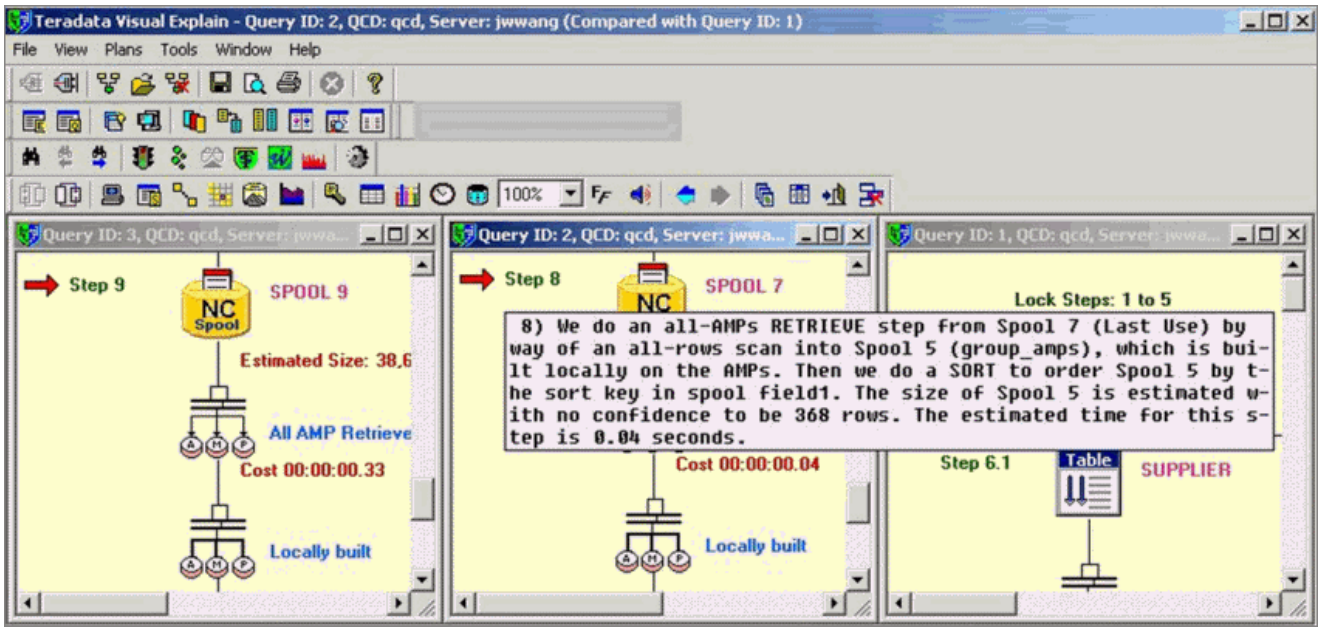
7 To select a base plan for the comparison, double-click on a row under Selected Execution Plan. The plan is moved to the top of the list and highlighted in green.

8 Select the check box in the Compare column next to the each plan to be compared.

9 Click Compare.

10 The differences are highlighted by red arrows next to the appropriate steps. Position the mouse pointer so that the cursor is over the red arrows. To see tool tips explaining the differences, see "Tool Tips and Pop-Up Menus" on page 48.

Figure 30: Execution Plans - Visual Compare



Textual Comparison

Use Teradata Visual Explain to generate a textual comparison of execution plans using reports.

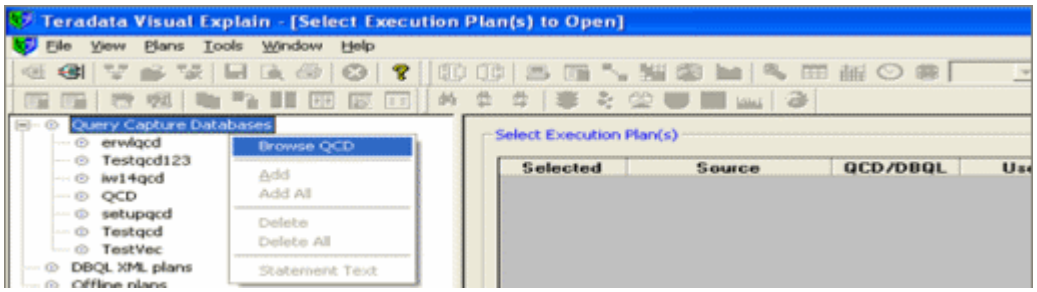
To compare execution plans textually

1 Click .

The **Select Execution Plans for Textual Compare** dialog box appears. The following selections are available:

- Plans from a database (go to step 2)
- Open plans (go to step 3)
- Offline plans (go to step 4)

Figure 31: Select Execution Plans - Textual Compare



2 [Optional] To select plans from a database, use one of the following options:
   - Right-click on the Query Capture Databases node to open the shortcut menu.
   - Select the Browse QCD menu item to display the available QCD names.
   - Right-click on a selected QCD to open the shortcut menu.
   - Select the Browse Plans menu item to display the Browse QCD Plans dialog box.

To Load ...	Do the following ...
The plans associated with a workload	In the Workload box, enter the workload name or click ... to get a list of available workloads.
A specific plan	In the Query Tag box, enter the name of the query tag. [Optional] In this field, the percent sign (%) and underscore (_) can be used as wildcard characters and can be used in any combination. The percent sign (%) represents any string of zero or more arbitrary characters. For example, query% would match query, query1, query12, and so on. The _ represents one arbitrary character; therefore, any single character is acceptable in the position in which the underscore character appears. – or – Enter one query ID number in the Query ID box.
A range of plans	In the Query ID box, enter a range. For example, enter `1-100` or `1,3,6-10`.
Plans from a different server	In the tree view, select Open Plan to open the shortcut menu. Click the Connect menu item. After entering logon information, open the Browse Plans menu item for QCD, Browse DBQL Plans menu item for DBQL, and menu item Browse Offline Plans for Offline Plans.
View XML plans	Select QCD XML Plans.

3 To select plans that are already open:
   a Click Browse Opened Plans.
   b Click Open to display the plans.
   The plans are listed under Selected Execution Plans.
4 To select offline plans:
   a Click Offline Plan in the tree view, and select Browse Offline Plans.
   b From the Open dialog box, choose one or more plans.
   When selecting a file, the description of the plan appears under Description. If the file contains more than one plan, the Description box identifies the number of plans the file contains.
   c Click Open. The plans are displayed in the Offline Plans node.
5 [Optional] To load one or more DBQL plans:
   a In the tree view, right-click on DBQL XML Plans to open the shortcut menu.
   Select the Browse DBQL Plans menu item to display the Select DBQL Criteria dialog box.

b Under Selection, use one of the following options:
- Select DBQL (default) to browse XML queries from the DBQL.
- Select External Database to browse XML queries from an external database.

c [Optional] Set DBQL filter options:

i DBQL Filter Options are present in the Select DBQL Criteria dialog box.

ii Click OK.

d Filtered DBQL plans populate the tree view under the DBQL XML Plan node.

e In the Available Execution Plans, right-click on the QueryID to open the shortcut menu and select the Add menu-item checkbox in the Available Execution Plans list.

f To select a base plan for the comparison, double-click on a row under Selected Execution Plan.

6 [Optional] To see complete statement text, select a statement and then click . For more information on the Statement Text dialog box, see "Viewing Statement Text" on page 77.

7 Select a base plan for the comparison by double-clicking on a row in the Selected Execution Plans list. This moves the plan to the top of the Selected Execution Plans list and the plan is highlighted in green.

8 Choose a plan to compare to the base plan by selecting the checkbox in the compare column next to the plan.

Note: Only two plans can be compared at a time.

9 Click Compare to display the reports. The differences in the cells are highlighted in red.

To view comparison reports, see "Chapter 10 Viewing Comparison Reports"

To print a textual compare report, see "Printing Textual-Compare Reports" on page 132.

Comparing Steps in Execution Plans

Use the Compare Steps feature to see the differences in particular steps of two or more execution plans. Unlike the visual and textual comparisons, the Compare Steps feature includes the ability to compare different steps between plans.

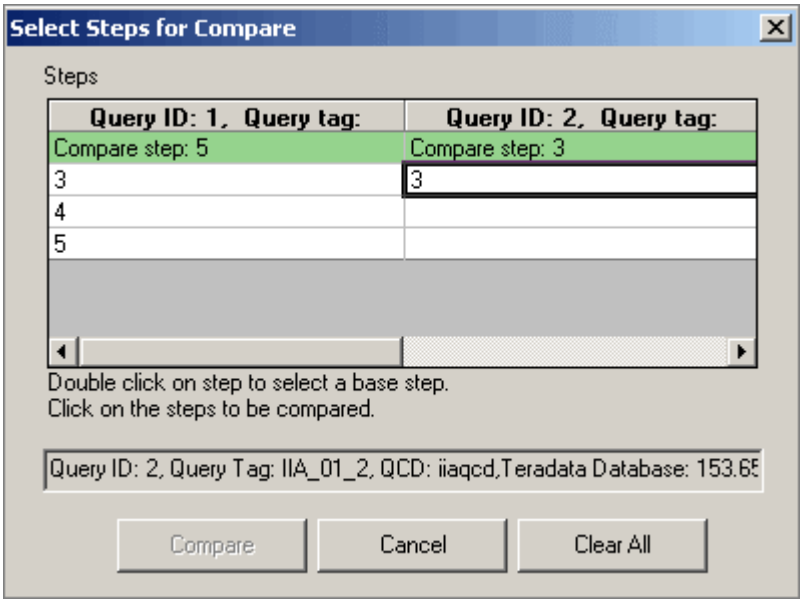
To use the Compare Steps feature

Load two or more execution plans before using the Compare Steps feature.

1 Click .

The Select Steps for Compare dialog box appears.

Figure 32: Select Steps for Compare



2 Double-click on a step in the left column to use as the basis for the comparison.

The step is moved to the top of the column, highlighted in green, and annotated as "Base Step."

3 In the right column, click on the steps to compare to the selected base step.

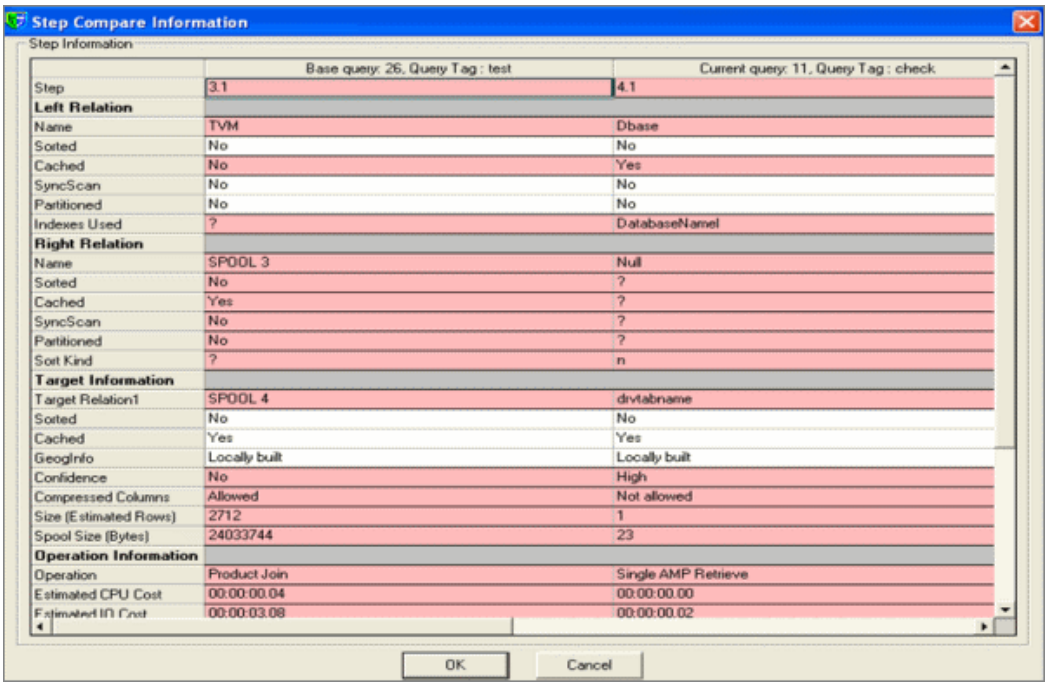
All selected steps are moved to the top of the column, highlighted in green, and annotated as "Compare Step."

4 [Optional] To remove all base and compare steps that are highlighted in green, click **Clear All**.

5 Click **Compare**.

The **Step Compare Information** window appears, and the differences between the base step and the compared steps are highlighted in red.

Figure 33: Step Compare Information



Step Compare Information

Step Information

	Base query: 26, Query Tag : test	Current query: 11, Query Tag : check
Step	3.1	4.1
Left Relation		
Name	TVM	Dbase
Sorted	No	No
Cached	No	Yes
SyncScan	No	No
Partitioned	No	No
Indexes Used	?	DatabaseName1
Right Relation		
Name	SPOOL 3	Null
Sorted	No	?
Cached	Yes	?
SyncScan	No	?
Partitioned	No	?
Sort Kind	?	n
Target Information		
Target Relation1	SPOOL 4	drvtabname
Sorted	No	No
Cached	Yes	Yes
GeogInfo	Locally built	Locally built
Confidence	No	High
Compressed Columns	Allowed	Not allowed
Size (Estimated Rows)	2712	1
Spool Size (Bytes)	24033744	23
Operation Information		
Operation	Product Join	Single AMP Retrieve
Estimated CPU Cost	00:00:00.04	00:00:00.00
Estimated IO Cost	00:00:03.08	00:00:00.02

OK Cancel

6 Do one of the following:

- Click **OK** to return to the **Select Steps for Compare** dialog box and perform another comparison.
- Click **Cancel** to close the comparison and return to the **Select Steps for Compare** dialog box.

Displaying Differences in Explain Text

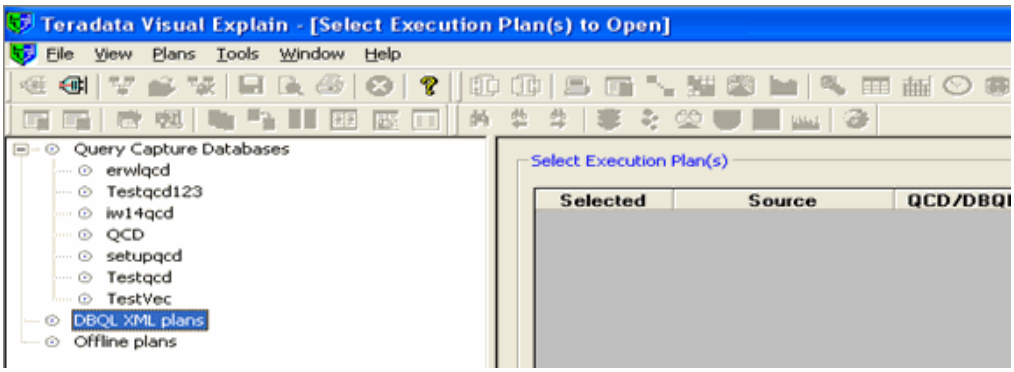
Use Teradata Visual Explain to display the differences in the Explain text for two plans.

To display Explain text differences

1 Click .

The **Select Execution Plans for Explain Text Differences** window appears.

Figure 34: Select Execution Plans for Explain Text Differences



2 From the **Open Plans** window, select one of the following options:

- Plans from a database (go to step 3)
- Open plans (go to step 4)
- Offline plans (go to step 5)
- DBQL plans (go to step 6)

3 To select plans from a database:

a In the tree view, right-click the **Query Capture Databases** node to open the shortcut menu.

b Select the **Browse QCD** menu item to list the available QCD names in the tree view.

c Select a specific QCD from the QCD names in the tree view.

d Right-click the QCD name to open the shortcut menu.

e Select the **Browse Plans** menu item to display the **Browse QCD Plans** dialog box.

To load ...	Do the following ...
The plans associated with a workload	In the **Workload** box, enter the workload name or click ... to get a list of all available workloads.

To load ...	Do the following ...
A specific plan	In the Query Tag box, enter the name of the query tag. [Optional] In this field, the percent sign (%) and underscore (_) can be used as wildcard characters and can be used in any combination. The percent sign (%) represents any string of zero or more arbitrary characters. For example, query% would match query, query1, query12, and so on. The _ represents one arbitrary character; therefore, any single character is acceptable in the position in which the underscore character appears. – or – Enter one query ID number in the Query ID box.
A range of plans	In the Query ID box, enter a range. For example, enter `1-100` or `1,3,6-10`.
Plans from a different server	In the tree view, select Open Plan to open the shortcut menu. Click the Connect menu item. After entering logon information, open the Browse Plans menu item for QCD, Browse DBQL Plans menu item for DBQL, and menu item Browse Offline Plans for Offline Plans.

4 To select plans that are already opened:

   a Select Opened Plan.

   b Click Browse Open Plan(s).

   The plans are placed in the Available Execution Plans list.

5 To select offline plans:

   a In the tree view, right-click Offline Plan to open the shortcut menu.

   b Select the Browse Offline Plan menu item to display the Open dialog box.

   c Choose the plans from the Open dialog box.

   When selecting a file, the description of the plan appears in the Description box. If the file contains more than one plan, the Description box includes the number of plans that the file contains.

   d Click Open to place the plans in the Offline Plan node in the tree view.

6 [Optional] To load one or more DBQL plans:

   a In the tree view, right-click DBQL XML Plans to open the shortcut menu.

   b Select the Browse DBQL Plans menu item to open the Select DBQL Criteria dialog box.

   c Under Selection, use one of the following options:

      • Select DBQL (default) to browse XML queries from the DBQL.
      • Select External Database to browse XML queries from an external database.

   d [Optional] Set DBQL filter options:

      i DBQL Filter Options are present in the Select DBQL Criteria dialog box.

      ii Click OK.

      iii The filtered DBQL Plans are listed below the DBQL XML Plan node in the tree view.

iv In the **Available Execution Plans** list, right-click on a **QueryID** to open the shortcut menu.

v Select the **Add** menu item to enter a check mark in the QueryID checkbox.

vi Enter data in the **Select DBQL Criteria** dialog box.

See "To view the execution plan graphically or textually from the DBQL" on page 71 for more information about completing the **Select DBQL Criteria** dialog box.

The **Select DBQL Criteria** dialog box closes and the **Select Plans from DBQL** dialog box appears.

7 [Optional] To see complete statement text, select a statement and then click .

For more information on the **Statement Text** dialog box, see "Viewing Statement Text" on page 77.

8 Select a base plan for the comparison by double-clicking on a row in the **Selected Execution Plans** list.

This moves the plan to the top of the **Selected Execution Plans** list and the plan is highlighted in green.

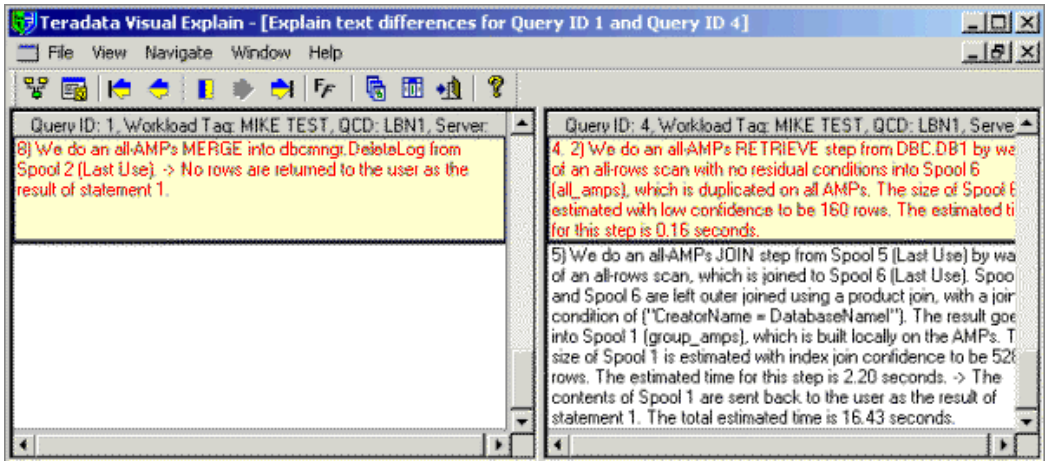
9 Choose a plan to compare to the base plan by selecting the checkbox in the compare column next to the plan.

Note: Only two plans can be compared at a time.

10 Click **Compare**.

The compare window opens. The differences in the cells are highlighted in red.

Figure 35: Compare Window



11 Use the commands on the **Navigate** menu or the corresponding buttons on the toolbar to move from one difference to another.

For more information about these commands, see "Navigate Menu Commands and Toolbar" on page 34.

12 Click **File > Close** to close the compare window.

Finding Differences in Compared Execution Plans

Using Teradata VE, it is easy to go to the next or previous difference in compared execution.

To find the differences in compared execution plans

Execution plans must be compared before differences can be identified. See "Visual Comparison" on page 113.

- To find:
  - A specific difference, click
  - The next step where there is a difference, click
  - The previous step where there is a difference, click

Comparing Multiple Plans Using Bulk Compare

Use Teradata Visual Explain to identify the differences in the execution paths for a set of plans across Teradata Database releases, as well as compare multiple plans as a one-batch operation. The bulk compare feature provides the facility to:

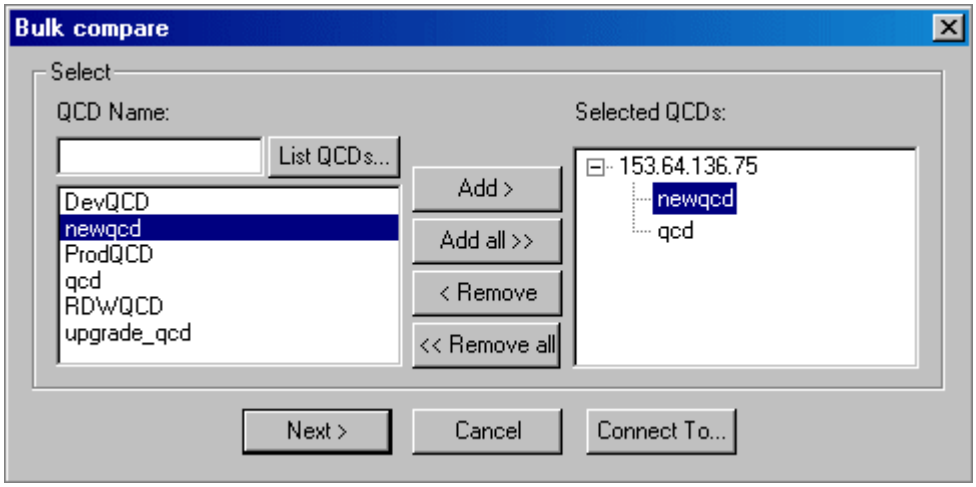
- Compare a set of plans across Teradata Database releases
- Perform multiple plan comparisons as a one-batch operation
- Restart a previously aborted bulk-compare operation
- Generate output reports
- View the current bulk compare results
- View the previous bulk compare results

Note: This feature is not available for XML plans captured into the QCD.

To compare multiple plans using the Bulk Compare feature

1 Click .

The Bulk compare dialog box appears.

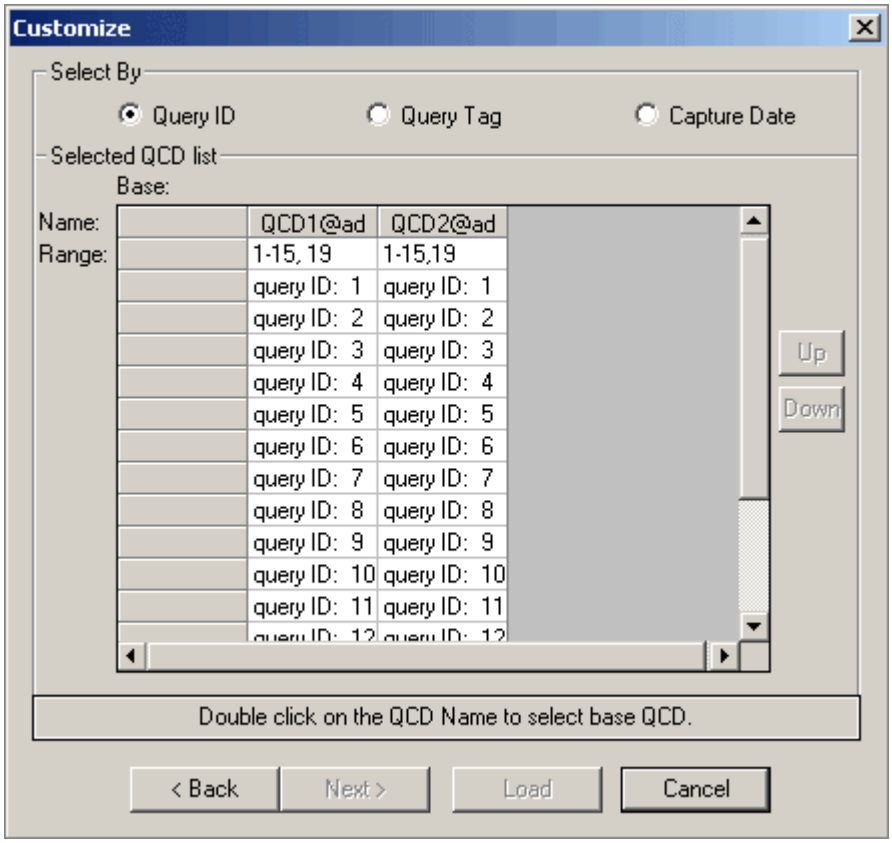


2 Select the QCDs containing the plans to be compared by doing one or more of the following:

- Specify a QCD name:
  - i Enter the name of a QCD in the **QCD Name** box.
  - ii Click **Add**.
- Choose from a list of available QCDs:
  - i Click **List QCDs**.
  - ii Select QCDs from the list.
  - iii Click **Add**.
- Add QCDs from a different server:
  - i Click **Connect To**.
  - ii Specify the new server name.
- Remove selected QCDs from the **Selected QCDs** list:
  - i Highlight the QCD name.
  - ii Click **Remove**.

3 Click **Next**.

The **Customize** dialog box appears.



4 Select a range type used to select the queries for comparison:

- Query ID
- Query Tag
- Capture Date

5 Under Selected QCD list, type a range in the Range fields.

Range Type	Range
Query ID	Enter the range of queries for each QCD. For example, enter `1-100` or `1,3,6-10`.
Query Tag	Enter the name of the query tag. The % and _ characters can be used as wildcards, in any combination. The % character represents any string of zero or more arbitrary characters. For example, query% would match query, query1, query12, and so on. The _ character represents one arbitrary character; therefore, any single character is acceptable in the position in which the _ character appears.
Capture Date	Enter the date in *yyyy/mm/dd* format.

6 Click Load to display the queries for the specified ranges.

7 [Optional] To view statement text, get query server or source information, or remove any of the displayed queries:

   a Right-click on the specific query cell.

   b Select the appropriate menu item.

8 [Optional] To reorder queries for comparison (remapping from the default one-to-one order):

   a Highlight the a query.

   b Click Up or Down to move the query within the query list.

   Note: To drag and drop queries, click the top left corner with the left mouse button.

9 Double-click on the top cell of the column containing the base OCD for the bulk compare.

   The selected column becomes the left-most column and is highlighted.

10 Click Next.

   The Save As dialog box appears.

11 [Optional] To save the results of each plan comparison in separate files, select Save as multiple files.

12 [Optional] To remove the control file after viewing the Bulk Compare results, select the Delete control file check box.

   During the Bulk Compare operation, a control file is created with a.*bcf* extension. This file is used to display the results in the View Report window.

13 Click Compare to start the bulk compare process.

   A confirmation message appears indicating the Bulk Compare operation completed.

14 Click OK to close the confirmation message.

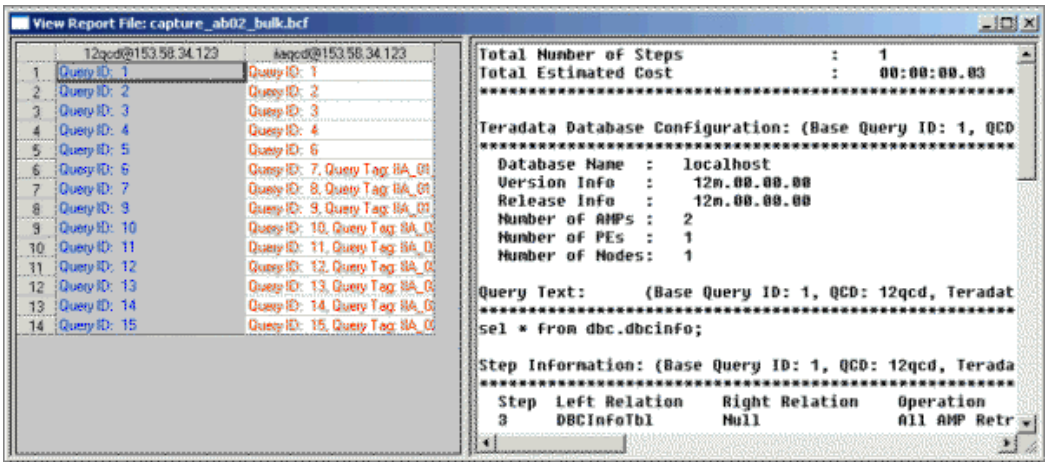
   When the Bulk Compare operation is complete, the Bulk Compare report viewer opens.

Viewing the Bulk Compare Results

The results of the bulk compare are saved in text-file format. Although any standard text editor can be used to view these files, Teradata Visual Explain provides a more flexible viewing method.

To view the results of the bulk compare

1 After the Bulk Compare operation finishes, the Bulk Compare report viewer opens.



2 [Optional] To select a previously saved bulk-compare file:

- Click **Plans > Bulk Compare > View Report**.

  The **Open** dialog box appears.
- Select a *.bcf* file
- Click **Open**.

3 Select an option from the **Filter By** list.

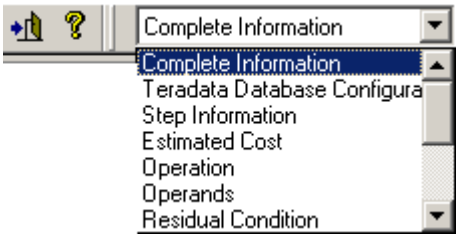
The default is **All**. However, the selection remains the same when the application is restarted or the Bulk Compare report viewer is opened subsequently.

Menu Option	Description
Unmatched	Displays only those queries that are different from the attribute selected from the **Compare Attribute** list.
Matched	Displays only the queries whose plans are matched with the base query plan.
All	Displays the matched and unmatched queries.

4 Select an option from the **Compare Attribute** list.

The default is **Complete Information**.

For example, selecting **Unmatched** from the **Filter By** list and **Join Condition** from the **Compare Attributes** list results in the display of queries that are unmatched on the Join Condition attribute. The queries appear in the left pane.



Menu Options	Description
Complete Information	Displays all reports
Teradata Database Configuration	Displays the Teradata Database Configuration report.
Step Information	Displays the Step Information report.
Estimated Cost	Displays the Estimated Cost report.
Operation	Displays the Operation report.
Indexes Used	Displays the Indexes Used report.
Index Condition	Displays the Index Condition report.
Residual Condition	Displays the Residual Condition report.
Join Condition	Displays the Join Condition report.
Join Order	Displays the Join Order report.
Source Attributes	Displays the Source Attribute report.
Target Attributes	Displays the Target Attributes report.

For information about reports, see Chapter 10: "Viewing Comparison Reports."

5 Select a query from the QCD list in the left pane.

The corresponding information displays in the text window in the right pane.

6 [Optional] From the Bulk Compare report viewer, double-click on the cells in the left pane to open the report window.

To control the number of windows displayed, click **View > Multiple Windows**. Multiple windows appear per query plan for the selected attributes. New windows are opened for each query plan. To use the same set of windows for the attributes of each plan, clear this command.

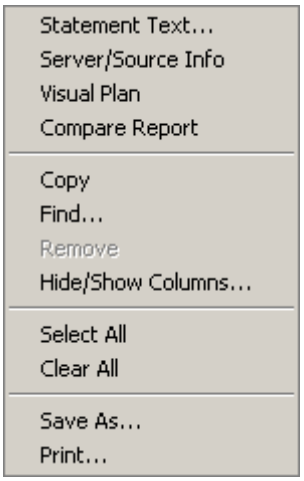
7 [Optional] To display the statement text, server/source information, or visual plan of any query:

a Right-click on the query.

The shortcut menu appears.

b Select a command.

Perform any of the commands listed on the shortcut menu, including printing or saving the information that is displayed in the text pane.



Setting the Variance Threshold

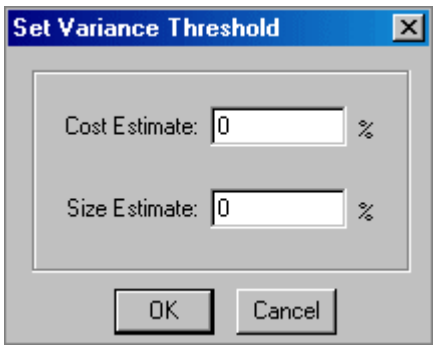
Set the variance threshold to ignore minor differences in estimated cost and spool size, making it easier to identify significant plan differences.

If the percent difference at any step is above the specified variance threshold, the cost and row size differences are highlighted. Plans where all of the differences are less than the threshold variance show up as matching plans.

To set the variance threshold

1 From the Bulk Compare report viewer, click .

The Set Variance Threshold dialog box appears.



2 Enter a percentage in the Cost Estimate box.

3 Enter a percentage in the Size Estimate box.

Restarting Bulk Compare

The Bulk Compare feature can be recovered from the last aborted or terminated state.

To restart a bulk compare

1 Click .

Note: If the previous bulk compare aborted or failed, a message appears indicating that the previous bulk compare was not successful.

2 Click OK to restore the previous bulk compare from the last checkpoint.

3 To start the regular bulk compare interface, click Cancel.

Printing

The print function is composed of the following features:

- Print execution plan
- Print report
- Preview

Printing an Execution Plan

An execution plan can be printed in a variety of formats, depending on selections made in the Print dialog box.

To print an execution plan

An execution plan must be open before performing this procedure.

1 Click .

The Print dialog box appears.

2 Under Print Options, select one or more of the following:

- Visual Explain, to print the graphical depiction of the execution plan (with icons)
- Statement Text, to print the statement text
- Explain Text, to print the Explain text
- Tool Tips, to print the tool tip information
- Visual Explain and Fit to one page, to print the execution plan with connectors on a single page

Note: Print preview is not available when printing textual compare reports.

Printing Textual-Compare Reports

Reports containing textual comparisons can be printed. However, if cells are resized, the following constraints apply:

- Only the text in the cells that is visible on screen is printed.

- Hidden columns are not printed.
- Print preview is not available for textual compare reports.

To print textual-compare reports

Generate a report comparing execution plans before printing. See "Textual Comparison" on page 117.

1 Click .

The Print dialog box appears.

2 Under Print Options, do one of the following:

- Select Complete Information to print all textual compare reports for all windows currently open.
- Select one or more check boxes to print reports with specific attributes. For example, to print only cost reports, select Cost. The options selected apply only to open textual compare windows.

3 To set page orientation, do one of the following:

- To automatically determine the page orientation (landscape or portrait), select Smart Print.

  Note: Any page orientation settings set previously in the Properties dialog box are ignored.

- To set the page orientation manually:

  i Clear Smart Print.

  ii Click Properties.

  iii From the Properties dialog box, set the page orientation

  iv Click OK to close the Properties dialog box.

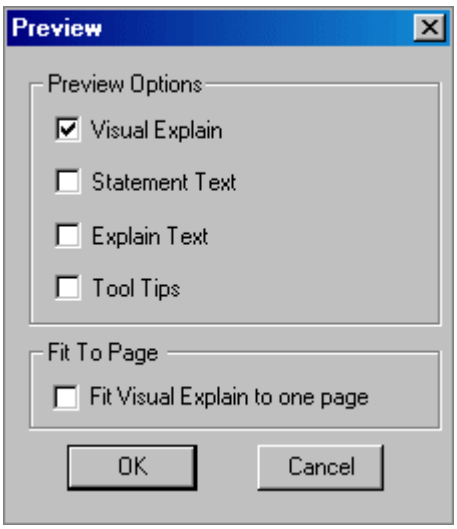
Previewing Before Printing

Use print preview to view a plan as it will appear when printed. Display multiple pages of a plan in a reduced size to see page breaks before printing the document.

To display a print preview

1 Click .

The Preview dialog box appears.



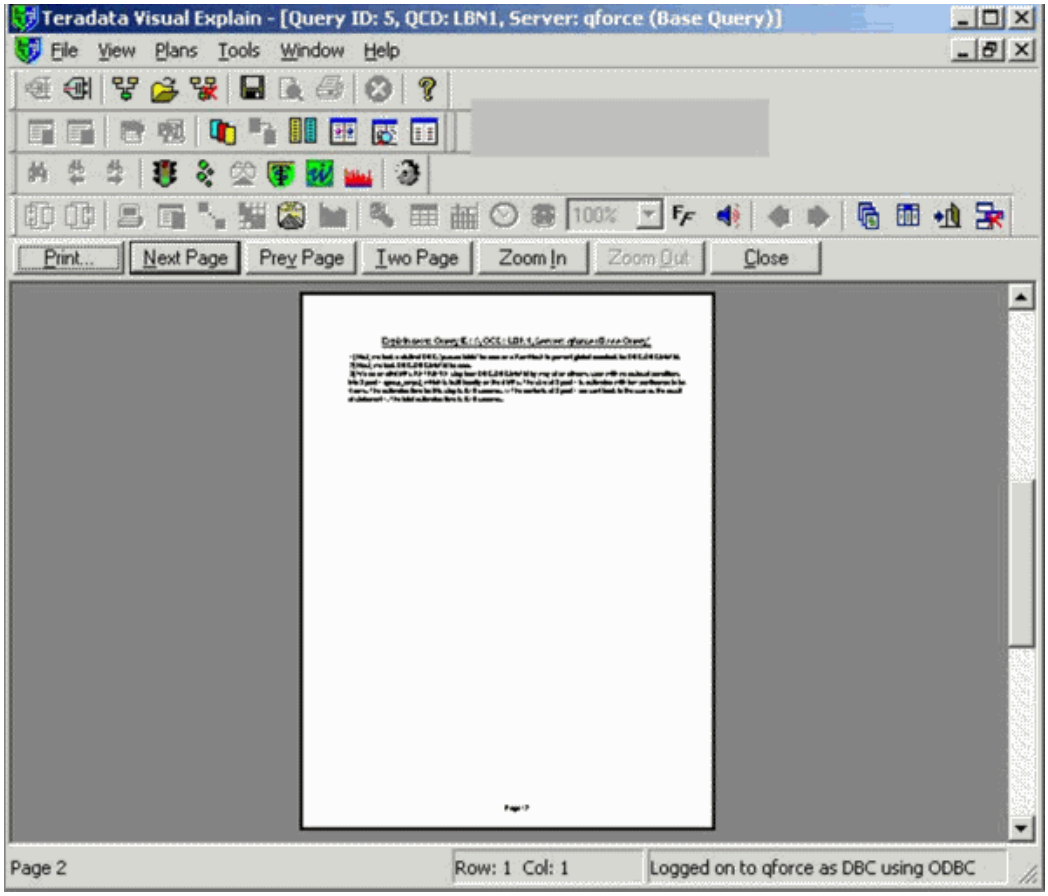
2 Select one or more Preview Options check boxes.

3 [Optional] To fit the plan on one page, select Fit Visual Explain to one page check box.

4 Click OK.

A print preview of the plan appears.

Figure 36: Print Preview



5 Do one of the following:

- Click **Print** to print the plan.

  The **Print** dialog box appears. For more information, see "Printing an Execution Plan" on page 132.
- Click **Close** to close this window and return to the plan without printing.

Saving an Execution Plan

One or more execution plans can be saved to a file (*.vec*). Saving an execution plan creates what is referred to as an *offline* plan. Optionally, the corresponding visual plan, Explain text, statement text, or tool tips can be saved.

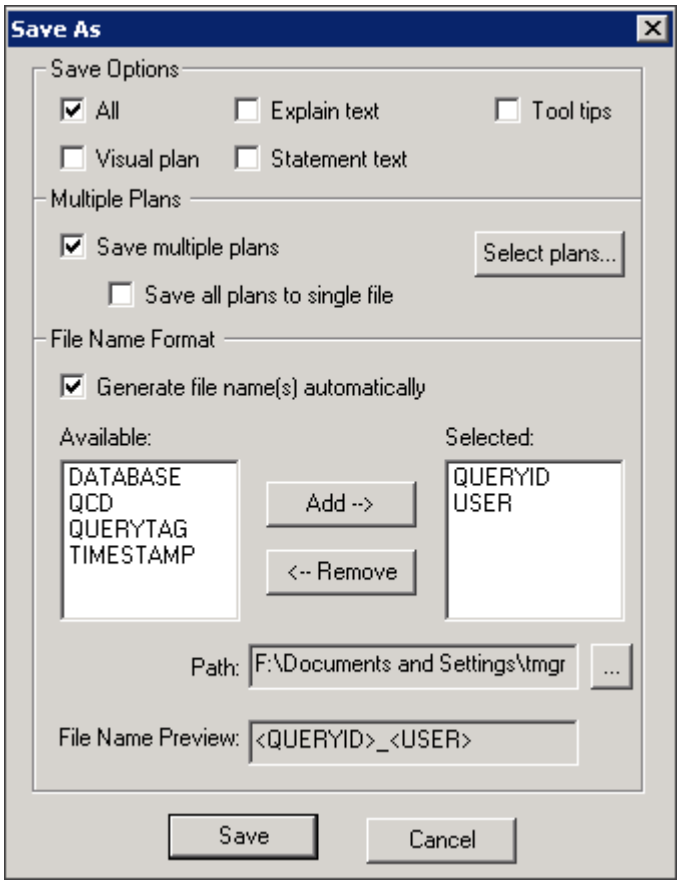
When saving the plan as a file, a file name can be entered or automatically generated. To generate a file name automatically, one or more parameters must be selected. The parameters form the foundation of the file name. For example, selecting the USER and SERVER parameters results in a plan saved as *john_server1.vec*. (See "To save an execution plan" on page 136, step 5, for more information.)

To save an execution plan

Before saving, open one or more execution plans.

1 Click .

The Save As dialog box appears.



2 Under Save Options, do one of the following:

- Select one or more check boxes to define the type of information saved.
- Select the All check box (default setting) to save all options.

Check Box	Description
All	Saves the visual plan, Explain text, statement text, and tool tips in separate files.
Explain text	Saves the Explain text of the plan in a text file (*.txt*).
Tool tips	Saves the tool tips that are displayed in the visual plan in a text file (*.txt*).

Check Box	Description
Visual plan	Saves the plan details into a binary file (*.vec*). Also saves the Explain text, statement text, and tool tips in the *.vec* file so they are available when the offline plan is opened.
Statement text	Saves the query text of the plan in a text file (*.sql*).

3 [Optional] Under **Multiple Plans**, click **Select plans** to select open plans to save.

4 [Optional] Under **Multiple Plans**, select check boxes to further define how the selected plans are saved.

Check Box	Description
Save multiple plans	Save all plans selected in step 3. **Note:** If the check box is clear, only the active plan is saved. Click **Select plans** to specify plans to save.
Save all plans to single file	Save all plans selected in step 3 to a single file. The file name must be specified in the **File Name** box. **Note:** Automatic file-name generation is unavailable when this check box is selected, and the **File Name Preview** box becomes the **File Name** box. If the **Save all plans to single file** check box is clear, plans are saved to separate files.

5 Under **File Name Format**, do one of the following:

- To create the file name manually:
  - i Clear the **Generate file name(s) automatically** check box.
  - ii Type a file name in the **File Name** box.
- To generate the file name automatically:
  - i Select the **Generate file name(s) automatically** check box (default setting).
  - ii In the **Available** box, select one or more parameters.
  - iii Click **Add**.

Parameter	Information Included in File Name
USER	Name of the user who captured the plan
SERVER	Name of the server where the plan was captured
QUERYTAG	Query tag for the plan
QCD	QCD for the plan

Parameter	Information Included in File Name
QUERYID	Query ID for the plan
TIMESTAMP	Date and time the file is saved Note: The format is day, month, hour, minute, second (*ddmmm-hh-mm-ss*).

Selected parameter names appear in the File Name Preview box. Parameter names appear in the order in which they were selected and are separated by the _ symbol. For example, if the first parameter selected and moved to the Selected box is SERVER and the second parameter is USER, the file name is <SERVER>_<USER>.

6 [Optional] To remove parameters so they are not included in the file name:

   a In the Selected box, select the parameter to remove.

   b Click Remove.

7 [Optional] To change the default location at which the file is saved, do one of the following:

   - Type a location in the Path box.
   - Click ... to browse for a folder.

8 Click Save to save the execution plans.

CHAPTER 8

Working with Execution Plans

Use Teradata Visual Explain to load and display execution plans:

- Loading and Viewing an Execution Plan
- Locating a Step in an Execution Plan
- Navigating Execution Plans
- Locating Spool References
- Entering a Query and Viewing Its Execution Plan
- Setting or Resetting Random AMP Samples
- Setting or Resetting Cost Parameters
- Executing an SQL Query

Use Teradata Visual Explain to analyze plans:

- Analyzing Plans to Improve Performance

Use Teradata Visual Explain to do the following:

- Viewing Index Recommendations for Tables
- Viewing Current vs. Captured Indexes, Object Definitions, and Statistics
- Viewing Actual Versus Estimated Cardinalities
- Compare Actual to Estimated Cost
- Viewing Query Execution
- Viewing Teradata Database Configuration
- Viewing the Object Use Count Report

Loading and Viewing an Execution Plan

After an execution plan has been captured into the QCD (using the INSERT EXPLAIN statement), use Teradata Visual Explain to view the plan graphically or textually.

For more information, see:

- "To insert an execution plan into a QCD" on page 61
- "To view the execution plan graphically or textually from a QCD database (workload)" on page 62
- "To view the execution plan graphically or textually from the DBQL" on page 71
- "To load and view a saved execution plan" on page 140

To load and view a saved execution plan

1 Click .

The Open dialog box appears.

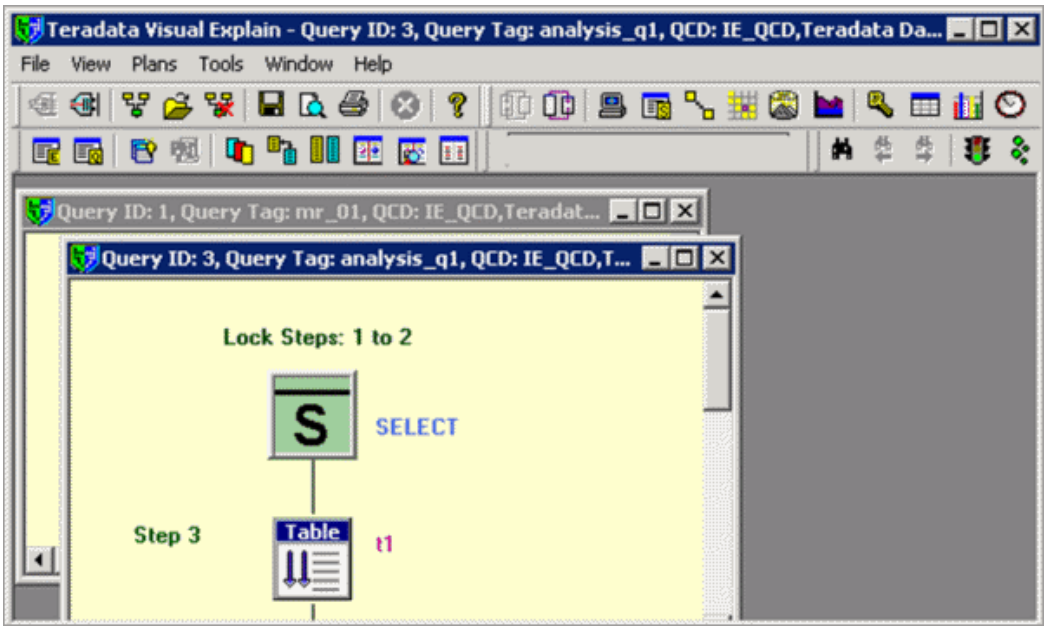
2 Select a plan (*.vec* file).

When a file is selected, the plan description appears in the Description box. If the file contains more than one plan, the Description box identifies how many plans the file contains.

3 Click Open.

The plans appear in the main window.

Figure 37: Load and View Execution Plans



Locating a Step in an Execution Plan

Use this procedure to quickly find a specific step in an execution plan.

To locate a step

Before locating a step, load an execution plan. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 From the menu bar, choose **Tools** > **Go To Step**.

   The **Go To Step** dialog box appears.

2 Enter a step number.

Navigating Execution Plans

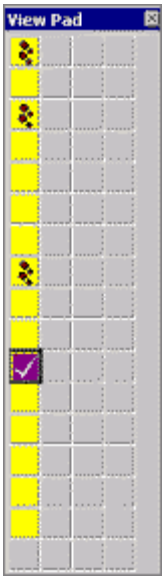
Use the view pad to move from one part of an execution plan to another without having to use the scroll bars.

To navigate execution plans

An execution plan must be loaded before using this procedure. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 Click .

   The **View Pad** appears.



2 Click the squares on the **View Pad** to navigate the plan:

- The yellow squares indicate pages that contain plan details.
- The gray squares indicate empty pages.
- The “check” graphic indicates the relative position in the plan.
- The “footprint” graphic identifies parts of the plan already viewed.
- The small red square means that the page contains a difference.

Locating Spool References

Use Teradata Visual Explain to move quickly from one spool reference to another.

To locate spool references

Before locating spool references, load an desired execution plan. For more information, see “Loading and Viewing an Execution Plan” on page 139.

1 Right-click on any displayed spool reference.
2 Click Spool Reference.
3 Double-click on any spool step number to locate it in the execution plan.

   Depending on the plan specifics, the following icons become available on the toolbar to quickly navigate between spool references:

   - (Previous Spool Reference)
   - (Next Spool Reference)

Entering a Query and Viewing Its Execution Plan

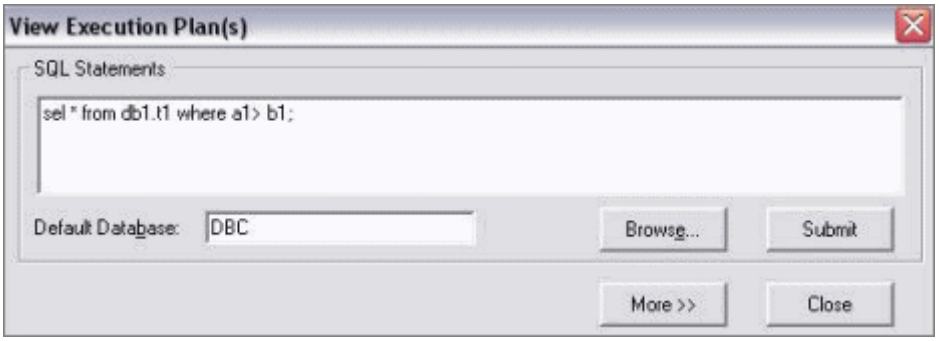
Enter a query from Teradata Visual Explain and display the resulting execution plan.

To enter a query and view its execution plan

1 Click .

   The View Execution Plan(s) dialog box appears.

Figure 38: View Execution Plan(s) - for Queries

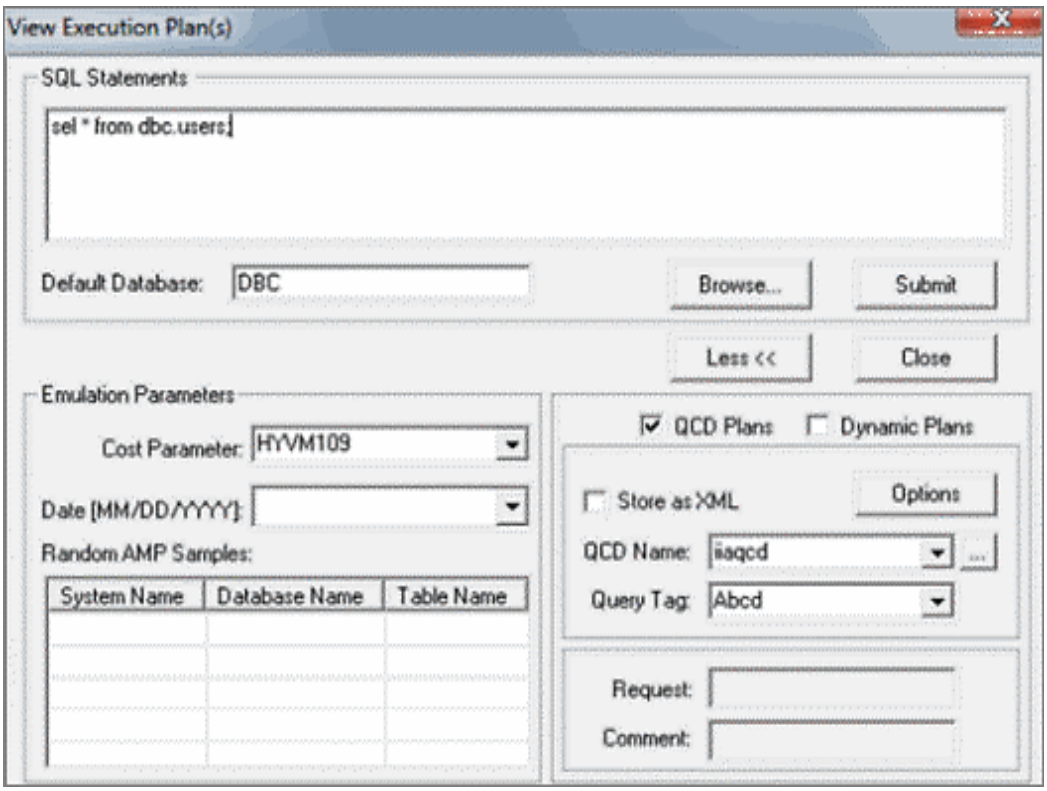


2 Use one of the following options:

- The default option is Explain in XML, which generates an execution plan but does not store plan information in the database.
- To insert execution plan information into the specified QCD, click the More button and select the Capture in QCD checkbox.
- To submit INSERT EXPLAIN statements with XML SQL and capture XML plans in the QCD, select the Store as XML checkbox.

  Note: This option is enabled only if the Capture in QCD checkbox is selected.

Figure 39: View Execution Plan - Expanded



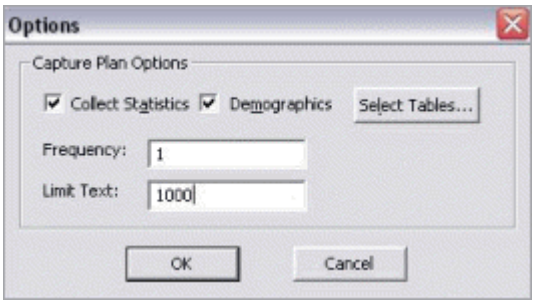
3 [Optional] To place a limit on the size of the query and DDL text captured in the QCD:

a Click the Options button to open the Options window.

b Enter **Limit Text** in the **Limit Text** checkbox and click **OK**.

If no value is entered, 0 is the default.

Figure 40: Options window



4 [Optional] Use the following steps to specify the number of times an SQL statement is typically performed within its identified workload:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Enter a value in the **Frequency** box in the **Options** window and click **OK**.

This value is used to weight the respective benefits of each column analyzed for inclusion in the index recommendation computed by Teradata Index Wizard. Any positive integer up to 4 B is valid. If a value is not specified, frequency defaults to 1.

5 In the **QCD Name** box, enter the name of the QCD database.

Note: This field is enabled only when **Capture in QCD** is selected.

6 In the **Query Tag** box, enter a name used to identify the query.

7 Do one of the following:

- To enter the statement text, type the text in the **Query** window.
- To open a file containing statement text, click **Browse** to select a *.sql* or *.txt* file.

8 [Optional] To collect statistics on tables involved in the plan:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Select the **Collect Statistics** checkbox in the **Options** window.

Statistics are collected on columns that are index candidates and for all columns and indexes referenced explicitly with values in the SQL statement when the **Collect Statistics** checkbox is selected. A candidate index column is defined as a column for which the value or join range access frequencies stored in the **Field** table of the specified QCD are greater than 0.

9 [Optional] To collect demographics on some tables involved in the plan:

a Ensure that the **Capture in QCD** checkbox is selected in step 2.

b Click the **Options** button to open the **Options** window.

c Select the **Demographics** checkbox.

d Click **Select Tables**.

10 If not already done, click **Select Tables** to specify the set of tables for which data demographics and statistics details are either to be included or excluded for analysis.

11 To emulate a target environment at various levels or scopes.

Note: Only the DBC or SystemFE user can set costs at the IFP and SYSTEM levels. An error occurs when these conditions are not met.

For more information, see "Setting or Resetting Random AMP Samples" on page 145 or "Setting or Resetting Cost Parameters" on page 146.

12 Click **Submit** to create the execution plan.

Setting or Resetting Random AMP Samples

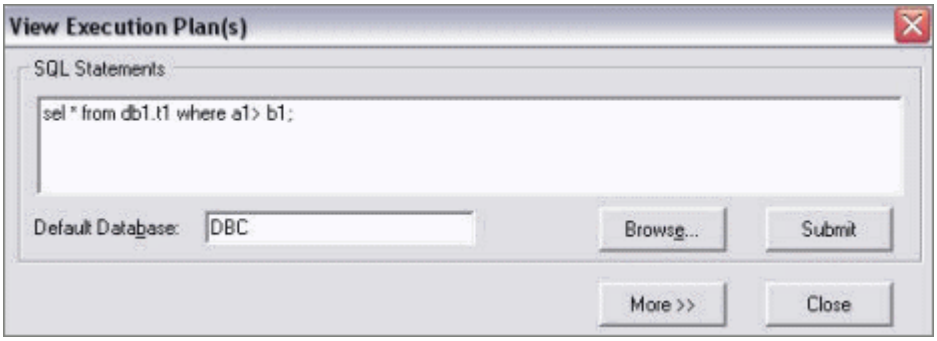
Use Teradata Visual Explain to set or reset random AMP samples for this session.

To set or reset random AMP samples

1 Click .

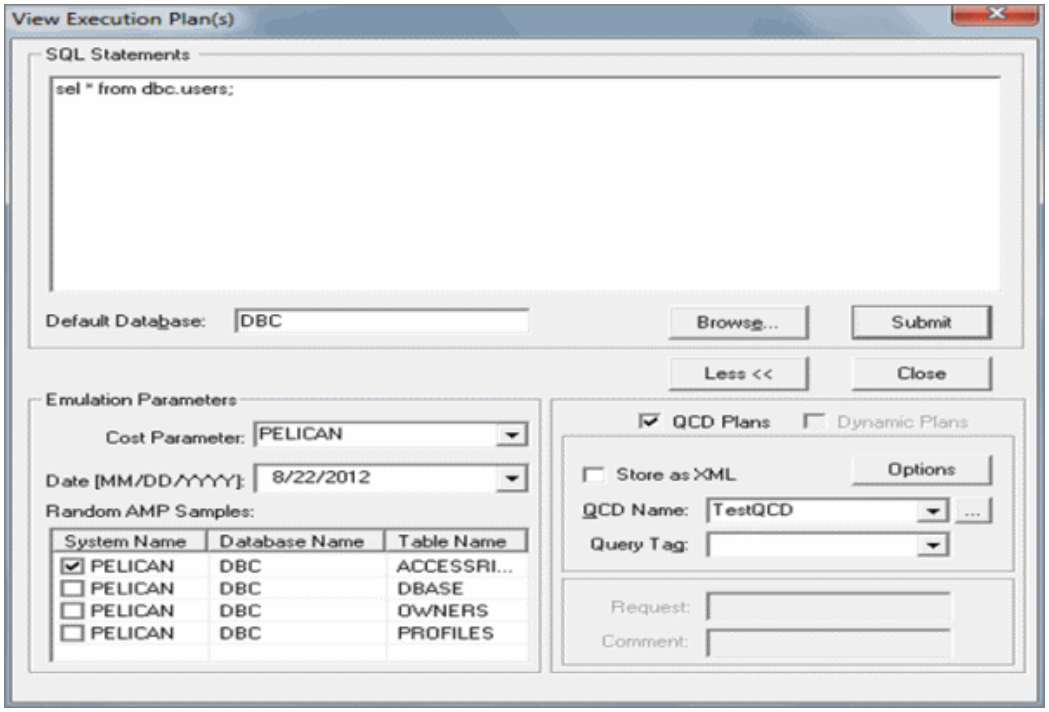
The **View Execution Plans** dialog box appears.

Figure 41: View Execution Plans - AMP Samples



2 Click **More** to show QCD Options and Emulation parameters. Random AMP samples for all tables are set at session level, which are reset after submitting the requests.

Figure 42: View Execution Plans Expanded - AMP Samples



3 Select the check box, under System Name, to select the RAS table.

Refer to the following publications for more information on random AMP samples.

- *Teradata System Emulation Tool User Guide* B035-2492.
- *SQL Request and Transaction Processing* B035-1142 (Target Level Emulation).

Setting or Resetting Cost Parameters

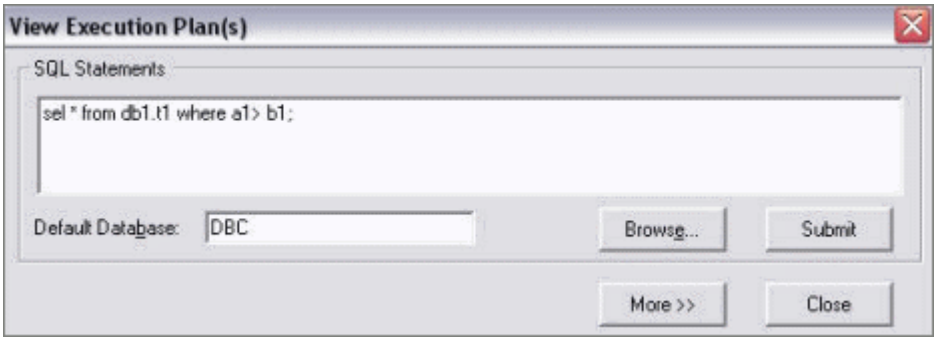
Use Teradata Visual Explain to set or reset cost parameters for the session.

To set or reset cost parameters

1 Click .

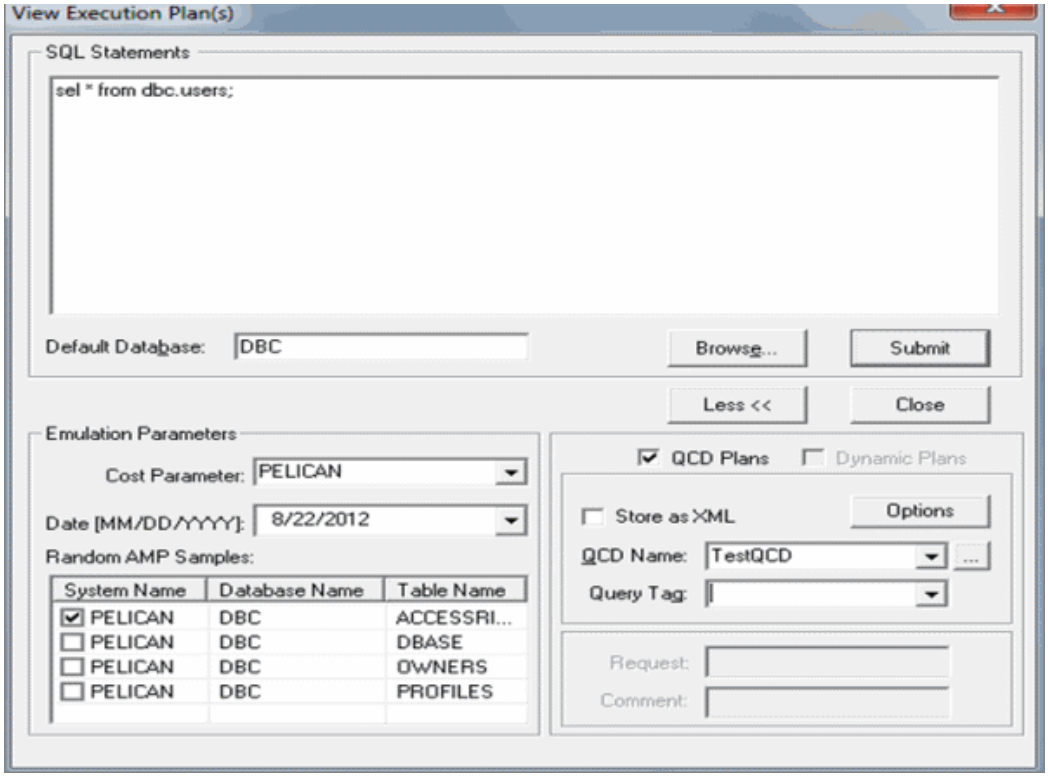
The View Execution Plans dialog box appears.

Figure 43: View Execution Plans - Cost Parameters



2 Click More to show QCD Options and Emulation parameters. The Cost Parameter is set at session level, which is reset after submitting the request.

Figure 44: View Execution Plan(s) Expanded - Cost Parameters



3 Select the Cost tag from the combination box.

Note: To set the Cost tag at system level, invoke TSET and use the Set TLE feature.

Refer to the following publications for more information on emulating cost parameters:

- *Teradata System Emulation Tool User Guide* B035-2492.
- *SQL Request and Transaction Processing* B035-1142 (Target Level Emulation).

Executing an SQL Query

The interface that is displayed for executing SQL depends on whether Use SQL Assistant or Execute SQL Window is selected. Access this option by clicking Tools > Options. Then, from the Options dialog box, select the General tab. For more information, see "Setting General Options" on page 84.

If Teradata SQL Assistant is selected as the interface for executing SQL, Teradata SQL Assistant is started. For more information about Teradata SQL Assistant, refer to *Teradata SQL Assistant for Microsoft Windows User Guide* B035-2430.

Note: When connected to Teradata Database using CLIv2, the Teradata Visual Explain native interface, Execute SQL, is used.

To execute an SQL query

1 Click .

The Execute SQL window opens.

2 Do one of the following:

- To select a query to be executed from a file, click File > Open Query.
- To execute the query, press F5.
- Type the query text in the Query window.
- Select a statement in the History window.

3 Click Execute.

The results of the query appear in one or more separate Result Set windows.

Table 25 describes all of the commands that can be used in the Execute SQL window.

Table 25: Execute SQL Window Commands

Menu	Command	Task
File	New Query	Clear the contents of the Query window.
File	Open Query	Select a query to be executed from a file.
File	Execute Query	Execute the query.
File	Save	Save the contents of the active window.
File	Print	Print the contents of the active window.
File	Close	Close the Execute SQL window.
Edit	Undo	Revert the previous operation done in the Query Text window.
Edit	Cut	Remove the selected text from the Query Text window.
Edit	Copy	Copy the selected contents of the active window.

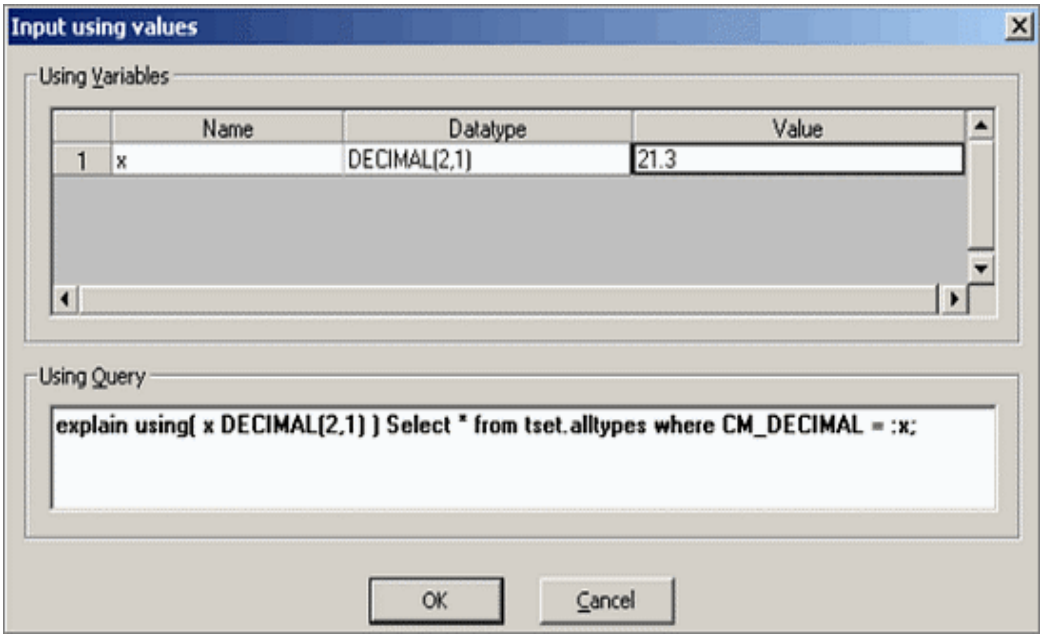
Table 25: Execute SQL Window Commands (continued)

Menu	Command	Task
Edit	Paste	Paste the contents of clipboard to the Query Text window.
Edit	Select All	Select everything in the active window.
Edit	Find	Find a string in the active window.
Edit	Find Next	Find the next occurrence of a string in the Query Text window.
Edit	Replace	Perform the replace operation in the Query Text window.
View	Toolbar	Toggle the toolbar of the Execute SQL window.
View	Status Bar	Toggle the status bar of the Execute SQL window.
View	Show History	Toggle the History window to show or hide.
Window	Cascade	Cascade the child windows in the Execute SQL window.
Window	Tile	Tile the child windows in the Execute SQL window.
Window	History	Make the History window active.
Window	Query	Make the Query window active.
Window	Result Set	Make a particular Result set active.
Help	Help Topics	View the Help topics.
Help	Using Help	View how to use Help.
Help	About	View the version and copyright information.

Input USING Values

When a query containing the USING clause is submitted in the execute SQL window, the Input using values screen displays. See Figure 45 on page 150.

Figure 45: Input USING Values



1 The Using Query display box shows the query containing the USING clause.
2 The Using Variables display box shows the values for the Name and Datatype variables.
3 For the Value variable, enter the data values in the Value column.
4 Click OK or Cancel:
   - Click OK to pass the variable data to the query during execution.
   - Click Cancel to prevent passing variable data to the query.

Analyzing Plans to Improve Performance

Use Teradata Visual Explain to improve plan performance by analyzing a set of queries (known as a workload) for index and partition recommendations. The recommendations are saved in the QCD tables.

Index and partition analysis can be performed separately or together on a current plan or on a group of plans.

Index recommendations for specific tables can be viewed after analysis. For more information, see "Viewing Index Recommendations for Tables" on page 153.

Analyzing Plans

Use this procedure to analyze a current plan or a group of plans, performing index and partition analyses separately or simultaneously.

Note: This feature is available only for plans captured in QCD.

To analyze a current plan or a group of plans

1 Select a plan or group of plans for analysis:

To analyze a plan:

a Load the plan.

b Right-click anywhere on the background.

c Click **Index Analysis** from the shortcut menu.

To analyze a group of plans:

a Load the plans.

b Click .

c In the **Select Plans for Analysis** dialog box, under **Available Execution Plans**, select the plans to analyze.

d Add selected plans to the **Selected Execution Plans** list.

e Click **Next**.

The **Define** dialog box appears. A unique workload name and index recommendation tag are generated internally, by default.

Figure 46: Index Analysis

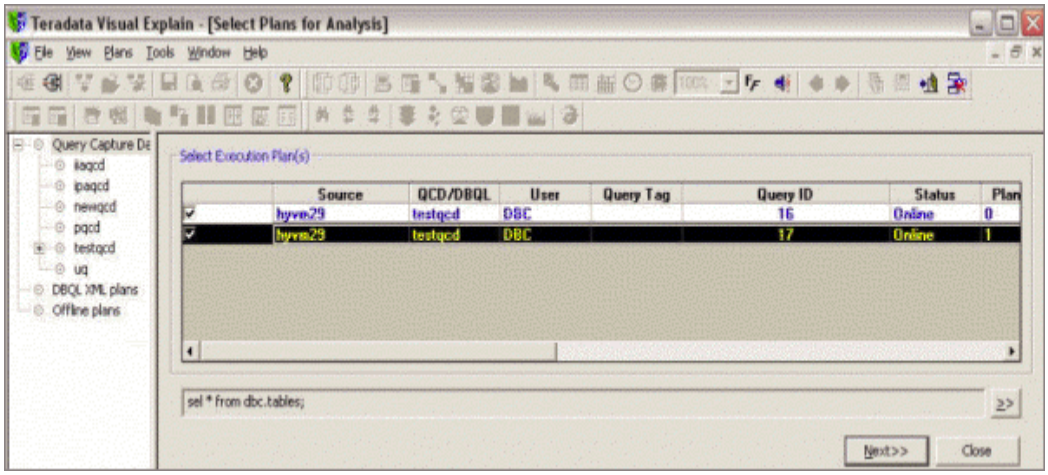
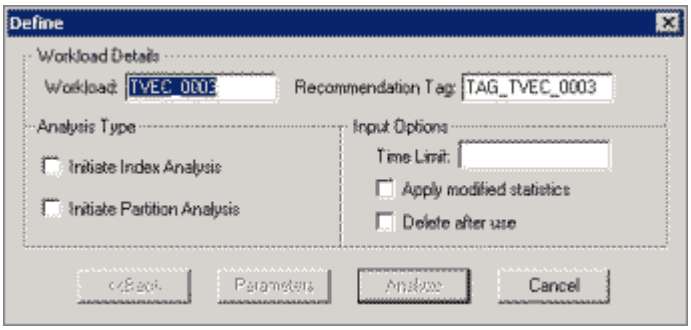


Figure 47: Define Workload



2 [Optional] To change the default, type a different workload name and index recommendation tag.

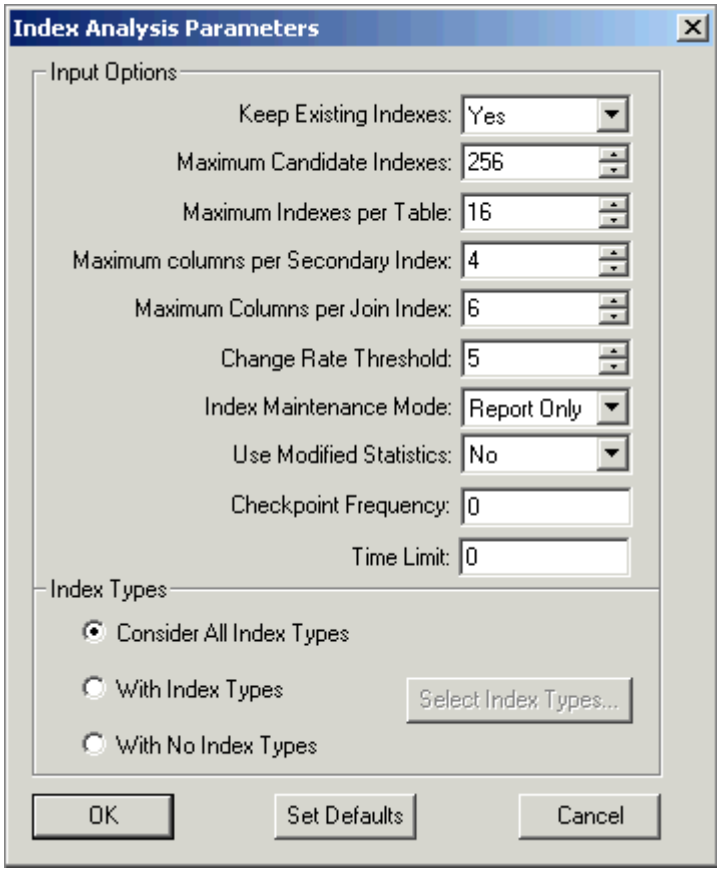
3 Under Analysis Type, select each type of analysis to perform:

- Initiate Index Analysis
- Initiate Partition Analysis

4 [Optional] If Initiate Index Analysis is selected in step 3, click Parameters to further define the index analysis.

The Index Analysis Parameters dialog box appears.

Figure 48: Index Analysis Parameters



5 [Optional] Set parameters to determine index recommendations.

For information on these parameters, refer to *Teradata Index Wizard User Guide* B035-2506 and read the topic Advanced Index Parameters.

6 Click OK to close this dialog box and return to the Define dialog box.

7 [Optional] Enter a number in the Time Limit box.

8 [Optional] Select the Apply modified statistics check box to use the modified statistics information during the analysis.

Note: The Apply modified statistics check box is available only when the Initiate Index Analysis check box is selected.

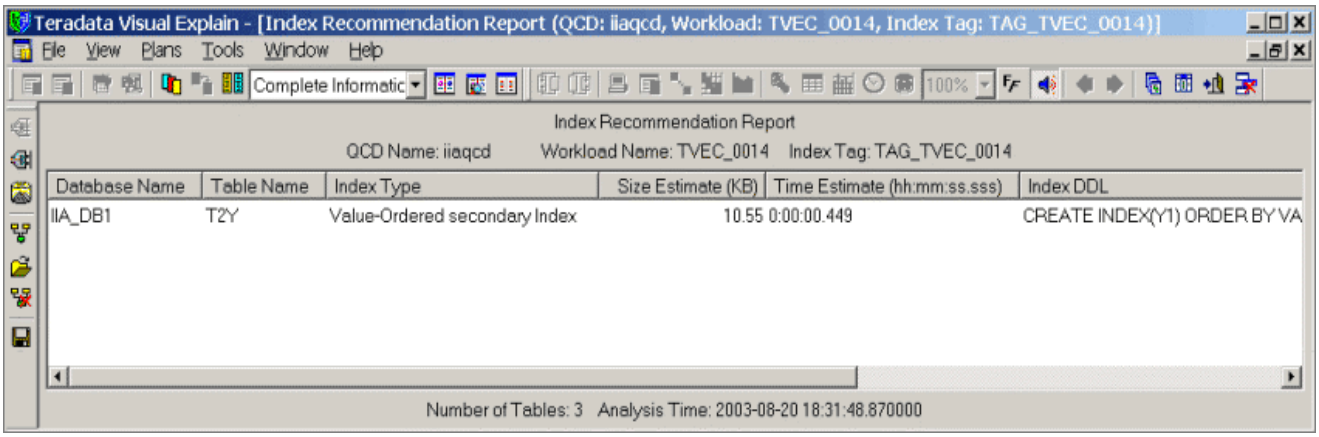
9 [Optional] Select the **Delete after use** check box to remove the workload and any recommendations after the analysis.

10 Click **Analyze**.

All selected analysis types are performed. A progress indicator appears showing the elapsed time since the start of the analysis. To terminate the analysis, click **Abort**.

The analysis is performed for the current plan. The resulting recommendations are shown in an Index Recommendation report. The index analysis provides the recommended secondary index for each table and the partition analysis provides partition recommendations for each table when a recommendation can be made. This report provides decision-making information to determine whether a recommendation is appropriate. The report also suggests indexes that can be dropped. For more information about the columns in this report, refer to *Teradata Index Wizard User Guide* B035-2506 and read the topic Index Recommendation report.

Figure 49: Index Recommendation Report



Viewing Index Recommendations for Tables

Use Teradata VE to view index recommendations information for a table from a previous index analysis operation.

To view index recommendations for tables

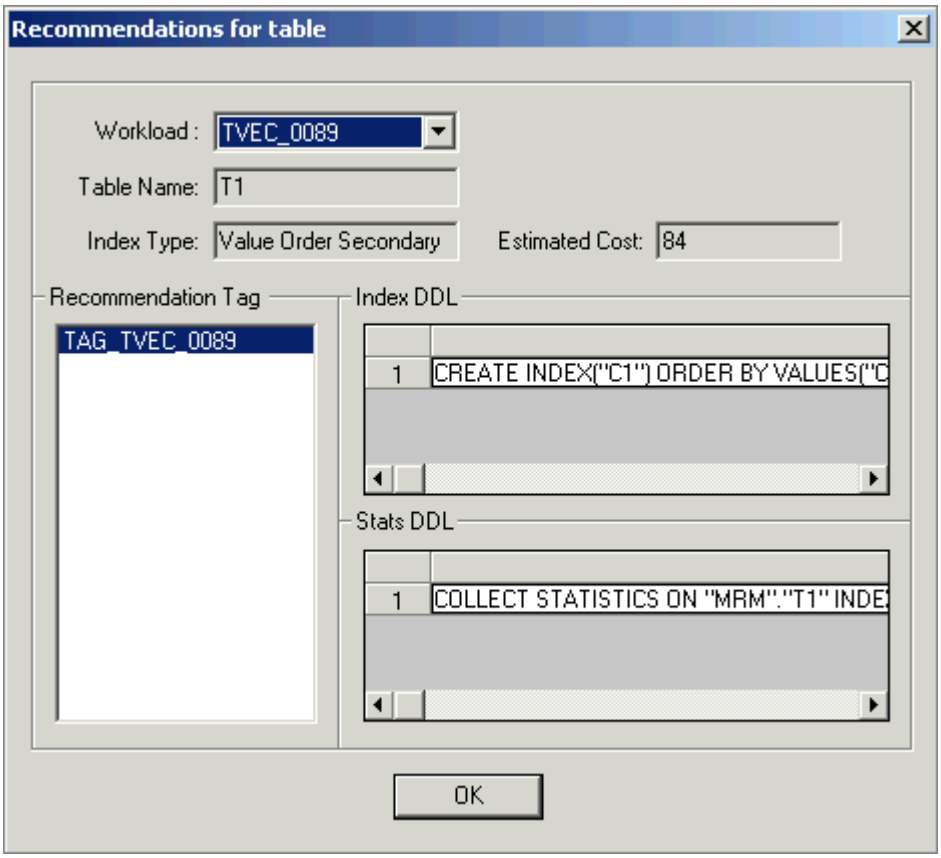
After loading and analyzing a plan or group of plans:

1 Right-click on in a plan.

2 Select **Recommendations** from the shortcut menu.

The **Recommendations for table** dialog box appears.

Figure 50: Recommendations for Table



Box Name	Description
Workload	Name of the workload used to obtain the index recommendations
Table Name	Table name
Index Type	Recommended-index type
Estimated Cost	Estimated cost to create the index in the database
Recommendation Tag	Index-name tag used while analyzing the indexes
Index DDL	DDL text used to create or drop the recommended indexes
Stats DDL	DDL statements used to collect the statistics recommended for improving the plan performance

Viewing Current vs. Captured Indexes, Object Definitions, and Statistics

Use Teradata Visual Explain to view table indexes, statistics, and objects used in an execution plan.

Viewing Indexes in the Tables

Use this procedure to view the indexes present in the tables referenced in the active query.

Note: Captured indexes, object definitions, and statistics are not available for plans opened from DBQL.

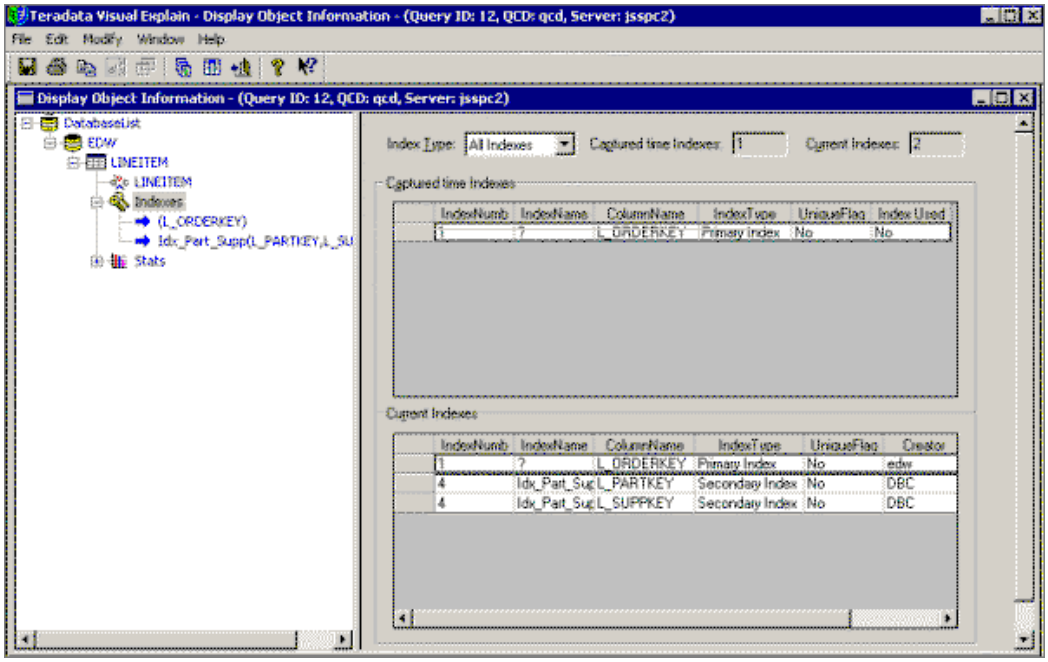
To compare current indexes to captured indexes

Load an execution plan before viewing the indexes. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 From the menu bar, choose View > Current vs. Captured > Indexes or from the toolbar, click .

The main window is transformed. The left pane displays a hierarchical tree of the objects referenced in the query.

Figure 51: Display Object Information



2 Highlight the index in the database list located in the left pane.

The indexes that existed at the time the execution plan was captured are listed in the right pane under Captured time Indexes. If the table still exists and the indexes were changed, the indexes are listed in the right pane under Current Indexes.

Note: The number of captured-time and current indexes is listed in the upper right corner and might be helpful for comparison.

3 From the Index Type list, select the type of indexes to view.

By default, All Indexes is selected.

Index Type	Description
Hash Ordered (ALL)	Display hash ordered covering secondary indexes. Note: ALL represents the syntax element specifying that the NUSI retain the row ID pointers for each logical row of a join index (as opposed to only the compressed physical rows).
Join Index	Display only join index types.
Primary Key	Display only primary-key index types.
Hash Index	Display only hash index types.
Value Ordered (ALL)	Display the value ordered covering secondary indexes. Note: ALL represents the syntax element specifying that the NUSI retain the row ID pointers for each logical row of a join index (as opposed to only the compressed physical rows).
Primary	Display only nonpartitioned, primary index types.
Partitioned	Display only partitioned, primary index types.
Secondary	Displays only secondary index types.
Unique	Display only unique-constraint, primary index types.
Value Ordered	Display only value-ordered secondary indexes.

Viewing Objects Used in the Execution Plan

To compare current objects to captured objects

Load an execution plan before viewing objects. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 Click .

The main window is transformed. The left pane displays a hierarchical tree of the objects referenced in the query.

2 To view the definition that existed at the time the execution plan was created, highlight an object in the database list.

If the object still exists, the current definition appears in the right pane.

Viewing Statistics for Tables Used in the Execution Plan

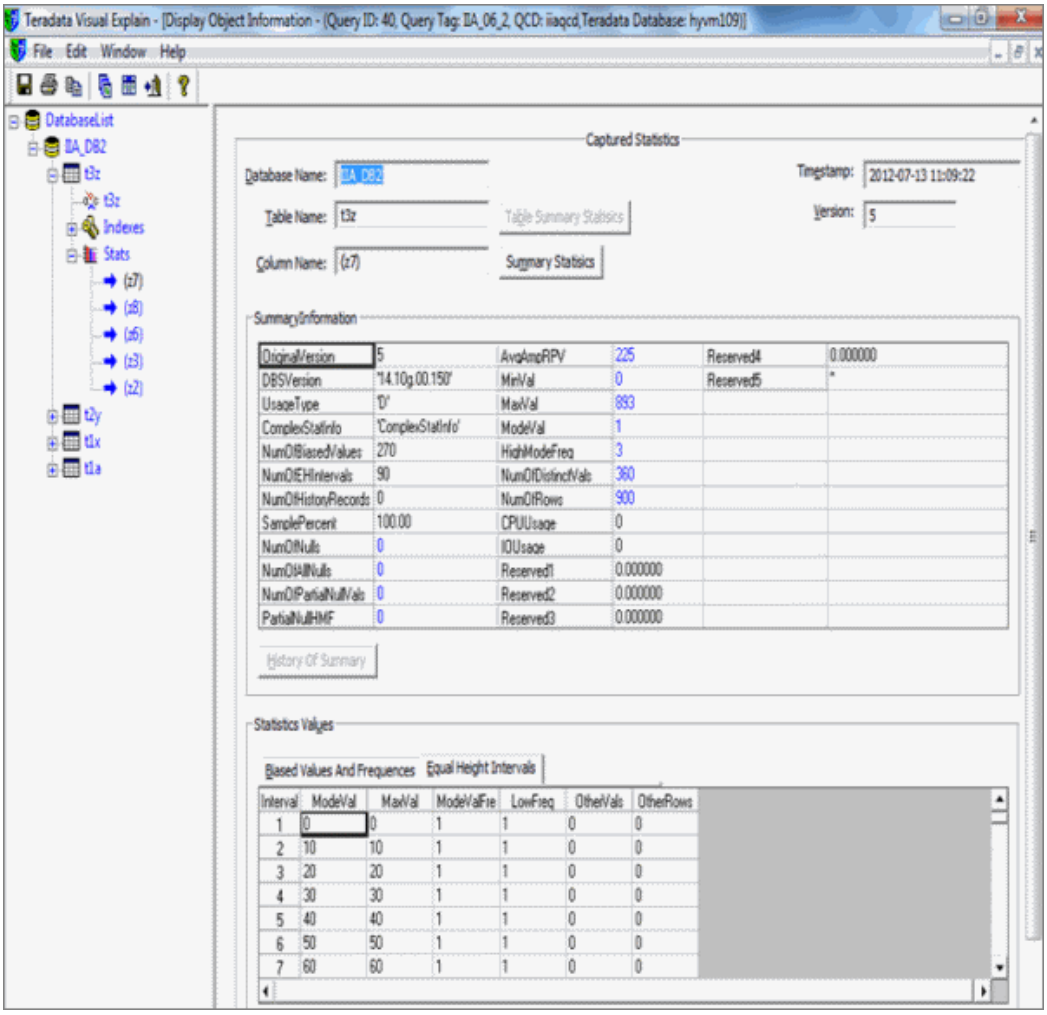
To compare current table statistics to captured-table statistics

Load an execution plan before viewing statistics. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 Click .

The Display Object Information window opens. The left pane displays a hierarchical tree of the objects referenced in the query.

Figure 52: Display Object Information



2 To view the statistics that existed at the time the execution plan was created, highlight a table in the database list (left pane).

If the table still exists, the current statistics appear in the right pane.

Window Areas/Items	Description
Statistics Information	
Database Name	Displays the name of the database on which the table is present.
Table Name	Displays the name of the table on which the column is present.
Column Name	Displays the name of the column on which the interval statistics are displayed.
Interval Type	Indicates the type of interval: • High-Biased Intervals • Equal-Height Intervals By default, ALL is selected.
Version	Displays the version number of the statistics structure in effect when the statistics were collected.
Timestamp	Displays the time the statistics were collected.
Summary Information	
Min Value	Displays an estimate of the smallest value for the specified column or index in the specified table. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Mode Value	Displays an estimate of the most frequently occurring value or values for the column or index in the specified table. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Sampled	Indicates whether the statistics are sampled.
Mode Frequency	Displays an estimate of the number of rows in the interval that have a modal value for the column or index. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Number of Nulls	Displays the number of rows for which one of the stats collected fields is null.
Sample Percent	Displays the percentage of statistics sampled if sampled statistics were collected on the column.
Number of Rows	Displays the number of rows in the table. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.

Window Areas/Items	Description
Number of Uniques	Displays the number of unique values for the column on which statistics were collected. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Number Of Intervals	Displays the number of intervals in the frequency distribution for the column or index. For Teradata Database versions earlier than 12.00.00, the maximum number of intervals is 100. For Teradata Database versions 12.00.00 or later, the maximum number of intervals is 200.
Number of All Nulls	Displays the number of rows having all the stats collected fields being nulls.
Average AMP RPV	Displays the average AMP-local RPV obtained by averaging the average RPV from each AMP.
Number of Amps	Displays the number of AMPs in the connected Teradata Database.
One AMP Sample Est	Displays one AMP sampling estimate of the table cardinality at the statistics collection time.
All AMP Sample Est	Displays all AMP sampling estimates of the table cardinality at the statistics collection time.
Note: The following values are displayed in the spreadsheet for all 100 or 200 intervals.	
Interval	A bounded, non-overlapping set of attribute values.
Mode Value	Displays an estimate of the most frequently occurring value or values for the column or index in the specified table. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Max Value	Displays an estimate of the largest value for the column or index in the interval. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Mode Frequency	Displays an estimate of the number of rows in the interval that have a modal value for the column or index. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Non-Modal Value	Displays the number of distinct non-modal values (values that are not the most frequently used) in the interval. Note: If the non-modal value is -1, there is one loner in the interval. If the non-modal value is -2, there are two loners in the interval. The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Non-Modal Rows	Displays the total number of rows for all the non-modal values in the interval. Note: The description relates to the columns listed under the Sampled Stats tab as a whole. This is known as *interval 0* information. Interval 0 contains global statistics for the column or index.
Original Version	Displays previous statistics version.

Window Areas/Items	Description
DBS Version	Displays database version statistics collected on.
Usage Type	Displays Usage Type of the statistics. Valued values are 'S' for Summary mode and 'D' for Detailed mode.
Complex Stat Information	Displays the information for Statistics on Complex Types such as Geospatial.
Number of History Records	Displays the Number of History Records.
Number of Partial Null Values	Displays the Number of Partial Null Values.
Partial Null HMF	Displays the highest frequency of values having partial nulls of the Expression List.
IO Usage	Displays the amount of IO consumed.
Number of Biased Values	Displays the number of Biased Values.
Number of EH Intervals	Displays the number of EH Intervals.
Number of Distinct Values	Displays the number of Distinct Values in the table.
CPU Usage	Displays the amount of CPU Consumed.

Viewing Actual Versus Estimated Cardinalities

When comparing actual cardinalities to estimated cardinalities, Teradata Visual Explain loads the actual cardinality information from the DBQL for the most recently logged execution of the query.

Note: This feature is available only for plans captured using QCD.

To compare actual cardinality to estimated cardinality

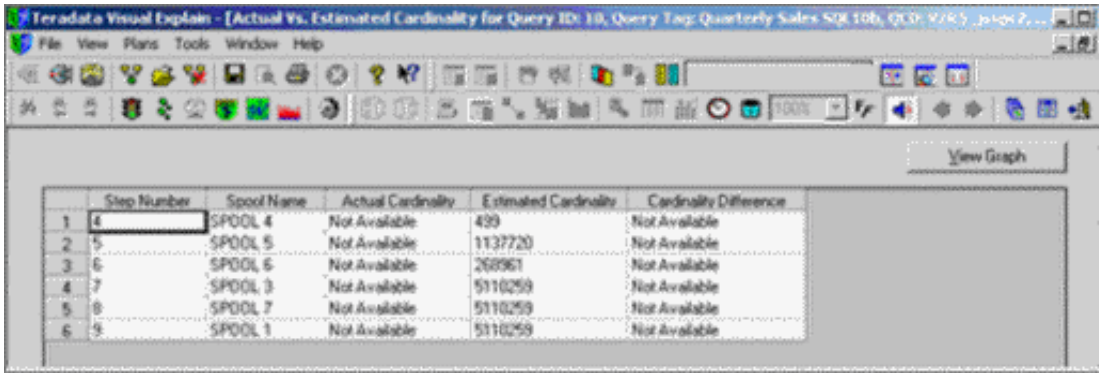
Load an execution plan before viewing cardinalities. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 Click .

The Actual Versus Estimated Cardinality window opens.

Note: If the query is not present in the DBQL, only the estimated cardinality appears. If the query is found, the actual and estimated cardinalities appear.

Figure 53: Actual Versus Estimated Cardinality



Column Name	Description
Step Number	Step number in the query plan.
Spool Name	Name of the spool in the query plan.
Actual Cardinality	Actual number of rows in the spool.
Estimated Cardinality	Estimated number of rows in the spool.
Cardinality Difference	Difference between actual versus estimated cardinality.

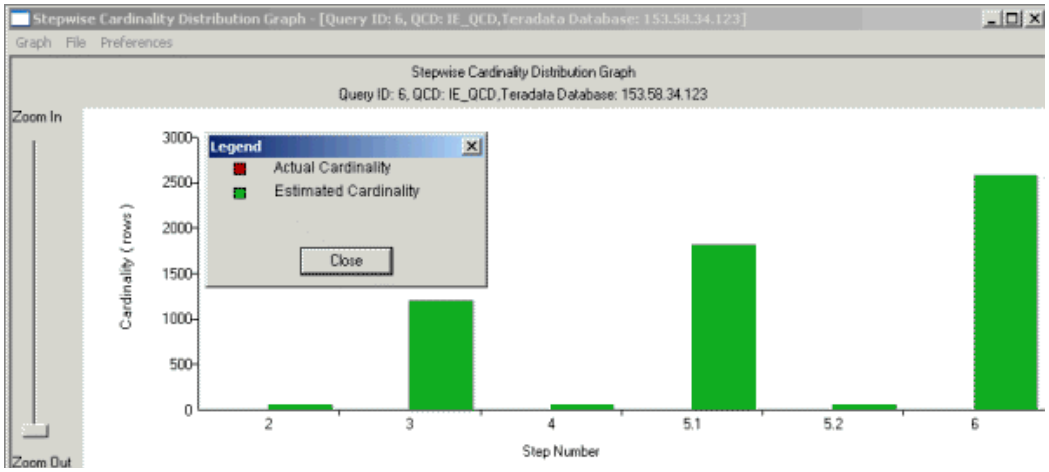
2 [Optional] Double-click the column headings to change the sort order.

For example, double-click the **Estimated Cardinality** column heading to list the steps with the highest cardinality estimates first.

3 To view the graphical layout of the stepwise cardinalities, click **View Graph**.

The Stepwise Cardinality Distribution Graph appears and displays the actual cardinality compared to the estimated cardinality in a graph format.

Figure 54: Stepwise Cardinality Distribution Graph



Compare Actual to Estimated Cost

When comparing actual to estimated cost, use Teradata Visual Explain to load the actual cost information from the DBQL for the most recently logged execution of the query.

Note: This feature is available only for plans captured using QCD.

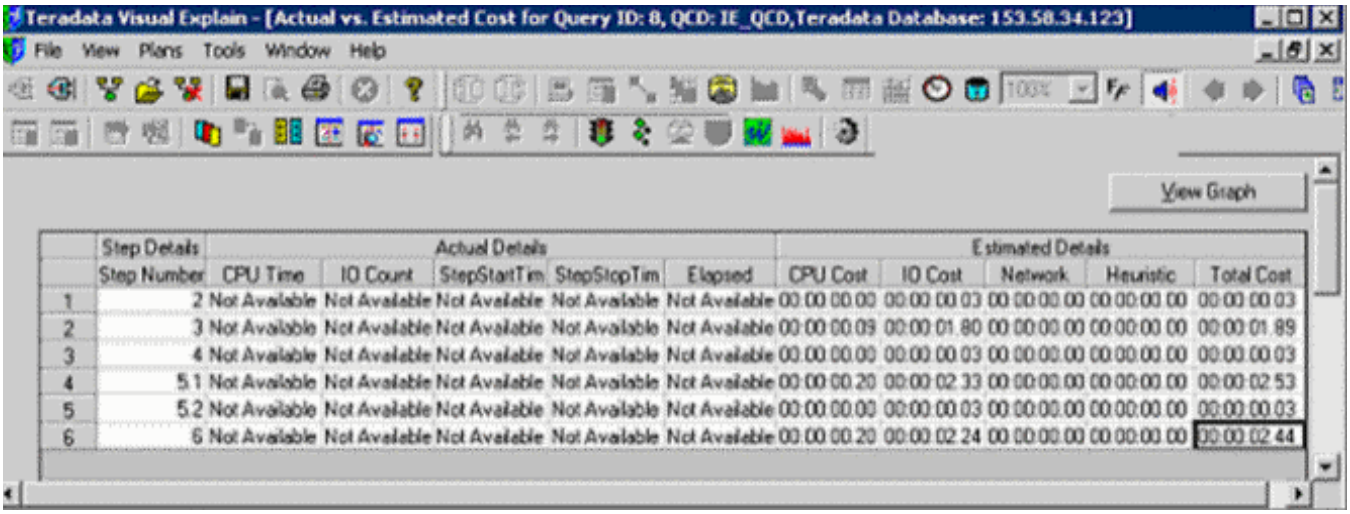
To compare actual to estimated cost

Load an execution plan before viewing cost. For more information, see "Loading and Viewing an Execution Plan" on page 139.

1 Click .

The Actual Versus Estimated Cost window opens.

Figure 55: Actual Versus Estimated Cost for Query ID



Column Name	Description
Step	Step number in the query plan
CPU Time	CPU the step uses
IO Count	I/O the step uses
StepStart Time	Timestamp, to the nearest microsecond, when the step was sent to the AMP
StepStop Time	Timestamp, to the nearest microsecond, when the step returned from the AMP
Elapsed Time	Difference between StepStop Time and StepStart Time
CPU Cost	Estimated CPU time required to execute the step
IO Cost	Estimated I/O time required to execute the step
Network Cost	Estimated network time required to execute the step

Column Name	Description
Total Cost	Total estimated time required to execute the step

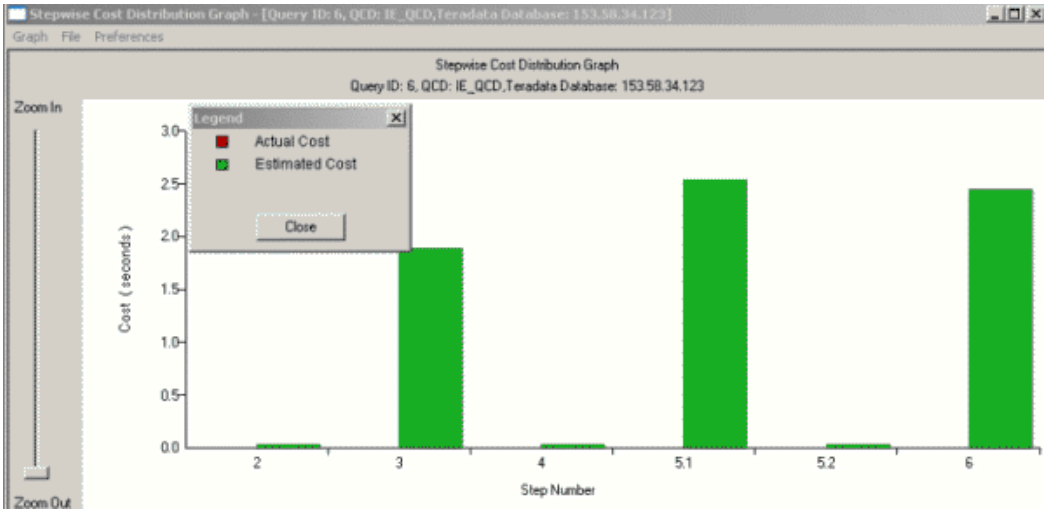
Note: If the query is not present in the DBQL, only the estimated cost appears. If the query is found, the actual and estimated stepwise costs appear.

2 [Optional] In the Actual vs. Estimated Cost window, double-click the column headings to change the sort order.

For example, double-click the CPU Cost column heading to list the steps with the highest cost estimates first.

3 To view the graphical layout of the stepwise costs, click View Graph in the table.

Figure 56: Stepwise Cost Distribution Graph for Query ID



Viewing Query Execution

Monitor and retrieve information about a query in real time, when connected using CLIv2.

To view the query execution

Connect to Teradata Database using CLIv2 before viewing query execution.

1 Click .

The Monitor SQL window opens and displays all the steps in the execution plan along with the actual cost, estimated cost, and cardinality information.

Note: The Monitor SQL feature is available only for SELECT statements.

2 To access common functions, display the shortcut menu by right-clicking on the header row of the information grid, then choosing Copy, Find, Hide/Show Columns, Select All, Clear All, Save As, or Print.

3 To view a graph comparing actual cardinality to estimated cardinality, click **View Cardinality Graph**.

4 To view a graph comparing actual cost to estimated cost, click **View Cost Graph**.

The query is resubmitted to Teradata Database to get the actual stepwise cost and cardinality. Only SELECT statements are resubmitted.

Viewing Teradata Database Configuration

View the Teradata Database configuration where the execution plan is captured.

To view the Teradata Database configuration

Load an execution plan before viewing the Teradata Database configuration. For more information, see "Loading and Viewing an Execution Plan" on page 139.

- Click .

  The **Teradata Database Configuration** dialog box appears. Information in this dialog box cannot be altered. The table describes the information displayed in this dialog box.

Box Name	Description
Database Name	Name of the database to which the user was connected when the execution plan was captured.
Number of AMPs	Number of AMPs on the system where the plan is captured.
Number of PEs	Number of PEs on the system where the plan is captured.
Number of Nodes	Number of nodes on the system where the plan is captured.
Release	When the plan is captured, the release level of the software loaded on Teradata Database. For example: • V2R.*nn.nn.nn.nn* for releases earlier than 12.00.00 • *nn.nn.nn.nn* for release 12.00.00 and later **Note:** This attribute comes from the DBC.DBCInfoTbl system table.
Version	When the plan is captured, the version of the software on Teradata Database. For example, 05.01.02.03 represents the: • Major release number (05) • Minor release number (01) • Maintenance release number (02) • E-fix or patch release number (03) **Note:** These numbers come from the DBC.DBCInfoTbl system table.

Viewing the Object Use Count Report

Use this procedure to:

- Analyze how often objects are used in a plan
- Identify tables with the potential for performance improvements
- Analyze use of a specific object in a plan
- Delete obsolete or unused objects to improve system performance

To display the Object Use Count report for a plan or plan icon

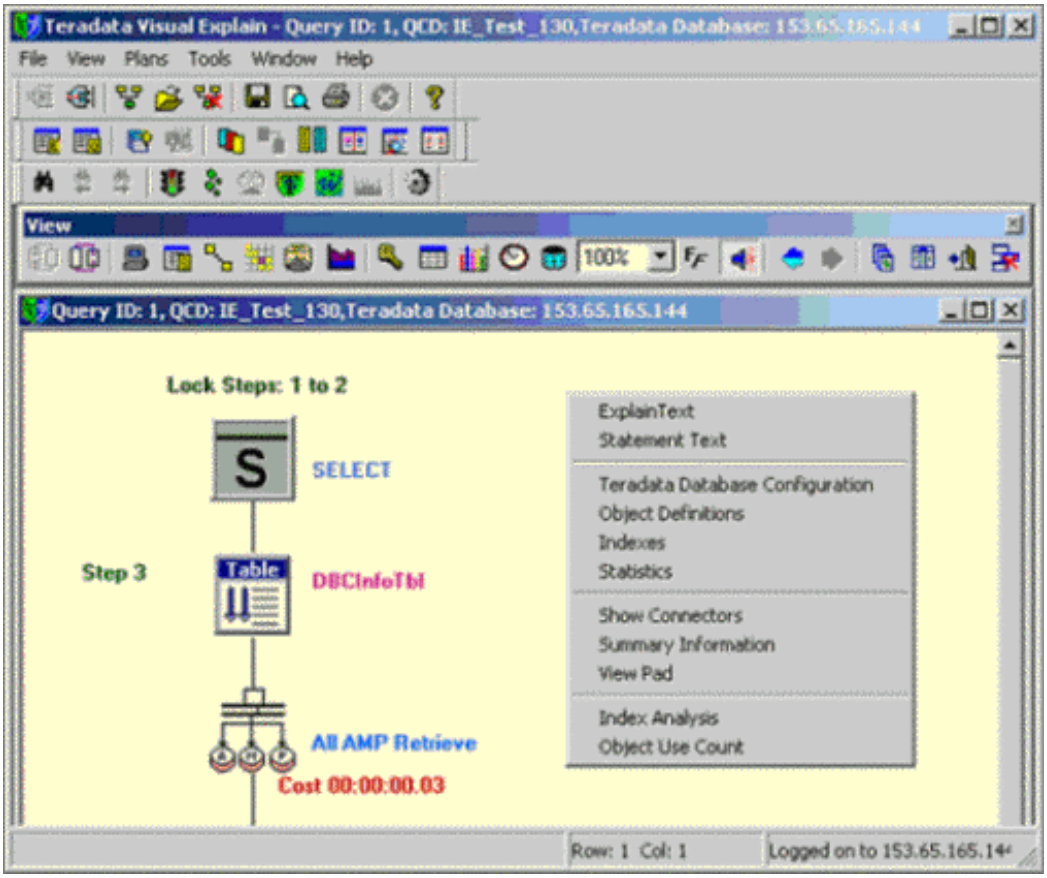
1 Open an execution plan, online or offline.

2 Do one of the following:

To view details about all of the objects used in the plan:

a Right-click anywhere on the plan background.

Figure 57: Object Details



b Click **Object Use Count** from the shortcut menu.

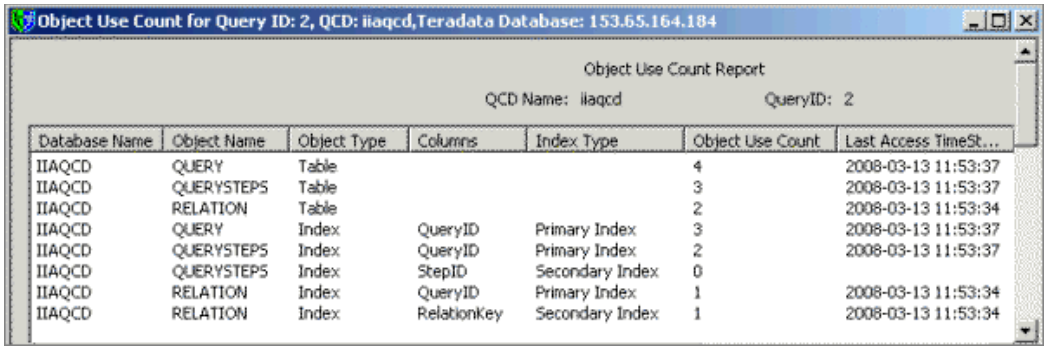
To view details about a specific object:

a Right-click on an icon.

b Click **Object Use Count** from the shortcut menu.

The **Object Use Count** report appears.

Figure 58: Object Use Count Report



Object Use Count for Query ID: 2, QCD: iiaqcd,Teradata Database: 153.65.164.184

Object Use Count Report

QCD Name: iiaqcd QueryID: 2

Database Name	Object Name	Object Type	Columns	Index Type	Object Use Count	Last Access TimeSt...
IIAQCD	QUERY	Table			4	2008-03-13 11:53:37
IIAQCD	QUERYSTEPS	Table			3	2008-03-13 11:53:37
IIAQCD	RELATION	Table			2	2008-03-13 11:53:34
IIAQCD	QUERY	Index	QueryID	Primary Index	3	2008-03-13 11:53:37
IIAQCD	QUERYSTEPS	Index	QueryID	Primary Index	2	2008-03-13 11:53:37
IIAQCD	QUERYSTEPS	Index	StepID	Secondary Index	0	
IIAQCD	RELATION	Index	QueryID	Primary Index	1	2008-03-13 11:53:34
IIAQCD	RELATION	Index	RelationKey	Secondary Index	1	2008-03-13 11:53:34

Column Name	Description
Database Name	The name of the database that owns the object.
Object Name	The object name.
Object Type	The object type: index, table, or view.
Columns	The comma-separated columns that make up the index. **Note:** For Table and View object types, this column is empty.
Index Type	The type of index (for example, Primary Index).
Object Use Count	The object use count of the objects available from the DBC.tables and DBC.indices views.
Last Access TimeStamp	A timestamp indicating the last time the object was accessed.
Insert Use Count	Displays the Insert Use Count of the objects available from. **Note:** This column is available in the report when connected to database version 14.10 or later.
Update Use Count	Displays Update Use Count of the objects. **Note:** This column is available in the report when connected to database version 14.10 or later.
Delete Use Count	Displays Delete Use Count of the objects. **Note:** This column is available in the report when connected to database version 14.10 or later.

CHAPTER 9

Data Demographics

The Data Demographics window visually depicts the demographics of a selected table, its sub-tables, and statistics in its stat fields. The demographics option is available only for online visual plans.

Demographic information is presented in a graphical format such as bar graphs or line graphs and in a report form.

For more information on data demographics, refer to *Teradata Index Wizard User Guide* B035-2506.

View Data Demographics

The Data Demographics window shows several different graphs of information about the selected table. For a list of specific graphs, see "Data Demographics Graph Types" on page 172.

To learn how the Demographics window works, see "Data Demographics Window Controls" on page 170

To control the number of windows that open when viewing Data Demographics. see "Control the Number of Open Windows" on page 167.

View demographics for one or more tables using the following options:

- "View Data Demographics from the Visual Plan" on page 168
- "View Data Demographics from the View Menu" on page 168

Control the Number of Open Windows

To control the number of open windows

Use this procedure to control the number of open windows displaying demographics information.

1 Click Tools > Options.

2 Click the General tab.

3 Do one of the following:

- To display one window, clear the Use multiple windows for demographics display check box.

- To display a separate window for each plan:
  - i Select the Use multiple windows for Demographics Display check box.
  - ii Clear the Open separate window for each table check box.
- To display a separate window for each table:
  - i Select the Use multiple windows for Demographics Display check box.
  - ii Select the Open separate window for each table check box.

View Data Demographics from the View Menu

To open the demographics window from the View menu

1. Load an online plan in Teradata Visual Explain.
2. Click View > Show Demographics.

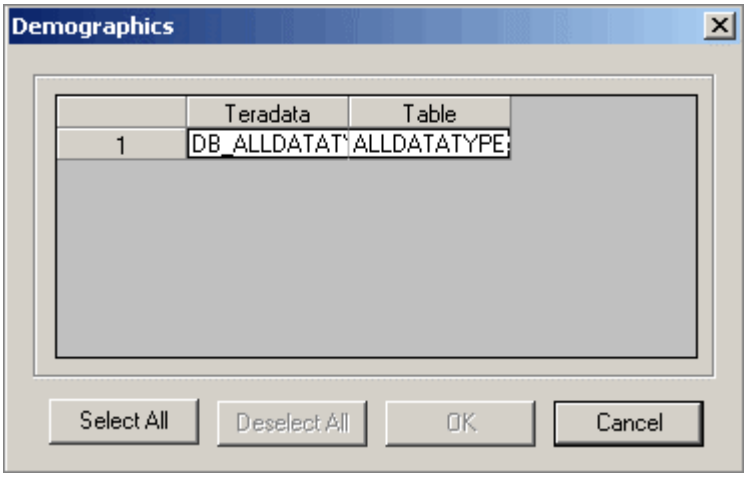
   The Data Demographics window appears.

View Data Demographics from the Visual Plan

To open the demographics window from the visual plan

1. Load an online plan in Teradata Visual Explain.
2. Right-click on a table icon in the visual plan.
3. Click Table Demographics.

   The Demographics dialog box opens.



Note: You can also click to open the Demographics dialog box.

4 Select a table or multiple tables, and then click OK.

The Data Demographics window appears.

Note: You can also right click on a table in the visual plan and select Show Demographics to open the Data Demographics window.

Data Demographics Menu Options

The Data Demographics window displays information about a selected table in graphical or report form.

The window gives you several ways to view and manage the information.

The following table describes how to use the Data Demographics menu options.

Table 26: Data Demographics Menu Descriptions

Menu Item	Description/ Function
File	Use to print, save as, print preview, close.
View	• Set Font- sets the fonts for reports. • Restore Default Preferences- restores the default preferences. • Change Graph Settings- changes the type of graph displayed. • Change Graph Perspective- change the perspective of a 3D graph. • Fit in Window- fit the graph in the current window size. • Actual Size- shows the graph in its actual size.
	Use these controls to move from view to view. • First Page • Previous Page • Current Page • Next Page • Last Page
	• Fit in Window- fit the graph in the current window size. • Actual Size- shows the graph in its actual size.
Graphs	Selects the type of graph you want to see.
Report	View each type graph in report form.
Preferences	Sets the preferences for each type of graph.

Data Demographics Toolbar Icons

Click the icons on the Data Demographic window toolbar to access graphs and reports.

Table 27: Data Demographics Toolbar Icons

Icon	Description
	Opens the Disk Space Utilization graph.
	Opens the Disk Space Utilization graph in a report form.
	Opens the Spool Space Utilization graph.
	Opens Spool Space Utilization graph in a report form.
	Opens the Row Count/Avg. Row Size graph.
	Opens the Row Count/Avg. Row Size graph in a report form.
	Opens the Blocks Distribution graph.
	Opens the Blocks Distribution graph in a report form.
	Opens the Statistics Details graph. The Summary Statistics Information graph opens by default. See "Statistics Details Graph" on page 180 for more icons that are available when viewing this graph.
	Opens the Statistics Details graph in a report form. The Summary Statistics Information report opens by default.

Data Demographics Window Controls

Use the controls in the Data Demographics window to manage the different types of information available. The controls are explained in the following topics:

- Selecting Bar or Line Graphs
- Move Through Table Pages

Selecting Bar or Line Graphs

Graphs can be presented using lines, 2D bars, or 3D bars. Use the Bar/Line drop-down menu to select the type you want.

Note: You can also change graph settings using the Change Graph Settings dialog box in the View menu.

Each type of bar graph and line graph has additional options that can be selected from the additional drop-down box, shown below.

Figure 59: Bar/Line drop-down menu

Graph options are shown in the following table.

Table 28: Bar and Line Graph View Options

Graph Type	Descriptions
Bar2D and Bar3D Bar Graphs	• barStacked- shows the graph bars stacked on top in each other. • barZClustered- shows the bar graphs side by side.
Line Graphs	• lineLines- shows simple line for the line graph. • lineSymbols- uses symbols to highlight graph points. • lineSticks- uses colored sticks to indicate graph points.

Move Through Table Pages

You can navigate through different pages of a graph by specifying the X-axis range in the edit box next to the page controls. Specify the range as follows: <Minivalue><Maxvalue>. For example, 2-3.

Move forward or backward through table graphs using the arrow buttons, shown in the following figure. The buttons allow you to move to the next page, previous page, last page and first page.

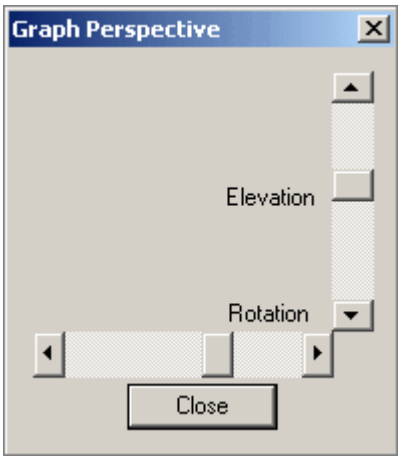
Figure 60: Navigation Controls for Table Graphs

To change the graphic perspective

When the Bar graph is set to 3D, you can change the perspective of the graph.

1 Click Perspective (as shown to the right) .

The Graph Perspective controller opens.



2 Use the sliders to change the graph perspective.

Note: You can also open the Graph Perspective controller by clicking View > Change Graph Perspective.

Data Demographics Graph Types

There are several types of information graphs available in the Data Demographics window. Each depicts information visually using bars or lines and in a report form. Each graph can be printed and saved for presentations or reports.

The type of information graphs include:

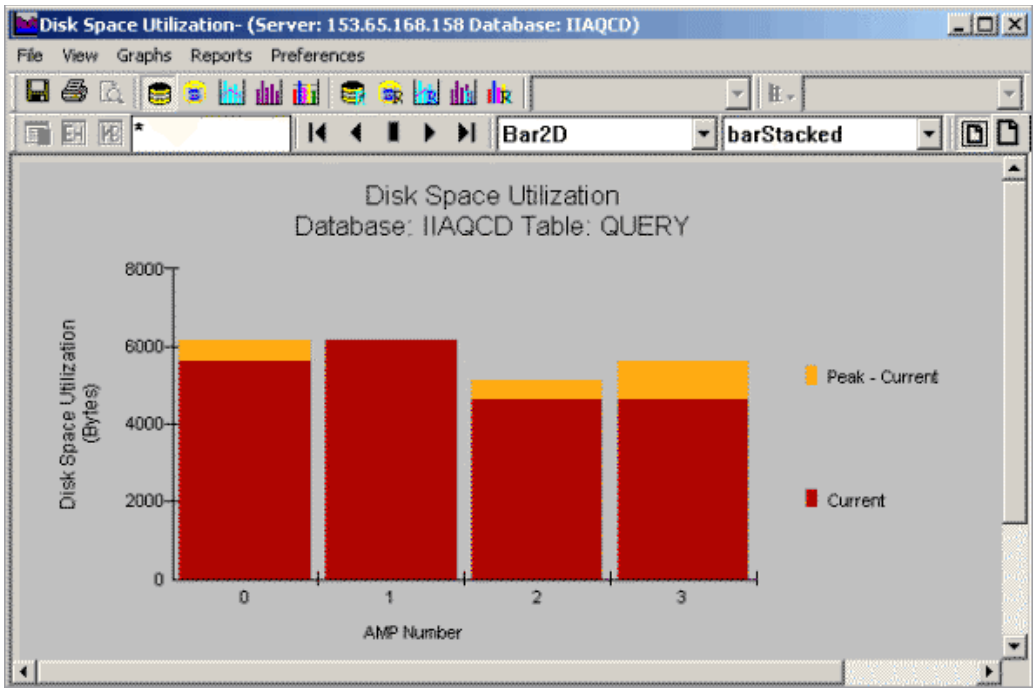
- Disk Space Utilization Graph
- Spool Space Utilization Graph
- Row Count/Average Row Size Graph
- Blocks Distribution Graph
- Statistics Details Graph

Disk Space Utilization Graph

The Disk Space Utilization graph shows the disk space utilization of the table on the AMPs in the system.

To open the disk space utilization graph

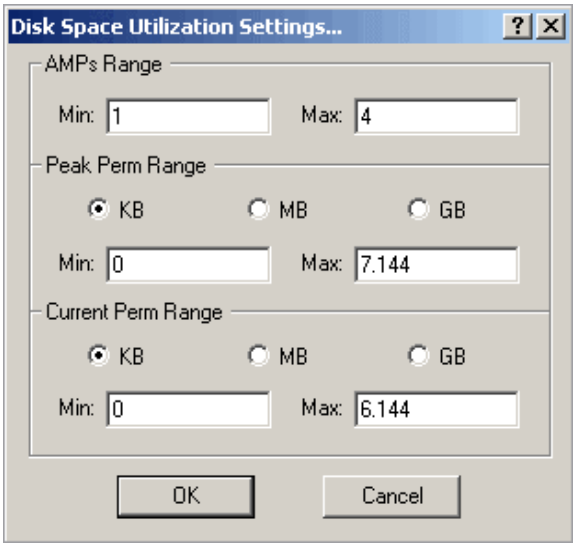
✔ In the Data Demographics window, click Graphs > Disk Space Utilization.



To change the settings for the disk space utilization graph

1 Click **Preferences > Disk Space Utilization.**

The **Disk Space Utilization Settings** dialog box appears.



2 Set the following options:

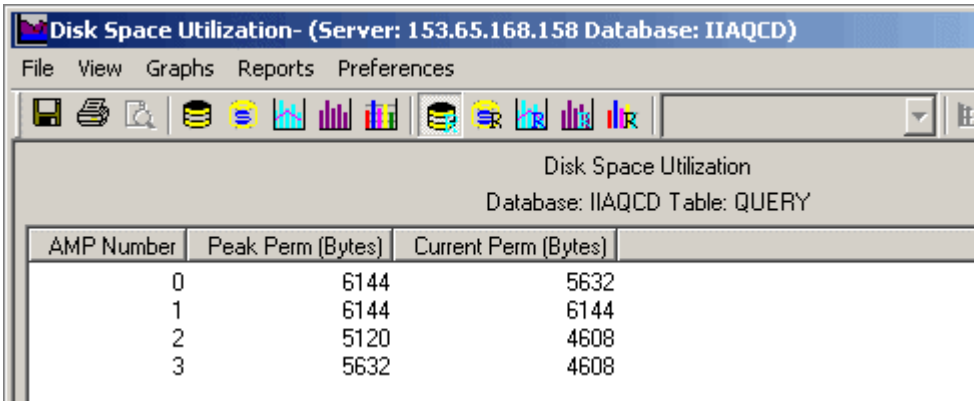
Option	Description
AMPs Range	Specify the minimum and maximum AMP range.

Option	Description
Peak Perm Range	Specify the peak permanent space range.
Current Perm range	Specify the current permanent space range.

3 Click OK.

To open the disk space utilization report

✔ Click Reports > Disk Space Utilization.



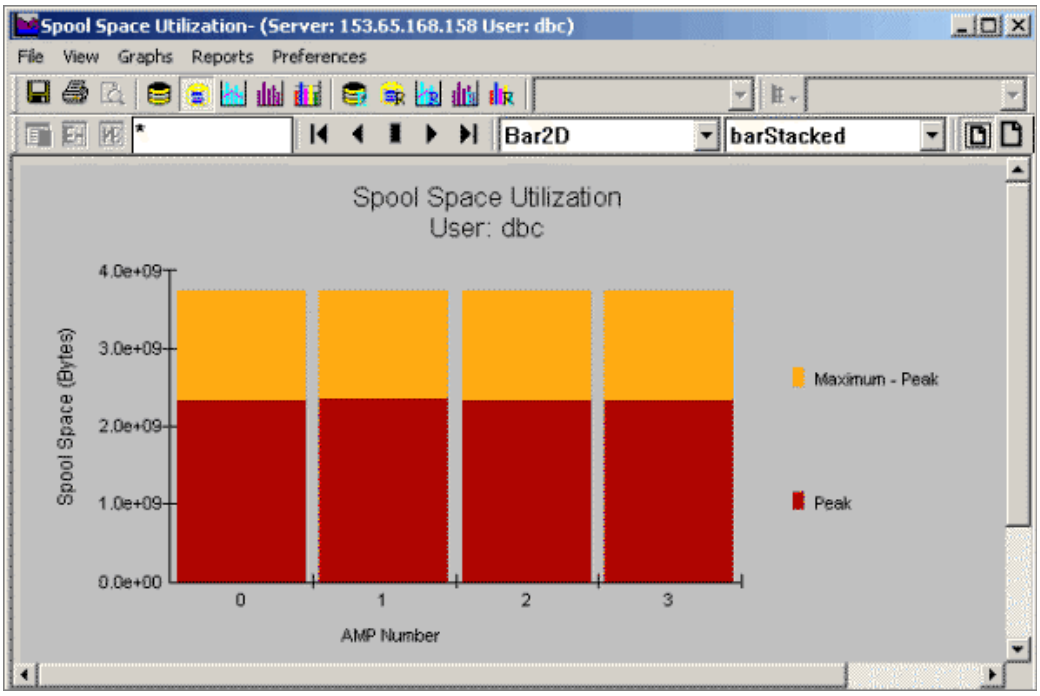
Spool Space Utilization Graph

The Spool Space Utilization graph shows the spool space use of the table over the AMPs in the system.

To open the spool space utilization graph:

- Click Graphs > Spool Space Utilization.

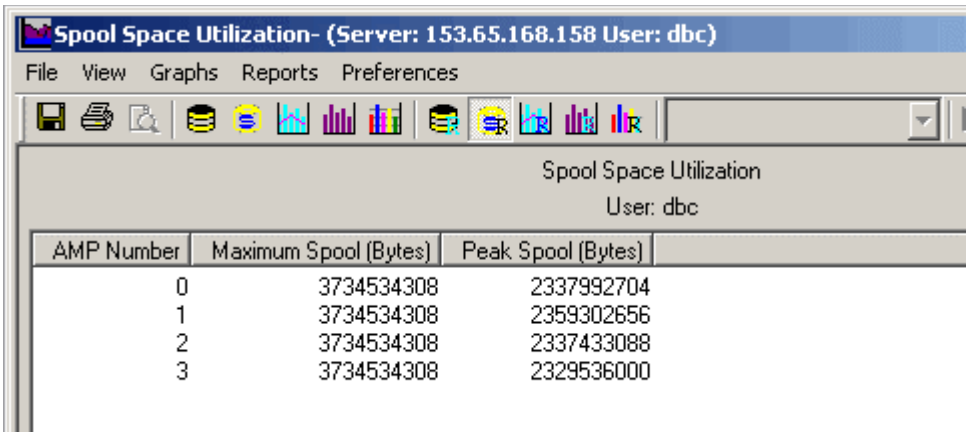
  The Spool Space Utilization graph appears.



To view the spool space utilization report

- Click Reports > Spool Space Utilization.

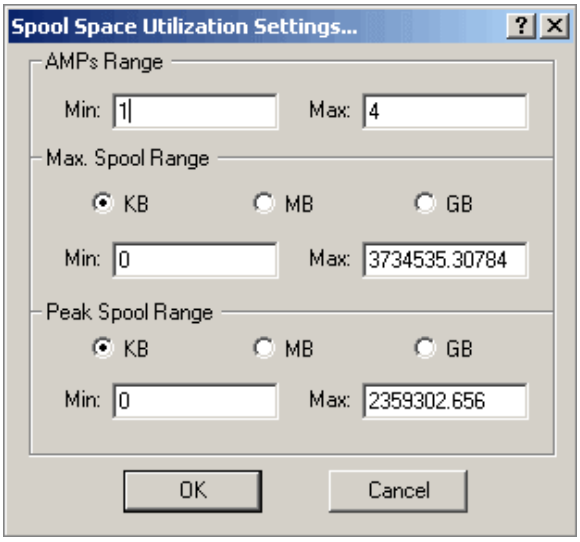
  The Spool Space Utilization report appears.



To change the settings of the spool space utilization graph

1 Click Preferences > Spool Space Utilization.

  The Spool Space Utilization Settings dialog box appears.



2 Set the following options:

Option	Description
AMPs Range	Specify the minimum and maximum AMP range.
Max. Spool range	Specify the total amount of temporary spool space range.

Option	Description
Peak Spool Range	Specify the range for the highest amount of temporary spool space.

3 Click OK.

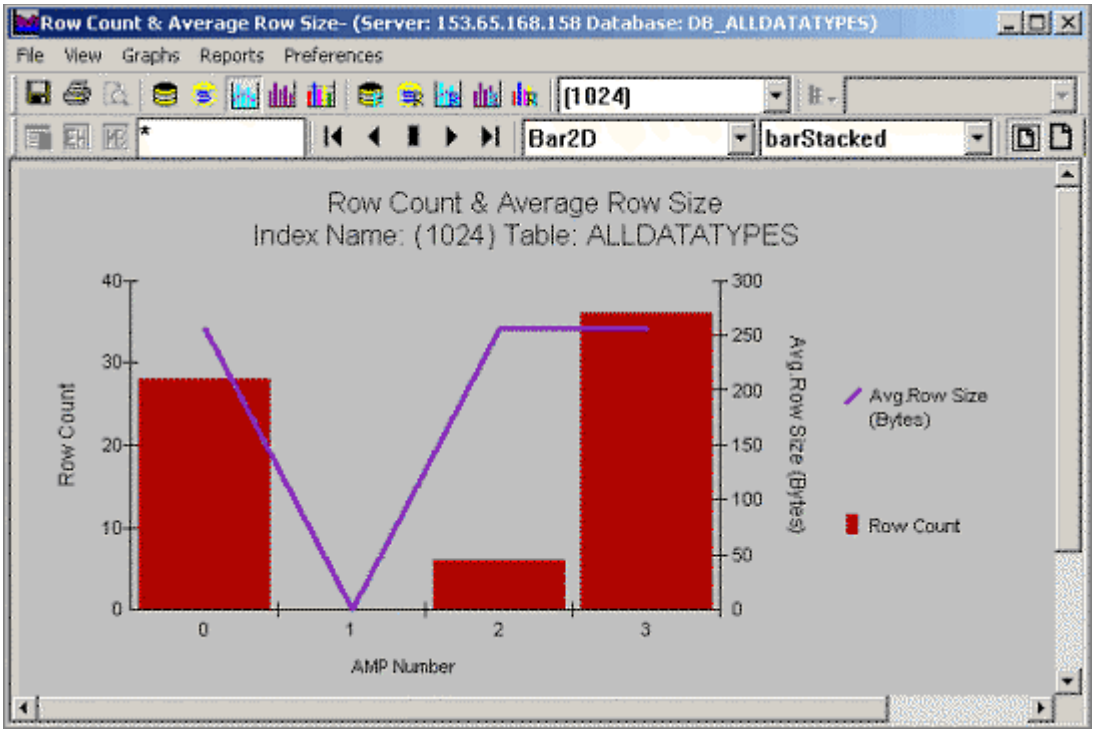
Row Count / Average Row Size Graph

The Row Count/Average Row Size graph row distribution of the tables (and sub-tables) over the AMPs in the system. The sub-table for the table can be selected from the Standard box.

To open the row count/average row size graph:

- Click Graphs > Row Count/Average Row Size.

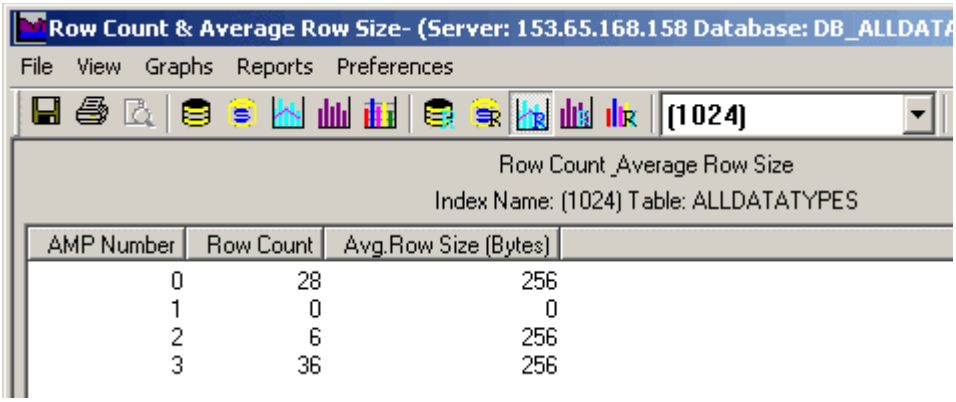
  The Row Count/Average Row Size graph appears.



To view the report form of this graph:

- Click Reports > Row Count/Average Row Size.

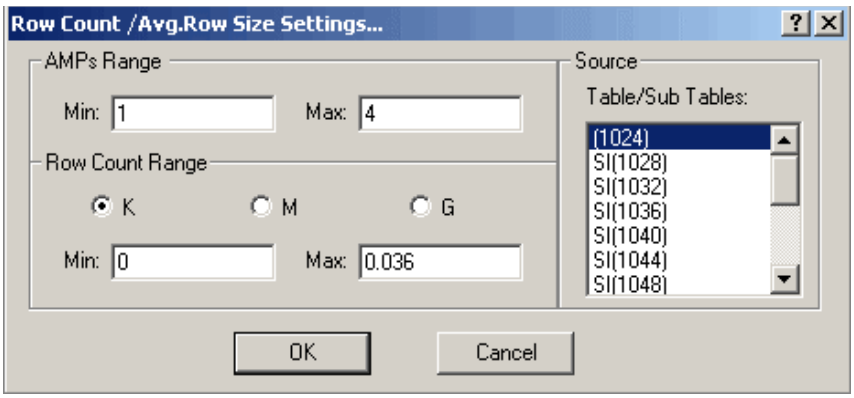
  The Row Count/Average Row Size report appears.



To change the settings of the row count/avg. row size graph

1 Click Preferences > Row Count/Avg. Row Size.

  The Row Count/Avg. Row Size dialog box appears.



2 Set the following options:

Option	Description
AMPs Range	Set the minimum and maximum AMP range.
Row Count Range	Set the minimum and maximum row count of a table or an index sub table for different AMPs.
Source: Tables/Sub Tables	Select the row distribution for the base table or for any of the available index sub tables.

3 Click OK.

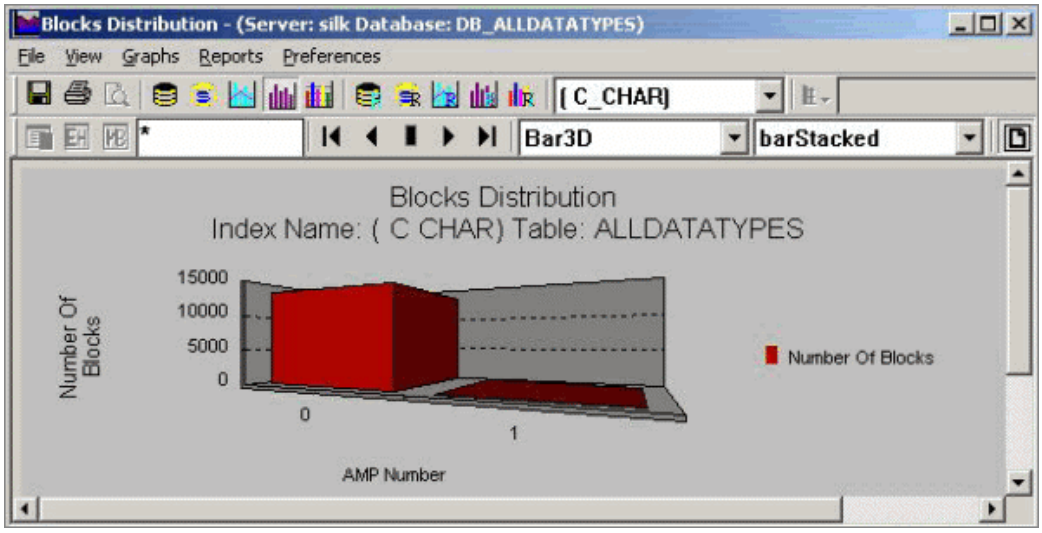
Blocks Distribution Graph

The Block Distribution graph shows block distribution of the selected tables (or sub-tables) over the AMPs in the system.

To open the blocks distribution graph

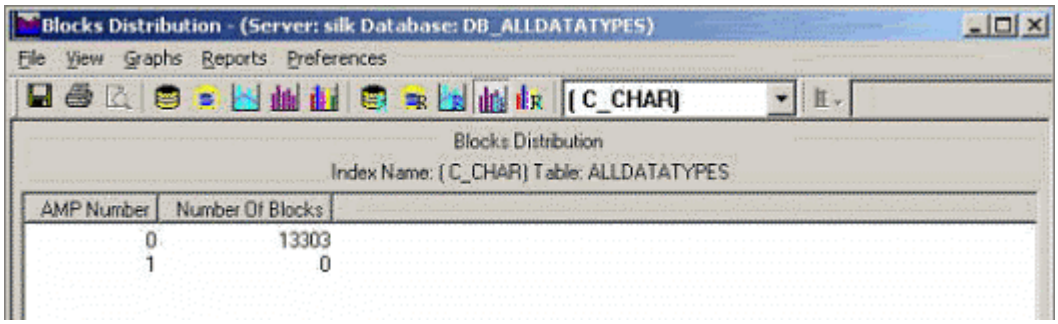
- Click Graphs > Blocks Distribution.

  The Blocks Distribution graph appears.



To view the report form of this graph

- Click Reports > Blocks Distribution.

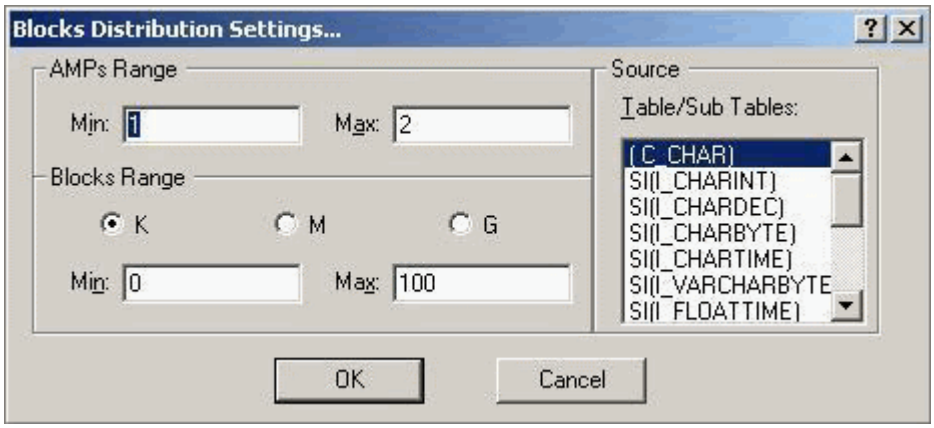


  The Blocks Distribution report appears.

To change the settings of this graph

1 Click Preferences > Blocks Distribution.

  The Blocks Distribution Settings dialog box appears.



2 Set the following options:

Option	Description
AMPs Range	Specify the minimum and maximum AMP range.
Blocks Range	Block Distribution of a table or an index sub table over different AMPs. Specify the minimum and maximum range.
Source: Table/Sub Tables	Select the row distribution for the base table or for any of the available index sub tables.

3 Click OK.

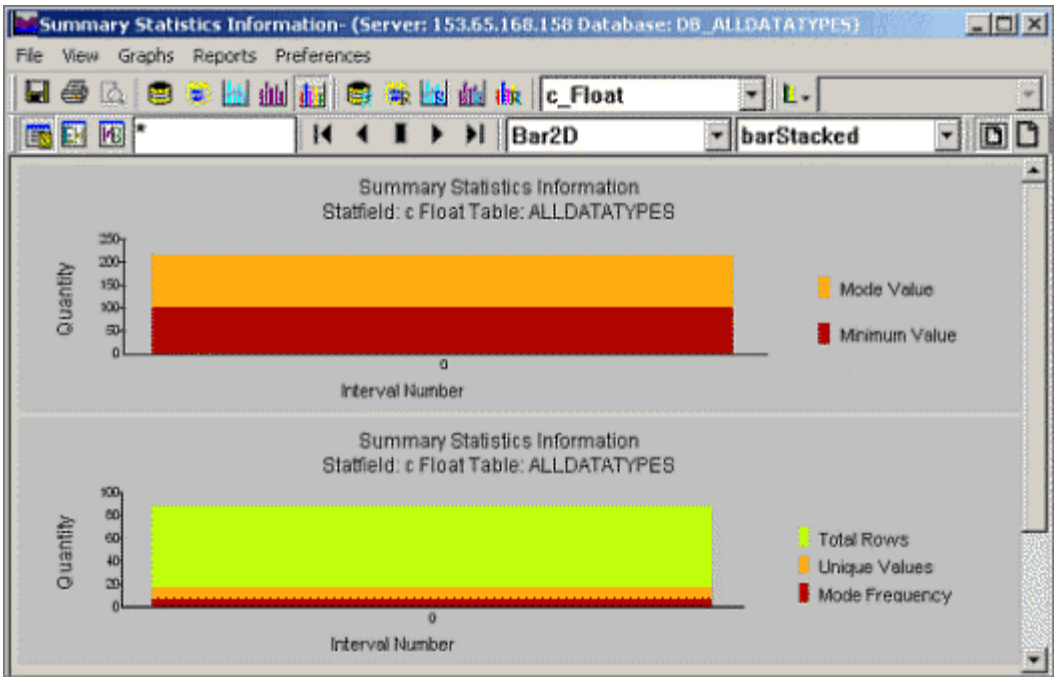
Statistics Details Graph

The Statistics Details graph shows the statistics of the specified column or index of the table.

To view the statistics collected on the table or sub-table

- Click Graphs > Statistics Details.

  The Summary Statistics Information (Statistics Details) graph appears.



The Statistics Details graph shows three levels of details of the selected table. The following table shows the icons used for each level and explains the level.

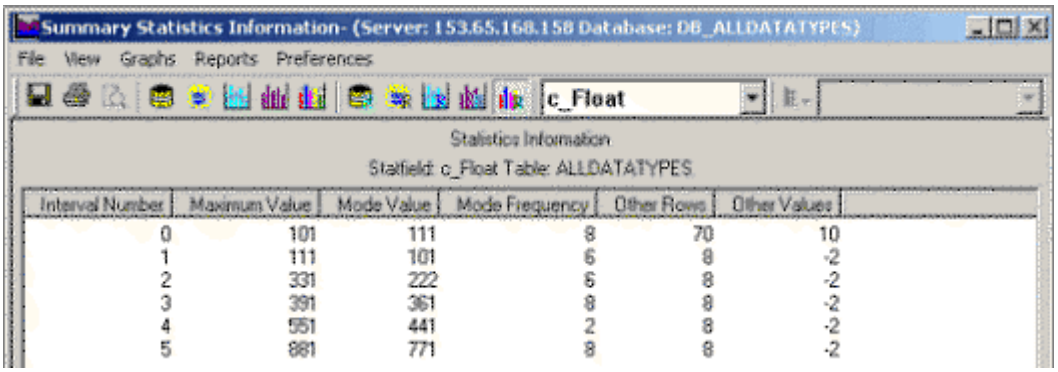
Table 29: Statistics Details Graph Icon Descriptions

Type of Detail	Icon	Description
Summary Statistics Information		Shows a summary of the statistics on the selected tables.
Equal Height Statistics Information		Shows equal heights statistics.
High-Biased Column Statistics		Shows high bias column statistics.

To view the report form of this graph

- Click Reports > Statistics Details.

  The Summary Statistics Information report appears.

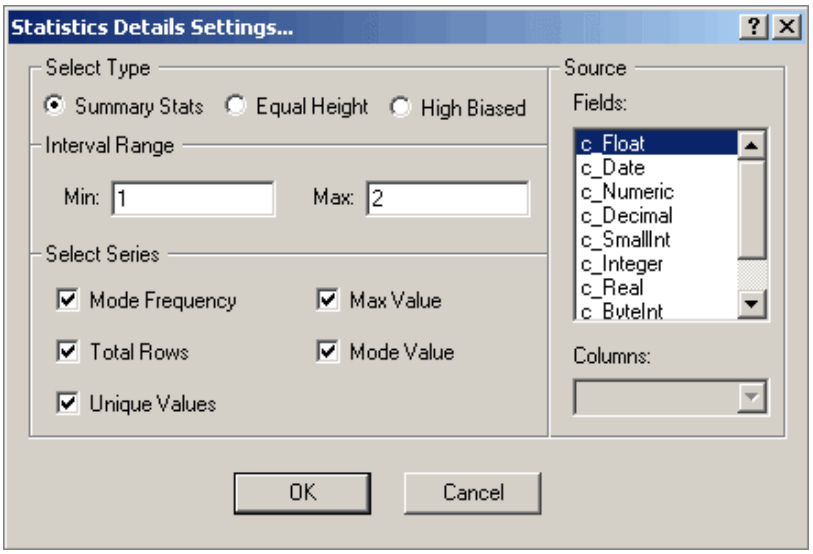


Interval Number	Maximum Value	Mode Value	Mode Frequency	Other Rows	Other Values
0	101	111	8	70	10
1	111	101	6	8	-2
2	331	222	6	8	-2
3	391	361	8	8	-2
4	551	441	2	8	-2
5	881	771	8	8	-2

To change the settings of the statistics details graph

1 Click Preferences > Statistics Details.

  The Statistics Details Settings dialog box appears.



2 Set the following options:

Option	Description
Select Type	• Summary Stats: The summary of the statistics on the selected tables. • Equal Height: Specifies equal heights statistics. • High Bias: Specifies high biased column statistics.

Option	Description
Interval Range	Set the minimum and maximum range of intervals in the frequency distribution for the column or index.
Mode Frequency	Check to display the number of rows in the interval that have a modal value for the column or index.
Total Rows	Check to display the total number of rows in the table.
Unique Values	Check to show the unique values for the column on which statistics were collected.
Max Value	Check to show the highest value for the specified column or index in the specified table.
Mode Value	Check to show the most frequently occurring value or values for the column or index in the specified table.
Source	• Fields: Select from the list of fields on which the statistics are available. • Columns: Select the columns that contain the index.

3 Click OK.

Save or Print Data Demographics

To save data demographics

- To save the graphs or report, click .

To print data demographics

- To print the graphs or report, click .

CHAPTER 10

Viewing Comparison Reports

This chapter explains how to view comparison reports and provides information about the reports. For more information, see:

- Display a Report
- Report Descriptions

Use Teradata Visual Explain to compare the execution plans across different QCDs or across different Teradata Database systems. Teradata Visual Explain creates a number of reports comparing various attributes of the plans. These reports help identify the differences in the plans. Attributes with differences are highlighted in the report.

There are two methods for comparing reports:

1. Compare the response time or join plans of queries:
   - On systems with different configurations
   - Across two different Teradata Database releases
   - With changing demographics
   - By adding an index and dropping an existing index
2. Compare reports of two plans using either:
   - An online comparison of any two plans opened in the Teradata Visual Explain interface.

     To learn how to select plans for textual comparison, see "Textual Comparison" on page 117.
   - The comparison reports of a previous bulk compare operation.

     To generate the comparison, double-click on a cell in the Bulk Compare report viewer.

The following textual reports can be generated:

- Complete Information Report
- Teradata Database Configuration Report
- Step Information Report (method 2)
- Estimated Cost Report (method 1)
- Operation Report
- Operands Report (method 1)
- Residual Condition Report
- Join Condition Report
- Join Order Report

- Source Attributes Report (method 2)
- Target Attributes Report
- Indexes Used Report
- Index Condition Report

Each report provides information on the base query and current query.

A *base query* is the query against which the current query is compared:

- If the report is opened through online comparison of plans, the base query is the one that is selected in the first row under Selected Execution Plans.

  For more information on selecting a base query, see "Textual Comparison" on page 117.
- If the report is opened through the Bulk Compare report viewer, the base query is available in the first column.

The *current query* is the query selected for comparison against the base query.

- If the report is opened through online comparison of plans, the current query is the one that is selected in a row other than the base query under Selected Execution Plans.

  For more information on selecting a current (or compare) query, see "Textual Comparison" on page 117.
- If the report is opened through the Bulk Compare report viewer, the current query is available in the cell that is double-clicked.

Display a Report

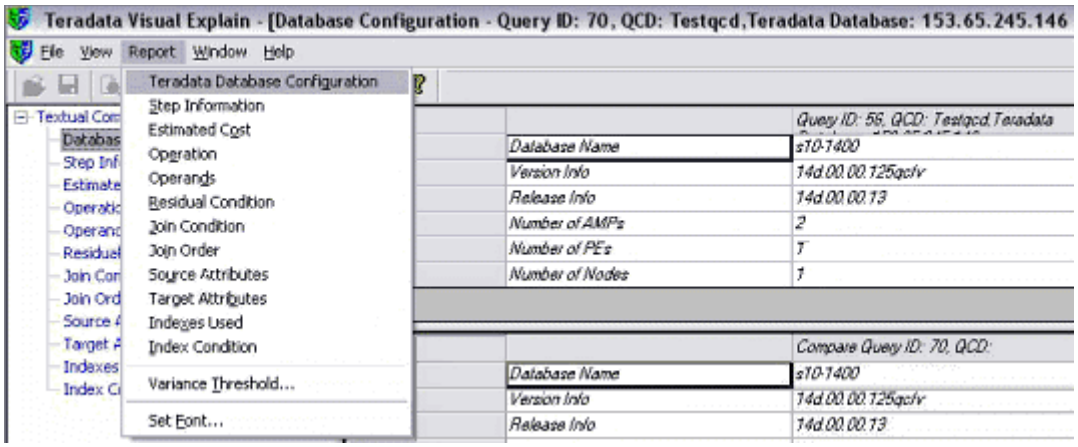
Compare execution plans before selecting a report type. For more information on performing a comparison, see "Textual Comparison" on page 117.

Note: To display all reports simultaneously, see "Complete Information Report" on page 187.

To display a report type

- In the textual compare report window, select a report name from the Reports menu or from the left pane of the tree view. The selected report appears in the right pane.

Figure 61: Reports Menu



The report selected displays.

Note: Steps containing differences are highlighted in red. To access the shortcut menu, right-click the row heading in any report. For more information, see "Shortcut Menus" on page 51.

Report Descriptions

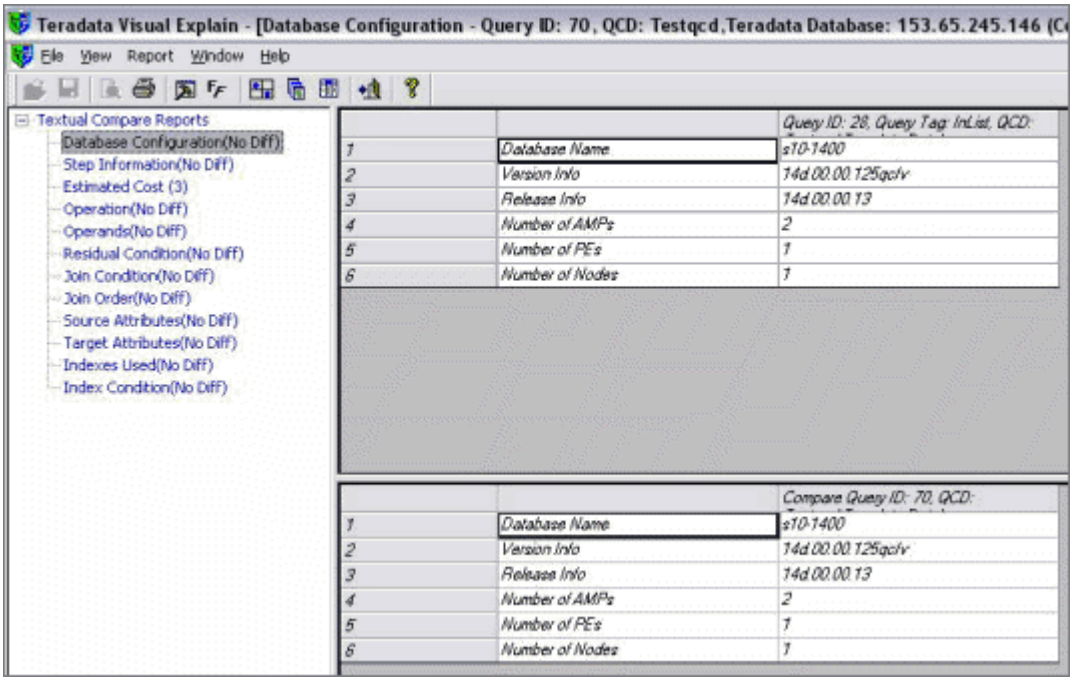
These topics contain information about the available report types.

- Complete Information Report
- Teradata Database Configuration Report
- Step Information Report
- Estimated Cost Report
- Operation Report
- Operands Report
- Residual Condition Report
- Join Condition Report
- Join Order Report
- Source Attributes Report
- Target Attributes Report
- Indexes Used Report
- Index Condition Report

Complete Information Report

Use the Complete Information report (see Figure 62) to view all of the reports simultaneously.

Figure 62: Example of Complete Information Report



Teradata Database Configuration Report

Use the Teradata Database Configuration report to compare the differences in the release, Teradata Database version, or the physical configuration of the Teradata Database system where the plans are captured. Figure 63 is an example of the report.

Figure 63: Example of a Teradata Database Configuration Report

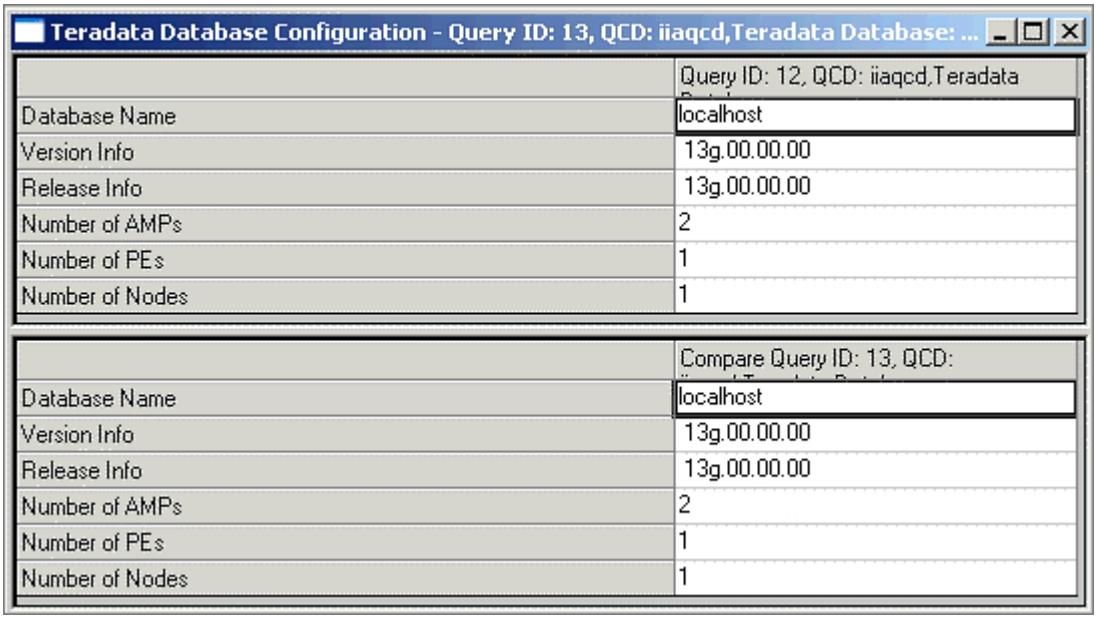


Table 30 describes the Teradata Database Configuration report.

Table 30: Teradata Database Configuration Report Description

Row	Description
Database Name	Name of the Teradata Database system where the query plan is captured.
Version Info	Version number of the Teradata Database where the query was captured. The value is defined in DBC.DBCInfo.
Release Info	Release number of the Teradata Database where the query was captured. The value is defined in DBC.DBCInfo.
Number of AMPs	Number of AMPs in the physical configuration.
Number of PEs	Number of PEs in the physical configuration.
Number of Nodes	Number of nodes in the physical configuration.

Step Information Report

The Step Information report displays the source relations used and the target relations built in a step along with the operation performed and cost estimates. Cost is displayed in *HH:MM:SS.ss* format.

Use this report to get information about the operands, operation, target, and estimated cost. The step attribute values display if they are available for any one of the plans. When there are multiple step attributes for a given step, the attributes are split across different rows. Figure 64 is an example of the report.

Figure 64: Sample Step Information Report

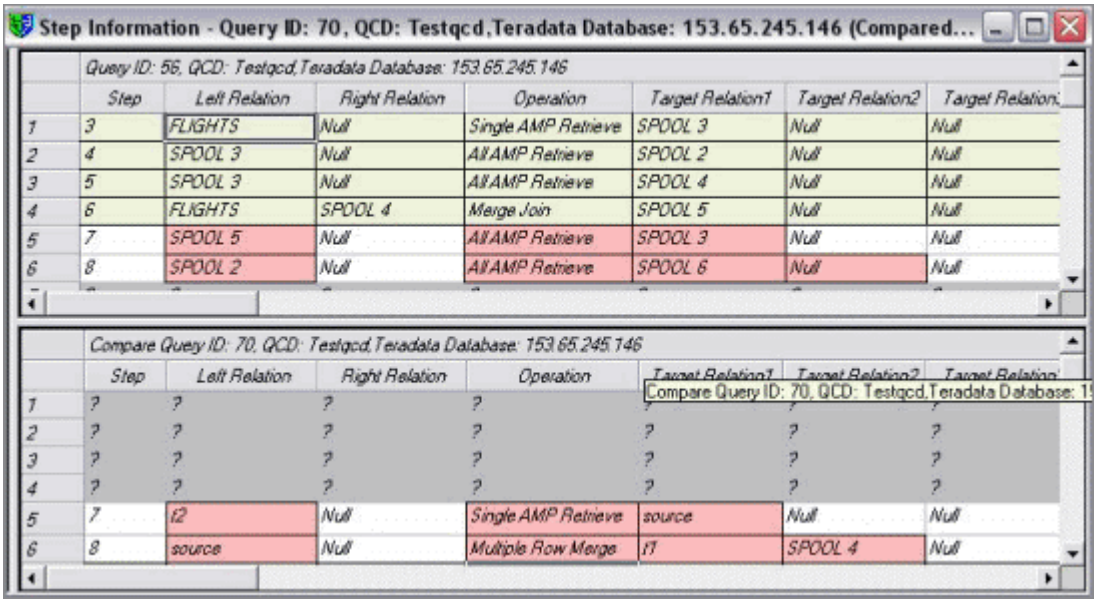


Table 31 describes the Step Information report.

Table 31: Step Information Report Columns and Descriptions

Column	Description
Step	The specific step number in the compare plan.
Left Relation	The name of the left relation in a step.
Right Relation	The name of the right relation in a step.
Operation	The type of operation performed. For example, Retrieve, Aggregate, or Join Method. For more information, see Table 13 on page 39, Table 18 on page 43, Table 19 on page 43, and Table 20 on page 46.
Target Relation1	The name of the first target relation in a step.
Target Relation2	The name of the second target relation in a step.
Estimated CPU Cost	The estimated CPU cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated IO Cost	The estimated I/O cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated Network Cost	The estimated network cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated Heuristic Cost	The estimated heuristic cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm)*.
Estimated Cost	The estimated cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS.ss*).
Attribute Name	Additional information for the step depending on the operation performed. For example: • Partition Count for hash join steps • Go to for recursive queries • Merge Mode for merge delete and merge update steps • Kind for sum, sample, or stat steps • Merge Type for Multiple Row Merge step Note: This column only appears if step attribute values are defined for at least one of the plans.
Attribute Value	The value of the step attribute type. For example: • Partition count • Recursive step number • UpdateInsert • UpdateOnly • InsertOnly Note: This column only appears if step attribute values are defined for at least one of the plans.

Estimated Cost Report

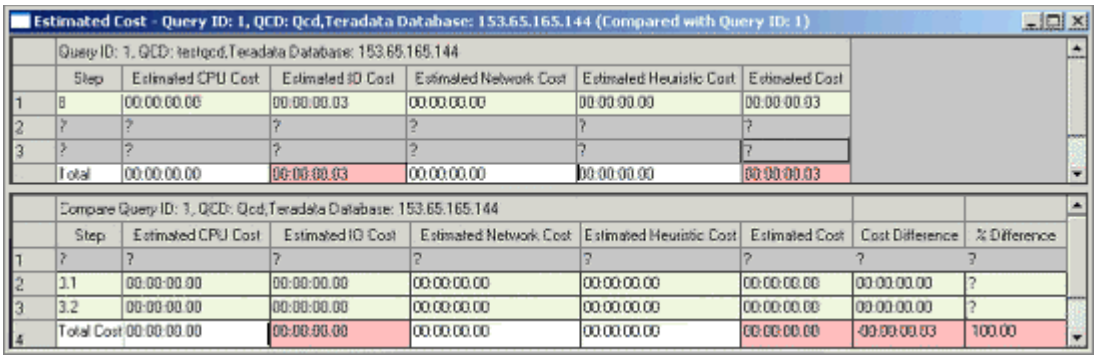
The Estimated Cost report provides cost estimates for each step. Use this report to analyze the cost estimate variations of two plans. For example, the cost differences of plans captured on two different configurations, or the differences of plans captured at two instances, or with changing demographics of tables.

To view the steps that take the most time or the least time to execute, double-click the column headings to sort the rows.

To emulate different physical configurations on one system and capture the plans, use the Set TLE and Reset TLE features of the Teradata System Emulation Tool application. For more information on target level emulation, refer to the *Teradata System Emulation Tool User Guide* B035-2492.

Figure 65 is an example of the report.

Figure 65: Sample Estimated Cost Report



Estimated Cost - Query ID: 1, QCD: Qcd,Teradata Database: 153.65.165.144 (Compared with Query ID: 1)

Query ID: 1, QCD: testqcd,Teradata Database: 153.65.165.144

	Step	Estimated CPU Cost	Estimated IO Cost	Estimated Network Cost	Estimated Heuristic Cost	Estimated Cost
1	8	00:00:00.00	00:00:00.03	00:00:00.00	00:00:00.00	00:00:00.03
2	?	?	?	?	?	?
3	?	?	?	?	?	?
	Total	00:00:00.00	00:00:00.03	00:00:00.00	00:00:00.00	00:00:00.03

Compare Query ID: 1, QCD: Qcd,Teradata Database: 153.65.165.144

	Step	Estimated CPU Cost	Estimated IO Cost	Estimated Network Cost	Estimated Heuristic Cost	Estimated Cost	Cost Difference	% Difference
1	?	?	?	?	?	?	?	?
2	3.1	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	?
3	3.2	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	?
4	Total Cost	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	00:00:00.00	-00:00:00.03	100.00

Table 32 describes the information displayed on the Estimated Cost report.

Table 32: Estimated Cost Report Column and Row Descriptions

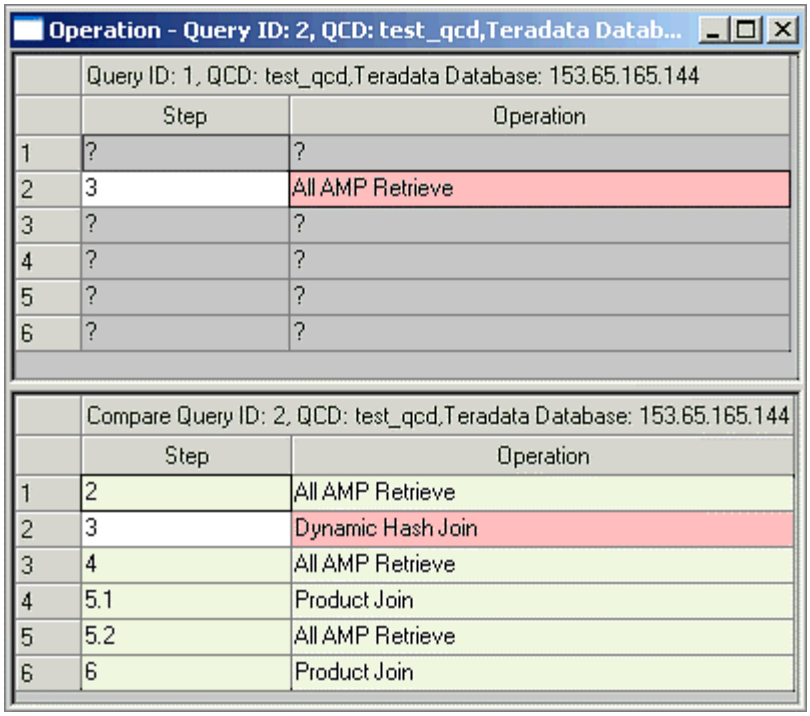
Column or Row	Description
Step	The number of a specific step in the plan.
Estimated CPU Cost	The estimated CPU cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated IO Cost	The estimated I/O cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated Network Cost	The estimated network cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated Heuristic Cost	The estimated heuristic cost of the step in hours, minutes, seconds, and milliseconds (*HH:MM:SS:mm*).
Estimated Cost	The estimated cost of the individual step in hours, minutes, seconds, and milliseconds (*HH:MM:SS.mm*).
Cost Difference	The absolute value of the cost difference for the specific step in hours, minutes, seconds, and milliseconds (*HH:MM:SS.mm*).
% Difference	The percent difference in the cost for the specific step. Steps that have a higher variance than the user-specified threshold value are highlighted in red.
Total Cost	Called the summary row, the bottom row of the report displays the overall cost variance between the plans. The format of this row differs from the other rows to indicate that it is a summary row.

Operation Report

The Operation report provides information about the operations performed while executing a query. Depending on the SQL statement, statistics available on the tables in the query and different operations (such as Join, Mergeupdate, and Aggregate step) are performed by Teradata Database. Refer to the query optimization section of *SQL Request and Transaction Processing* B035-1142 for more information.

Figure 66 is an example of the report.

Figure 66: Sample Operation Report



Operation - Query ID: 2, QCD: test_qcd,Teradata Datab...

Query ID: 1, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Operation
1	?	?
2	3	All AMP Retrieve
3	?	?
4	?	?
5	?	?
6	?	?

Compare Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Operation
1	2	All AMP Retrieve
2	3	Dynamic Hash Join
3	4	All AMP Retrieve
4	5.1	Product Join
5	5.2	All AMP Retrieve
6	6	Product Join

Table 33 describes the Operation report.

Table 33: Operation Report Columns and Descriptions

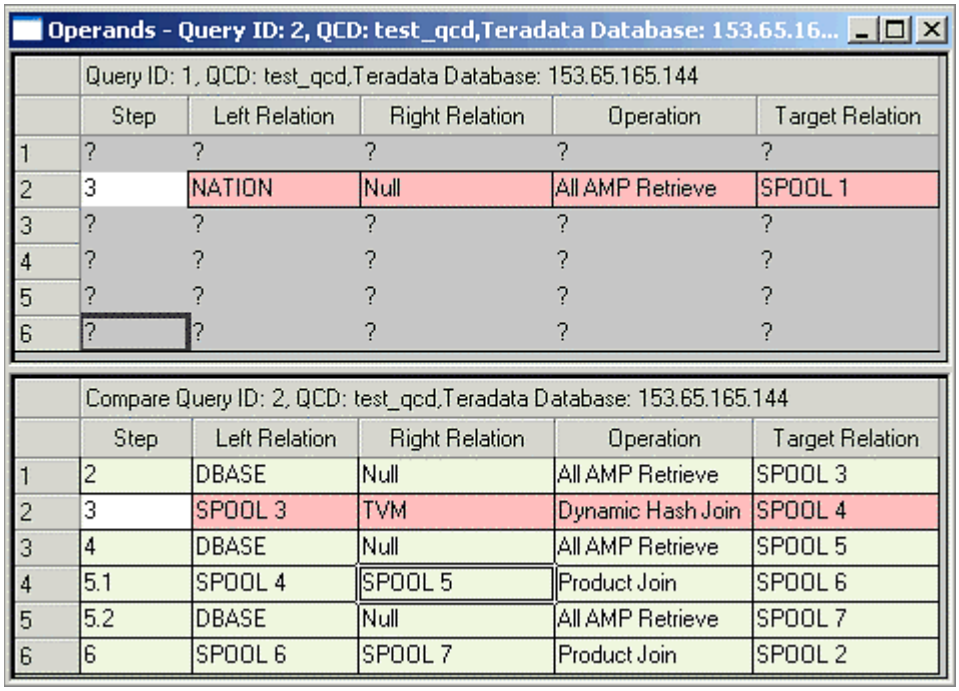
Column	Description
Step	The specific step number in the compare plan.
Operation	The type of operation performed. For example, Retrieve, Aggregate, or Join Method. For more information, see: • Retrieval Method Icons in Table 13 on page 39 • Aggregate Step Icons in Table 18 on page 43 • Join Method Icons in Table 19 on page 43 • Other Operation Icons in Table 20 on page 46

Operands Report

Each step is composed of one or more operands and a target. The Operand report provides information about the operands for each step. These are also referred to as *source relations*.

Figure 67 is an example of the report.

Figure 67: Example of an Operand Report



Operands - Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.16...

Query ID: 1, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Left Relation	Right Relation	Operation	Target Relation
1	?	?	?	?	?
2	3	NATION	Null	All AMP Retrieve	SPOOL 1
3	?	?	?	?	?
4	?	?	?	?	?
5	?	?	?	?	?
6	?	?	?	?	?

Compare Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Left Relation	Right Relation	Operation	Target Relation
1	2	DBASE	Null	All AMP Retrieve	SPOOL 3
2	3	SPOOL 3	TVM	Dynamic Hash Join	SPOOL 4
3	4	DBASE	Null	All AMP Retrieve	SPOOL 5
4	5.1	SPOOL 4	SPOOL 5	Product Join	SPOOL 6
5	5.2	DBASE	Null	All AMP Retrieve	SPOOL 7
6	6	SPOOL 6	SPOOL 7	Product Join	SPOOL 2

Table 34 describes the Operand report.

Table 34: Operand Report Columns and Descriptions

Column	Description
Step	The specific step number in the compare plan.
Left Relation	The name of the left relation in a step.
Right Relation	The name of the right relation in a step.
Operation	The type of operation performed. For example, All AMP Retrieve, Dynamic Hash Join, or Product Join. For more information, see: • Retrieval Method Icons in Table 13 on page 39 • Aggregate Step Icons in Table 18 on page 43 • Join Method Icons in Table 19 on page 43 • Other Operation Icons in Table 20 on page 46
Target Relation	Indicates the related spool.

Residual Condition Report

The Residual Condition report provides information about the predicate conditions applied on the table relations in a step. Only the steps where a residual condition is applied are displayed. Figure 68 is an example of the report.

Figure 68: Example of a Residual Condition Report

Residual Condition - Query ID: 13, QCD: iiaqcd,Teradata Database: na...

Query ID: 12, QCD: iiaqcd,Teradata Database: naren

	Step	PredicateKind	Residual Condition
2	3.2	Source Condition	IIAQCD.Relation.QueryID = 1
3	4.2	Source Condition	((IIAQCD.QuerySteps.RowType = 'A') AND ((IIAQCD.QuerySteps.StepAttributeType = 'JoinType ') AND (IIAQCD.QuerySteps.QueryID = 1))) OR ((IIAQCD.QuerySteps.RowType = 'G') AND (IIAQCD.QuerySteps.StepAttributeType = 'JoinType '))
4	5	?	?
5	?	?	?

Compare Query ID: 13, QCD: iiaqcd,Teradata Database: naren

	Step	PredicateKind	Residual Condition
2	?	?	?
3	4.2	Source Condition	(NOT (IIAQCD.QStep.StepNum IS NULL)) AND (IIAQCD.QStep.RowType = 'G')
4	5	Source Condition	(IIAQCD.p1.PredicateKind = 'Q') OR (IIAQCD.p1.PredicateKind = 'S')
5	7	Source Condition	(IIAQCD.p1.PredicateKind = 'Q') OR (IIAQCD.p1.PredicateKind = 'S')

Table 35 describes the Residual Condition report.

Table 35: Residual Condition Report Columns and Descriptions

Column	Description
Step	Specific step in the compare plan.
Predicate Kind	The predicate kind: • An additional join condition (A). • A range constraint (G). Used for value-ordered relations. • A join condition (J). • A condition on a left relation in a join (L). • A partition elimination (Q). Occurs for a source condition. A residual condition on the left or right table in a join or on a single-table retrieval. Partition elimination occurs prior to accessing the rows, so the condition applies only to rows retrieved from partitions that were not eliminated. • A condition on a right relation in a join (R). • A source condition (S). A residual condition on the left or right table in a join or on a single-table retrieval. No partition elimination occurs prior to accessing the rows.

Table 35: Residual Condition Report Columns and Descriptions (continued)

Column	Description
Residual Condition	A list of each predicate applied on a table in a step. A letter represents each predicate type: • A (additional join) • G (range constraint) • J (join condition) • L (condition on a left relation in a join) • Q (partition elimination) • R (condition on a right relation in a join) • S (source condition)

Join Condition Report

The Join Condition report provides information about the conditions on which the table or spool relations are joined. Only the steps where a join is performed are displayed. Figure 69 is an example of the report.

Figure 69: Example of a Join Condition Report

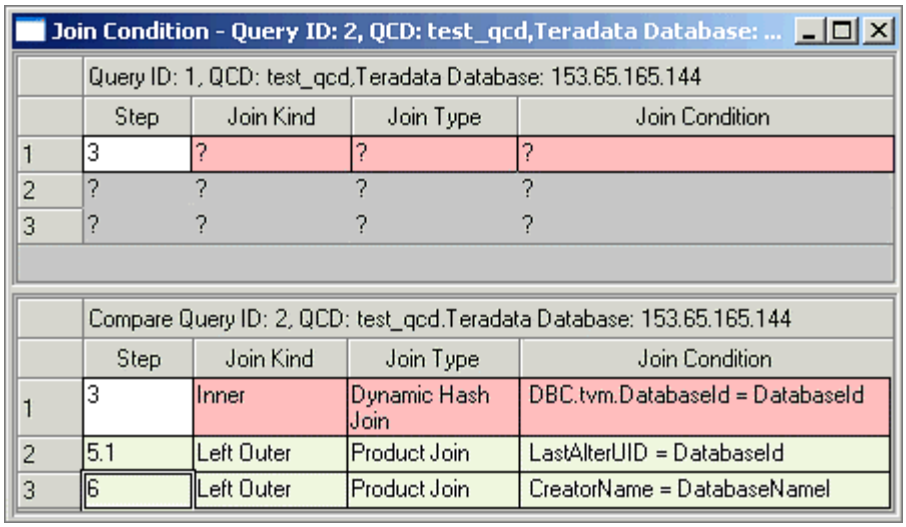


Table 36 describes the Join Condition report.

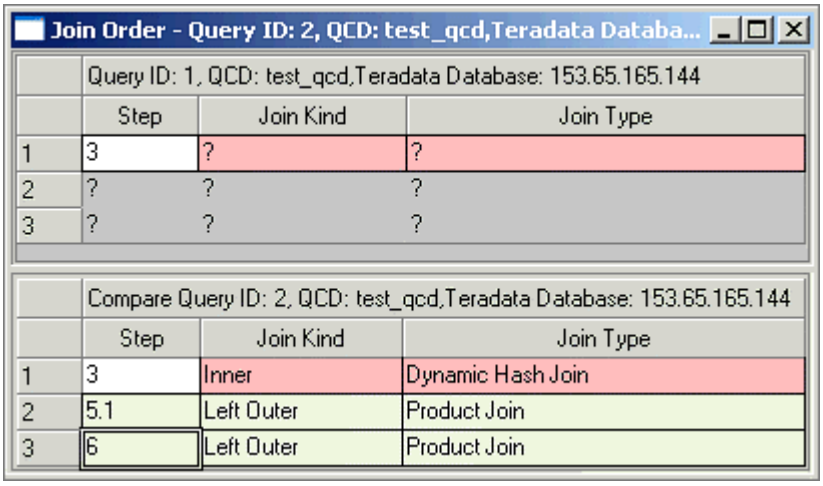
Table 36: Join Condition Report Columns and Descriptions

Column	Description
Step	The specific step in the compare plan.
Join Kind	The kind of join performed. For example, Inner, Left Outer, Right Outer, or Full Outer join.
Join Type	The type of join performed by Teradata Database. The join types that can be performed are further described in Table 19 on page 43.
Join Condition	The condition applied to join the table/spool relations.

Join Order Report

The Join Order report provides information about the order in which the joins are performed in a plan. Only the steps where a join is performed are displayed. Figure 70 is an example of the report.

Figure 70: Example of a Join Order Report



Join Order - Query ID: 2, QCD: test_qcd,Teradata Databa...

	Query ID: 1, QCD: test_qcd,Teradata Database: 153.65.165.144		
	Step	Join Kind	Join Type
1	3	?	?
2	?	?	?
3	?	?	?

	Compare Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.165.144		
	Step	Join Kind	Join Type
1	3	Inner	Dynamic Hash Join
2	5.1	Left Outer	Product Join
3	6	Left Outer	Product Join

Table 37 describes the Join Order report.

Table 37: Join Order Report Columns and Descriptions

Column	Description
Step	The specific step in the compare plan.
Join Kind	The kind of join performed. For example, Inner, Left Outer, Right Outer, or Full Outer join.
Join Type	The type of join performed by Teradata Database. The different join types that can be performed are further described in Table 19 on page 43.

Source Attributes Report

The Source Attributes report displays the attributes of the source operands at each step. Each step has two rows, one for the left relation and the other for the right relation. Figure 71 is an example of the report.

Figure 71: Sample Source Attributes Report

Source Attributes - Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.165.144 (Compared with Query ID: 1)

Query ID: 1, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Relation	Name	Sorted	Cached	SyncScan	Partitioned	Partition Count	Sort Kind	Sort Key
1	?	?	?	?	?	?	?	?	?	?
2	?	?	?	?	?	?	?	?	?	?
3	3	Left	NATION	No	No	Yes	No	?	?	?
4	3	Right	Null	?	?	?	?	?	?	?
5	?	?	?	?	?	?	?	?	?	?
6	?	?	?	?	?	?	?	?	?	?
7	?	?	?	?	?	?	?	?	?	?
8	?	?	?	?	?	?	?	?	?	?
9	?	?	?	?	?	?	?	?	?	?
10	?	?	?	?	?	?	?	?	?	?
11	?	?	?	?	?	?	?	?	?	?
12	?	?	?	?	?	?	?	?	?	?

Compare Query ID: 2, QCD: test_qcd,Teradata Database: 153.65.165.144

	Step	Relation	Name	Sorted	Cached	SyncScan	Partitioned	Partition Count	Sort Kind	Sort Key
1	2	Left	DBASE	No	No	Yes	No	?	?	?
2	2	Right	Null	?	?	?	?	?	?	?
3	3	Left	SPOOL 3	No	Yes	Yes	No	?	?	?
4	3	Right	TVM	No	No	Yes	No	?	?	?
5	4	Left	DBASE	No	Yes	Yes	No	?	?	?
6	4	Right	Null	?	?	?	?	?	?	?
7	5.1	Left	SPOOL 4	No	Yes	Yes	No	?	?	?
8	5.1	Right	SPOOL 5	No	Yes	Yes	No	?	?	?
9	5.2	Left	DBASE	No	Yes	Yes	No	?	?	?
10	5.2	Right	Null	?	?	?	?	?	?	?
11	6	Left	SPOOL 6	No	Yes	Yes	No	?	?	?
12	6	Right	SPOOL 7	No	Yes	Yes	No	?	?	?

Table 38 describes the Source Attributes report.

Table 38: Source Attributes Report Columns and Descriptions

Column	Description
Step	Identifies the specific step in the compare plan.
Relation	Identifies the type of relation (left or right).
Name	Identifies the name of the relation.
Sorted	Indicates whether the relation is sorted.
Cached	Indicates whether the relation is cached.
SyncScan	Indicates whether a relation is eligible for synchronized scanning.
Partitioned	Indicates whether a table or spool has a partitioned primary index.
Partition Count	Indicates the number of partitions on a table.

Table 38: Source Attributes Report Columns and Descriptions (continued)

Column	Description
Sort Kind	Identifies the way the relation is sorted: • Field1 sort • Field1 unique sort • Field1 Hash sort • FieldID sort • Field1 Hash unique sort • Field1 Hash min1 sort • Field1 Hash min2 sort • Field1 Hash max1 sort • Field1 Hash max2 sort • JoinIndex sort • Field1 min1 sort • Field 1 min2 sort • RowHash field1 sort • RowHashRow sort • RowHash sort • Unique field1 sort • Unique rowID sort • Unknown sort kind
Sort Key	If the relation is sorted, lists the sort information strings. The format of the individual SortKey strings is one of the following: • database_name.table_name.column_name • SPOOL number.column_name

Target Attributes Report

The Target Attributes report provides information about the target relations of a step.

Figure 72 is an example of the report.

Figure 72: Example of a Target Attributes Report

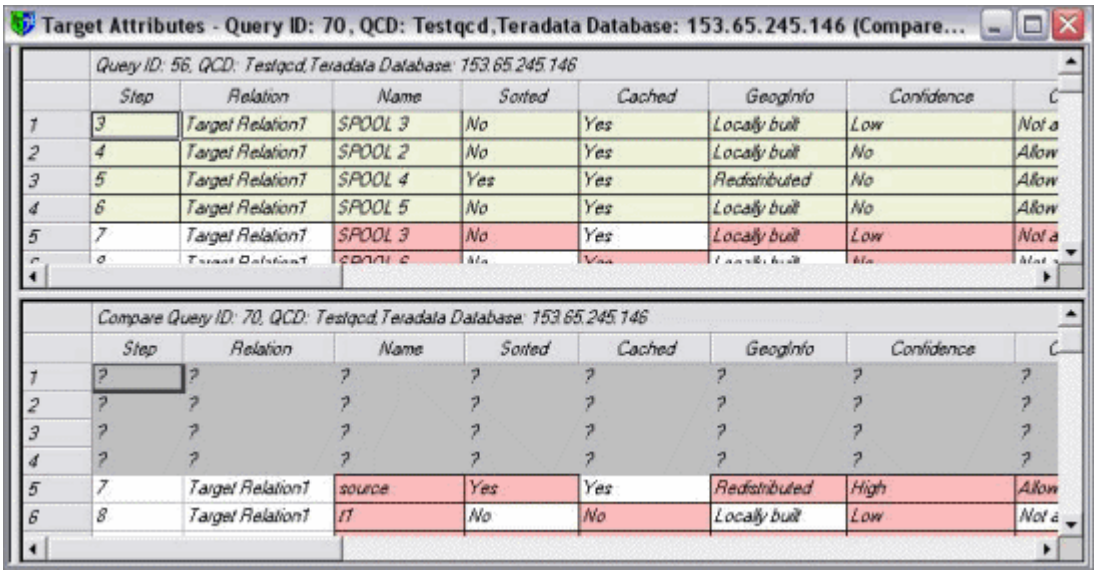


Table 39 describes the Target Attributes report.

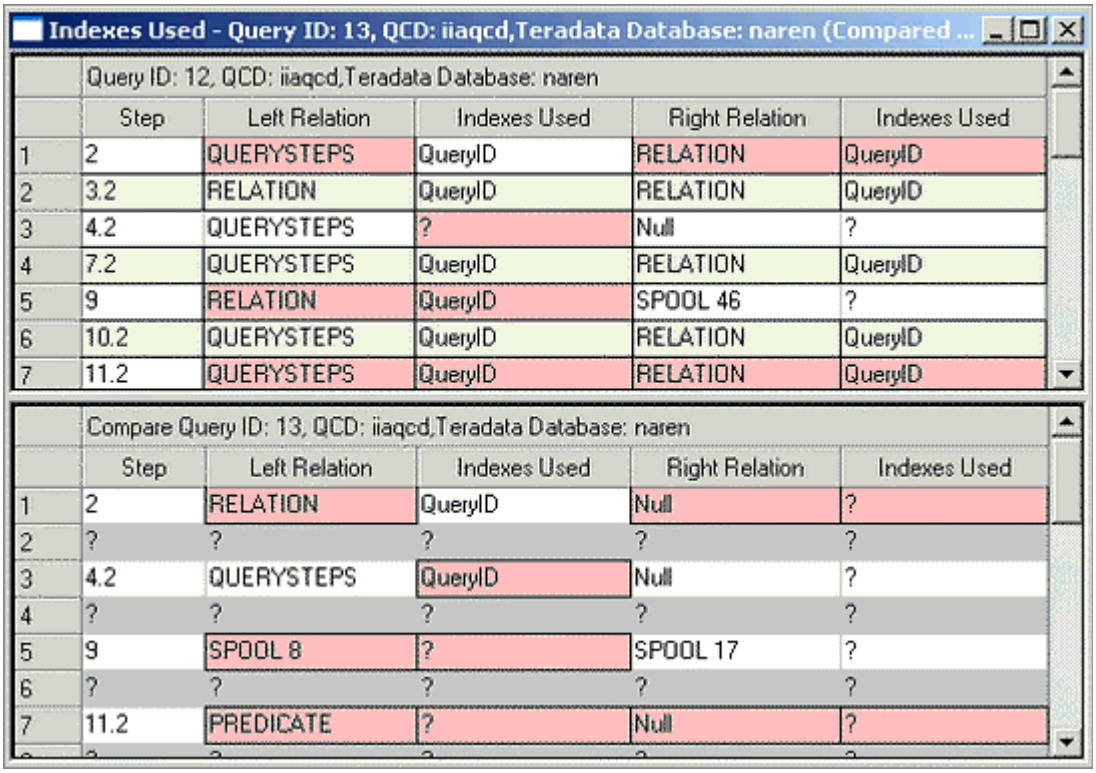
Table 39: Target Attributes Report Column Descriptions

Column	Description
Step	Identifies the specific step in the compare plan.
Relation	Identifies the type of relation (Target Relation1 or Target Relation2).
Name	Identifies the name of the relation.
Sorted	Indicates whether the target relation is sorted.
Cached	Indicates whether the target relation is cached.
GeogInfo	Defines the data redistribution method used in a step: • *Locally built* indicates the rows are locally built. • *Redistributed* indicates rows are redistributed on all AMPs. • *Duplicated* indicates the rows are duplicated on all AMPs.
Confidence	Specifies the confidence level for the estimated cardinality. The value in this column can be *Low*, *High*, *Index join*, or *No*.
Compressed Columns	Indicates whether the target spool can have compressed columns.
Size (Estimated Rows)	Indicates the cardinality estimate of the target spool.
Spool Size (Bytes)	Indicates the size of the spool in bytes.

Indexes Used Report

The Indexes Used report provides information about the indexes used in a plan. Only the steps where an index is used are displayed. Figure 73 is an example of the report.

Figure 73: Example of a Indexes Used Report



Indexes Used - Query ID: 13, QCD: iiaqcd,Teradata Database: naren (Compared ...

Query ID: 12, QCD: iiaqcd,Teradata Database: naren

	Step	Left Relation	Indexes Used	Right Relation	Indexes Used
1	2	QUERYSTEPS	QueryID	RELATION	QueryID
2	3.2	RELATION	QueryID	RELATION	QueryID
3	4.2	QUERYSTEPS	?	Null	?
4	7.2	QUERYSTEPS	QueryID	RELATION	QueryID
5	9	RELATION	QueryID	SPOOL 46	?
6	10.2	QUERYSTEPS	QueryID	RELATION	QueryID
7	11.2	QUERYSTEPS	QueryID	RELATION	QueryID

Compare Query ID: 13, QCD: iiaqcd,Teradata Database: naren

	Step	Left Relation	Indexes Used	Right Relation	Indexes Used
1	2	RELATION	QueryID	Null	?
2	?	?	?	?	?
3	4.2	QUERYSTEPS	QueryID	Null	?
4	?	?	?	?	?
5	9	SPOOL 8	?	SPOOL 17	?
6	?	?	?	?	?
7	11.2	PREDICATE	?	Null	?

Table 40 describes the Indexes Used report.

Table 40: Indexes Used Report Column Descriptions

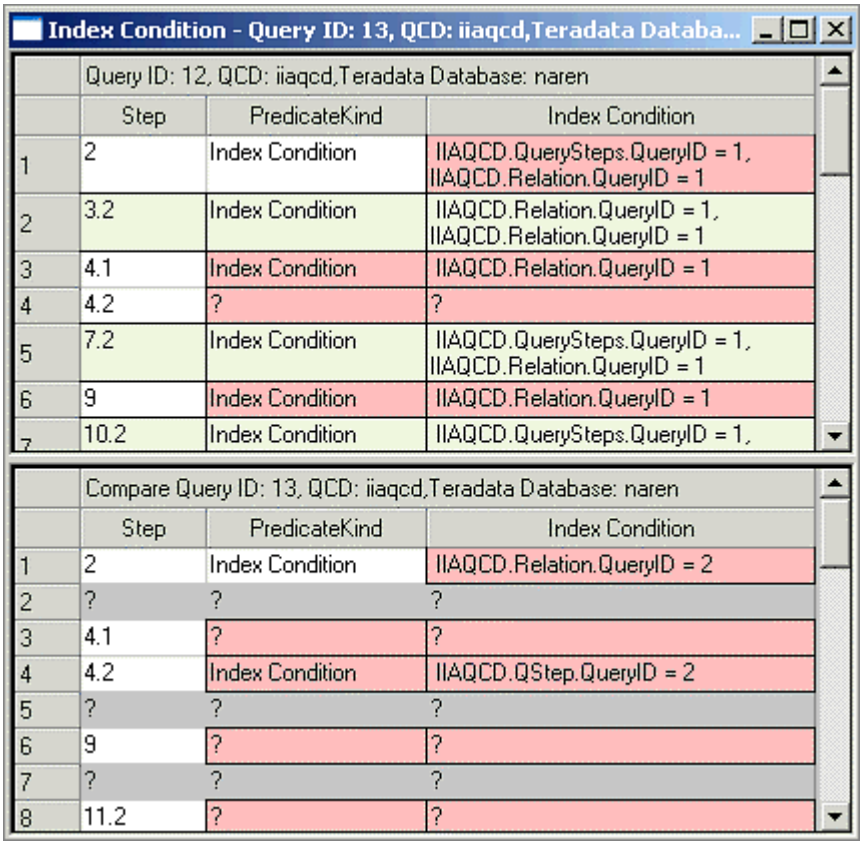
Column	Description
Step	Specific step in the compare plan.
Left Relation	Name of the left relation.
Indexes Used	Index name or index columns used at this step.
Right Relation	Name of the right relation.
Indexes Used	Index name or index columns used at this step.

Index Condition Report

The Index Condition report provides information about the predicate conditions applied on the table relations in a step. Only the steps where an index condition is applied are displayed.

Figure 74 is an example of the report.

Figure 74: Example of an Index Condition Report



Index Condition - Query ID: 13, QCD: iiaqcd,Teradata Databa...

Query ID: 12, QCD: iiaqcd,Teradata Database: naren

	Step	PredicateKind	Index Condition
1	2	Index Condition	IIAQCD.QuerySteps.QueryID = 1, IIAQCD.Relation.QueryID = 1
2	3.2	Index Condition	IIAQCD.Relation.QueryID = 1, IIAQCD.Relation.QueryID = 1
3	4.1	Index Condition	IIAQCD.Relation.QueryID = 1
4	4.2	?	?
5	7.2	Index Condition	IIAQCD.QuerySteps.QueryID = 1, IIAQCD.Relation.QueryID = 1
6	9	Index Condition	IIAQCD.Relation.QueryID = 1
7	10.2	Index Condition	IIAQCD.QuerySteps.QueryID = 1,

Compare Query ID: 13, QCD: iiaqcd,Teradata Database: naren

	Step	PredicateKind	Index Condition
1	2	Index Condition	IIAQCD.Relation.QueryID = 2
2	?	?	?
3	4.1	?	?
4	4.2	Index Condition	IIAQCD.QStep.QueryID = 2
5	?	?	?
6	9	?	?
7	?	?	?
8	11.2	?	?

Table 41 describes the Index Condition report.

Table 41: Index Condition Report Column Descriptions

Column	Description
Step	The specific step in the compare plan.
Predicate Kind	I (Index Condition)
Index Condition	Identifies one or more predicates applied on a table in a step.

CHAPTER 11

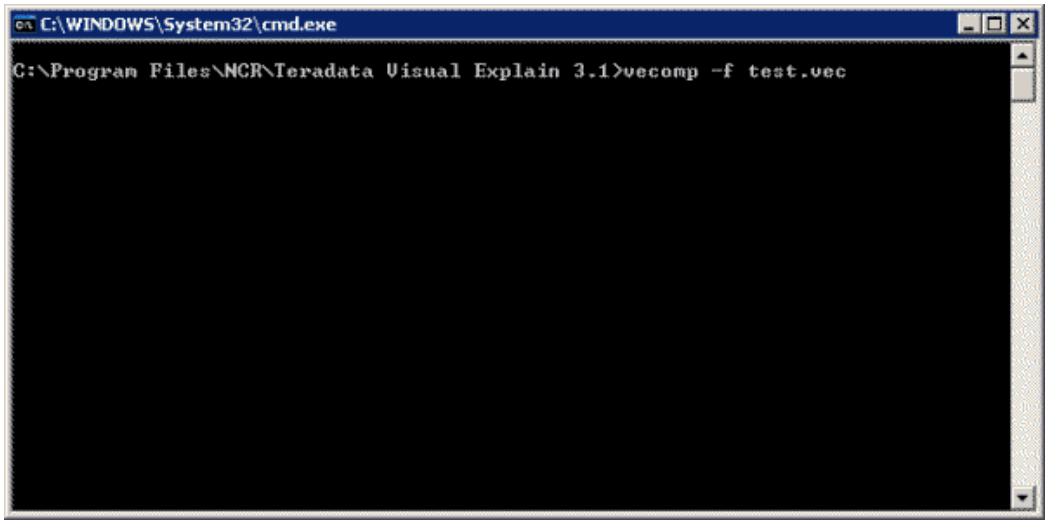
Using the Command Line Interface

Teradata Visual Explain provides a command line interface. Read the following topics for help using this interface:

- Command Line Switches
- Terms Used in Visual Explain
- Connecting to a Teradata Database
- View Execution Plan(s)
- Opening Offline Execution Plans
- Opening Online Execution Plans
- Comparing Execution Plans Visually
- Comparing Execution Plans Using Text Reports
- Displaying Differences in Explain Text of Execution Plans
- Comparing Multiple Execution Plans Using Bulk Compare

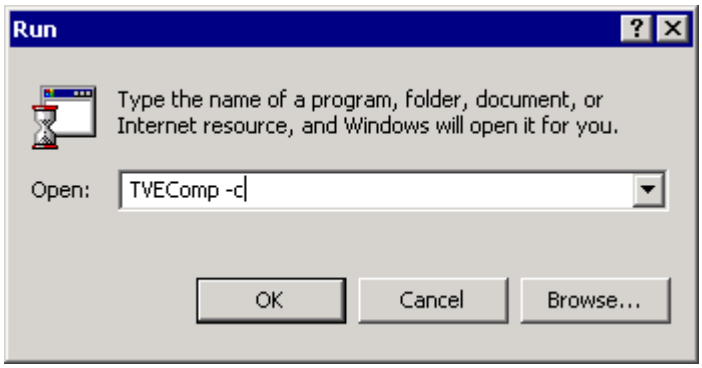
The command examples in this topic are executed from the command line interface. See Figure 75.

Figure 75: Command Line Interface



The same commands can be executed from the Run dialog box using TVEComp instead of VEComp. See Figure 76.

Figure 76: Run Window



Note: Command examples in this chapter use a "code-type" of font that displays the letter "l" (as in "letter") and the number "1" (one) using similar characters. For example, "`l`" is the letter and "`1`" is the number.

Command Line Switches

Table 42 describes the Teradata Visual Explain command line switches.

Table 42: Command Line Switches

Switch	Description
ux	To use DBC and QCD X views (on Teradata Database V2R6.0, V2R6.1, and V2R6.2) or DBC and QCD VX views (on Teradata Database 12.0 and later) while retrieving plans from a QCD. Note: The Use X views (DBC and QCD) check box must be selected.
unx	To use DBC and QCD non-X views (on Teradata Database V2R6.0, V2R6.1, and V2R6.2) or DBC and QCD V views (on Teradata Database 12.0 and later) while retrieving plans from a QCD. Note: The Use X views (QCD only) check box must be cleared.
uxq	To use only QCD X views (on Teradata Database V2R6.0, V2R6.1, and V2R6.2) or only QCD VX views (on Teradata Database 12.0 and later) while retrieving plans from the QCD. Note: The Use X views (QCD only) check box must be selected.
v	To specify expanded or compressed view mode for plan display. The default view mode is expanded.
p	To specify Open Online plans operation.
ie	To specify Insert and View Execution Plan operation.
vc	To specify Visual Compare of Execution Plans operation.
tc	To specify Textual Compare of Execution Plans operation.
ec	To specify Explain text differences of Execution Plans operation.
rc	To specify a range of queryID.
bc	To specify Bulk Compare of Execution Plans operation.

Table 42: Command Line Switches (continued)

Switch	Description
f	To specify a file name.
l	To specify CLIv2 connectivity.
c	To specify ODBC connectivity.
am	To connect to the server using a specific alternate logging mechanism.
d	To specify the QCD database name.
m	To specify output in multiple files. It is used in bulk compare.
t	To specify Query Tag. It is used in Insert and View Execution Plan operation to specify query tag while inserting the execution plan.
q	To specify query text. It is used in case of Insert and View Execution Plan.
db	To specify the default database name in case of command line INSERT EXPLAIN. The database supplied is only effective in case of INSERT EXPLAIN command line.
a	Used with textual compare to display all types of differences.
cs	Used with textual compare to display cost differences only.
ic	Used with textual compare to display index condition differences only.
jc	Used with textual compare to display join condition differences only.
rsc	Used with textual compare to display residual condition differences only.
jo	Used with textual compare to display join order differences only.
op	Used with textual compare to display operation differences only.
opd	Used with textual compare to display operand differences only.
sc	Used with textual compare to display Teradata Database configuration differences only.
scs	To connect to the server using a specified session character set.
ta	Used with textual compare to display target attribute differences only.
iex	To specify Insert Explain in XML and View Execution Plan captured in XML format in QCD. Note: Only available for Teradata Database version 13.10 and later.
xml	To view execution plan captured in xml format in QCD. Note: Only available for Teradata Database version 13.10 and later.
eix	To view execution plan generated from database using Explain In XML command.

Terms Used in Visual Explain

Table 43 describes the terms used in the command line feature.

Table 43: Terms and Syntax

Term	Syntax

Execution Plan	{
	{ Online Plan description} ||
	{ -f <Offline Execution Plan file name>}
	}

Online Plan Description	{ <Logon Specification> -d <database name> {-rc <Range condition> || -t <query tag> } { -ux || -unx } }
Online XML Plan Description	{ <Logon Specification> -d <database name> -xml {-rc <Range condition>} { -ux || -unx } } Note: Only available for Teradata Database version 13.10 and later.
Logon Specification	{ { -l <Server Name>\<User Id>\<Password> [-am<Authentication Mechanism>\<Authentication Parameter>] [-scs<Session Character Set Name>]} || { -c <ODBC DSN Name\password> [-am[<Authentication Mechanism>\]<Authentication Parameter>] [-scs<Session Character Set Name>]} }
Range Condition	{QueryId1 [{, || -} QueryId2….]}
Query Tag	{-t <querytag>}
Compare attribute	{ { -a } || {-cs} || {-sc} || {-jo} || {-jc} || {-ta} || {-op} || {-opd} || {-rsc} }
View Mode	{ [-v expanded || -v compressed] }

Connecting to a Teradata Database

Use the Teradata Visual Explain command line interface to log on to a Teradata Database.

To connect to Teradata Database, use the following command:

```
VEComp <Logon specification> [-d <qcdname>]
```

Table 44 describes commands used to connect to a Teradata Database.

Table 44: Examples for Connecting to a Teradata Database

Example	Command	Description
1	`VEComp -l`	Display the CLIv2 logon dialog box.
2	`VEComp -l SysA\dbc\dbc`	Use CLIv2to log on to server SysA.
3	`VEComp -l SysA\dbc\dbc -d qcd`	Use CLIv2 to log on to server SysA database qcd.
4	`VEComp -c`	Display the ODBC Data source selection dialog box.
5	`VEComp -c SysA`	Use the ODBC Data Source SysA to log on.
6	`VEComp -l SysA\dbc\dbc -am td2\xxx`	Use CLIv2 to log on to server SysA. The authentication mechanism is td2. The authentication parameter is *xxx*.
7	`VEComp -l SysA\dbc\dbc -am NONE\NONE`	Use CLIv2 to log on to server SysA. Extended security mechanism is not used.
8	`VEComp -l SysA\dbc\dbc -am  td2`	Use CLIv2 to log on to server SysA. The authentication mechanism is td2. No authentication parameter is used.
9	`VEComp -l SysA\dbc\dbc -am td2\xxx -scs UTF16`	Use CLIv2 to log on to source SysA. The authentication mechanism is td2, and *xxx* is the authentication parameter, using UTF-16 as the session character set.
10	`VEComp -c SysA\pwd1 -am td2\xxx`	Use ODBC to log on to source SysA. Assumes the DSN is configured with the user name. Connects using td2 as the authentication mechanism and *xxx* as the authentication parameter.
11	`VEComp -c SysA\pwd1 -am td2\xxx -scs UTF16`	Use ODBC to log on to source SysA. Assumes the DSN is configured with the user name. Connects using td2 as the authentication mechanism and *xxx* as the authentication parameter, using UTF-16 as the session character set.
12	`VEComp -c SysA\pwd1 -am  xxx`	Use ODBC to log on to source SysA. Assumes the DSN is configured with the user name and the authentication mechanism. Connects using *xxx* as the authentication parameter.
13	`VEComp -c SysA\user1\pwd1 -am xxx`	Use ODBC to log on to source SysA. Assumes the DSN is configured with the authentication mechanism. Connects using *xxx* as the authentication parameter.
14	`VEComp -c SysA\pwd1 -am NONE\NONE`	Use ODBC to log on to server SysA. Assumes the DSN is configured with the user name. No extended security mechanism is used.

Note: The -am switch is optional and should be used in conjunction with the -l or -c logon switches. It is possible to connect to Teradata Database using the direct user authentication mechanisms without specifying the -am switch. However, to use multiple connections (which are required when using the bulk compare feature or loading multiple online plans), the authentication mechanism must be supplied for all connections or none of the connections. That is, the -am switch must be supplied for all of the connections or not used for any connection.

To use an alternate logon mechanism for one server and not use any alternate logon mechanism for another specific server, the -am switch must be supplied for the specific server as `-am NONE\NONE`.

View Execution Plan(s)

The View Execution Plan(s) feature can be accessed through the command line interface.

To capture and then display the execution plan for a query, use the following command:

```
VEComp -ie <Logon specifications> [-db <default database>]
[-d <qcdname>] <SQL Statement> [-t <query tag>]
```

SQL Statement

```
{
{-q <SQL test>} || {-f <SQL file> }   SQL file is assumed to be text
file.
}
```

If the -d option is not specified, then "qcd" is used by default.

To capture and then display the execution plan in XML for a query, use the following

command:

```
VEComp -iex <Logon specifications> [-db <default database>]
[-d <qcdname>] <SQL Statement> [-t <query tag>]
```

To display the execution plan in XML for a query, without storing on the database, use the following command:

```
VEComp -eix <Logon specifications> [-db <default database>] <SQL
Statement>
```

Table 45 describes commands used with View Execution Plan(s).

Table 45: Examples for View Execution Plan(s)

Example	Command	Description
1	`VEComp -ie -l SysA\dbc\dbc -db "mydb" -q "sel * from mytbl1"`	Inserts the execution plans for the given query into the qcd database on server SysA and then displays it.

Table 45: Examples for View Execution Plan(s) (continued)

Example	Command	Description
2	`VEComp -iex -l SysA\dbc\dbc -db "mydb" -q "sel * from mytbl1"`	Inserts the execution plans for the given query in XML format into the QCD database on server SysA and then displays it. Note: Only available for Teradata Database version 13.10 and later.
3	`VEComp -ie -l SysA\dbc\dbc -d "mydb" -q "sel * from mydb.statistics" -q "sel * from mydb.columns" -t "MyQuery1" -t "MyQuery2"`	Inserts the execution plans for the given queries into the database on server SysA and displays them. Query tag for the execution plans are "MyQuery1" and "MyQuery2" respectively.
4	`VEComp -ie -c SysA -d QCD -f SQLfile.txt -t "customer query1" -db "MyDb"`	SQL text is loaded from file SQLfile.txt. Execution plan for the SQL statement is inserted into the qcd database and displayed. MyDb is the default database for executing the query.
5	`VEComp -eix -l SysA\dbc\dbc -db “mydb” -q “sel * from mytbl1”`	Generates an XML document for the given query on server SysA and then displays it.

Opening Offline Execution Plans

Visual plans that have been saved offline to a file can be opened through the command line interface.

To open an offline plan, use the following command:

```
VEComp  [-f] <filename.vec> [-v expanded || compressed]
```

Where:

filename is the name of a *.vec* file.

Table 46 describes commands used to open an offline execution plan.

Table 46: Examples for Opening Offline Execution Plan

Example	Command	Description
1	`VEComp test.vec`	Opens the offline plan from file *test.vec*.
2	`VEComp -f test.vec`	Opens the offline plan from file *test.vec*.
3	`VEComp -f test1.vec -f test2.vec -f test3.vec`	Opens the offline plans from the files *test1.vec*, *test2.vec*, and *test3.vec*.
4	`VEComp test1.vec test2.vec test3.vec`	Opens the offline plans from the files *test1.vec*, *test2.vec*, and *test3.vec*.
5	`VEComp -f test.vec -v compressed`	Opens the offline plan from file *test.vec* and displays it in the compressed view.

Opening Online Execution Plans

Plans can be loaded from QCDs across servers using the command line interface.

To open the execution plan for a query, use the following command:

```
VEComp -p <Online plan description1,
[Online plan description2...] > [View Mode]
```

Online Plan Description

```
{<Logon Specification>
-d <database name> {-rc <Range condition> || -t <query tag>}
{ -ux || -unx}}
```

Logon Specification

```
{
{ -l <Server Name>\<User Id>\<Password>} ||
{ -c <ODBC DSN Name>}
}
```

Range Condition

```
{
{ QueryId1[{ {,} || {-}} QueryId2...]}
}
```

Online Plan Description for XML

Note: Only available for Teradata Database version 13.10 and later.

```
{<Logon Specification>}
-d <database name> -xml {-rc <Range condition>}
```

Table 47 describes commands for opening an online execution plan.

Table 47: Examples for Opening Online Execution Plan

Example	Command	Description
1	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73-78`	Opens execution plans whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity.
2	`Vecomp -p -l sysB\dbc\dbc -d qcd -xml -rc 73-78`	Opens execution plans captured in XML format whose queryid is in the range 73-78 from the database qcd on server SysB using CLI connectivity. This command is only available for Teradata Database 13.10 and later. Note: You cannot open both XML and normal execution plans at the same time.
3	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73,78`	Opens execution plans having queryID 73, 78 from the database qcd on server SysB using CLIv2 connectivity.
4	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73-78,51,52`	Opens execution plans having queryID in the range 73–78 and 51, 52 from the database qcd on server SysB using CLIv2 connectivity.

Table 47: Examples for Opening Online Execution Plan (continued)

Example	Command	Description
5	`VEComp -p -c sysB -d qcd -rc 73-78,51,52`	Opens execution plans having queryID in the range 73–78 and 51, 52 from the database qcd on server SysB using ODBC connectivity.
6	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73-78 -l sysA\dbc\dbc -d qcd -rc 75-80`	Opens execution plans having queryID in the range 73–78 from database qcd on server sysB, and having queryID in the range 75–80 from database qcd on server SysA using CLIv2 connectivity.
7	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73-78 -ux`	Uses QCD X views while retrieving execution plans whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity.
8	`VEComp -p -l sysB\dbc\dbc -d qcd -rc 73-78 -unx`	Uses QCD Non-X views while retrieving execution plans whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity.

Comparing Execution Plans Visually

Two or more execution plans can be compared visually through the command line interface.

To compare execution plans visually, use the following command:

```
VEComp -vc <Execution Plan1 [Execution Plan2...]> [View Mode]
```

Execution Plan

```
{
{ Online Plan Description} ||
{-f <Offline Execution Plan file name>}
}
```

The first plan specified in the command is the base plan for comparison.

Table 48 describes commands for comparing execution plans visually.

Table 48: Examples for Comparing Execution Plans Visually

Example	Command	Description
1	`VEComp -vc -l sysB\dbc\dbc -d qcd -rc 73-78 -l sysA\dbc\dbc -d qcd -rc 75-80`	Opens execution plans whose queryID is in the range 73–78 from the database qcd on server sysB, and whose queryID is in the range 75–80 from the database qcd on server sysA using CLIv2 connectivity. The plans are compared after they are opened. In this example, if a plan with a queryID of 73 exists, it is treated as the base plan. Otherwise, the first plan found that matches the given criteria is the base plan.
2	`VEComp - vc -l sysB\dbc\dbc -d qcd -rc 73-78`	Opens and then compares execution plans whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity.

Table 48: Examples for Comparing Execution Plans Visually (continued)

Example	Command	Description
3	`VEComp -vc -l sysB\dbc\dbc -d qcd -xml -rc 73-78`	Opens and then compares execution plans captured in XML format whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity. Note: Only available for Teradata Database version 13.10 and later. You cannot compare XML and normal execution plans at the same time using the command line option.
4	`VEComp -vc -l sysB\dbc\dbc -d qcd -rc 73 -l sysA\dbc\dbc -d qcd -rc 75-80`	Opens and then compares execution plans whose queryID is 73 from database qcd on server sysB, and whose queryID is in the range 75–80 from database qcd on server sysA using CLIv2 connectivity. In this example, plan with queryID.
5	`VEComp -vc -f ValidVECfile1.vec -f ValidVECfile2.vec -f ValidVECfile3.vec`	Opens and then compares execution plans from files ValidVECfile1.vec, ValidVECfile2.vec and ValidVECfile3.vec. The first plan is the base plan.
6	`VEComp -vc ValidVECfile1.vec ValidVECfile3.vec ValidVECfile3.vec`	Opens and then compares execution plans from files ValidVECfile1.vec, ValidVECfile2.vec and ValidVECfile3.vec. The first plan is the base plan.
7	`VEComp -vc -f ValidVECfile1.vec -l sysB\dbc\dbc -d qcd -rc 73`	Opens and then compares plans from file ValidVECfile1.vec, the online execution plan having queryID 73 from database qcd on server SysB. The plan opened from the file ValidVECfile1.vec is the base plan.

Comparing Execution Plans Using Text Reports

Two or more execution plans can be compared using reports through the command line interface.

To compare execution plans using reports, use the following command:

```
VEComp -tc <Execution Plan1 [Execution Plan2...]> <[compare attribute]>
```

Compare Attribute

```
{
{ -a}  || {-cs} || {-sc} || {-jo} || {-jc} || {-ta}
|| {- op} || {-opd} || {-rsc}
}
```

The first plan specified in the command is the base plan for comparison.

Table 49 describes the commands for comparing execution plans using textual reports.

Table 49: Examples for Comparing Execution Plans Using Text Reports

Example	Command	Description
1	`VEComp -tc  -l sysB\dbc\dbc -d qcd -rc 73-78 -l sysA\dbc\dbc -d qcd -rc 75-80`	Open execution plans whose queryID is in the range 73–78 from database qcd on server sysB, and whose queryID is in the range 75–80 from database qcd on server sysA using CLIv2 connectivity. Textual compare differences are displayed for complete information.
2	`VEComp -tc -l sysB\dbc\dbc -d qcd -rc 73-78 -l sysA\dbc\dbc -d qcd  -rc 75-80 -a`	Open execution plans whose queryID is in the range 73–78 from database qcd on server sysB, and whose queryID is in the range 75–80 from database qcd on server sysA using CLIv2 connectivity. Textual compare differences are displayed for complete information.
3	`VEComp -tc -l sysB\dbc\dbc -d qcd -rc 73-78`	Opens execution plans whose queryID is in the range 73–78 from database qcd on server sysB, using CLIv2 connectivity. Textual compare differences are displayed for complete information. In this example, plan with queryID.
4	`VEComp -tc -l sysB\dbc\dbc -d qcd -xml -rc 73-78`	Opens and then compares execution plans captured in XML format whose queryID is in the range 73–78 from the database qcd on server sysB using CLIv2 connectivity. Note: Only available for Teradata Database version 13.10 and later. You cannot compare XML and normal execution plans at the same time using the command line option.
5	`VEComp -tc -l sysB\dbc\dbc -d qcd -rc 73 -l sysA\dbc\dbc -d qcd  -rc 75-80 -sc`	Open execution plans whose queryID is 73 from database qcd on Server sysB, and whose queryID is in the range 75–80 from database qcd on server sysA using CLIv2 connectivity. Textual compare window is displayed only for Software configuration differences.
6	`VEComp -tc -f ValidVECfile1.vec -f ValidVECfile2.vec -ta`	Opens execution plans from files ValidVECfile1, ValidVECfile2.vec. Textual compare window is displayed only for target attribute differences.
7	`VEComp -tc -f ValidVECfile1.vec ValidVECfile2.vec -jc`	Opens execution plans from files ValidVECfile1, ValidVECfile2.vec.
8	`VEComp -tc -f ValidVECfile1.vec -l sysB\dbc\dbc -d qcd -rc 73 -rsc`	Opens execution plan from files ValidVECfile1and online execution plan whose queryID is 73 from database qcd on server sysB, and compares them. Textual compare window is displayed only for residual Condition differences.

Table 49: Examples for Comparing Execution Plans Using Text Reports (continued)

Example	Command	Description
9	`VEComp -tc -f ValidVECfile1.vec -l sysB\dbc\dbc -d qcd -rc 73 -cs`	Opens execution plan from files ValidVECfile1and online execution plan whose queryID is 73 from database qcd on server sysB, and compares them. Textual compare window is displayed only for cost differences.
10	`VEComp -tc -f ValidVECfile1.vec -l sysB\dbc\dbc -d qcd -rc 11 -ic`	Opens execution plan from files ValidVECfile1and online execution plan whose queryID is 11 from database qcd on server sysB, and compares them. Textual compare window is displayed only for index condition differences.

Displaying Differences in Explain Text of Execution Plans

Explain text differences of execution plans can be accessed through the command line.

To display Explain text differences, use the following command:

```
VEComp -ec <Execution Plan1 [Execution Plan2...]>
```

The first plan specified in the command is the base plan for comparison.

Table 50 describes the commands for displaying differences using Explain text.

Table 50: Examples for Displaying Differences in Explain Text

Example	Command	Description
1	`VEComp -ec -l sysB\dbc\dbc -d qcd -rc 73-78 -l sysA\dbc\dbc -d qcd -rc 75-80`	Opens plans whose queryID is in the range 73–78 from database qcd on server sysB, and whose queryID is in the range 75–80 from database qcd on server sysA using CLIv2 connectivity.
2	`VEComp -ec -l sysB\dbc\dbc -d qcd -rc 73 -l sysA\dbc\dbc -d qcd -rc 75`	Opens plans whose queryID is 73 from database qcd on server sysB, and whose queryID is 75 from database qcd on server sysA using CLIv2 connectivity.
3	`VEComp -ec -l sysB\dbc\dbc -d qcd -rc 73-78`	Open plans whose queryID is in the range 73–78 from database qcd on server sysB, using CLIv2 connectivity.
4	`VEComp -ec -l sysB\dbc\dbc -d qcd -rc 73 -f validVECFile.vec`	Open plans whose queryID is 73 from database qcd on Server sysB, and that is from a valid VECFile.vec.
5	`VEComp -ec -f ValidVECfile1.vec -f ValidVECfile2.vec`	Opens execution plans from files ValidVECfile1, ValidVECfile2.vec.

Comparing Multiple Execution Plans Using Bulk Compare

Multiple execution plans can be compared using Bulk Compare, which can be invoked through the command line.

To compare multiple execution plans using Bulk Compare, use the following command:

```
VEComp -bc <count of QCDs> [-m] <Online Plan description1 Online Plan
description2
 [Online Plan description3...] > -f <output file name>
Count of QCDs: Number of online plan descriptions involved in Bulk
Compare.
```

The first plan specified in the command is the base plan for comparison.

Table 51 describes the commands for comparing multiple execution plans using the Bulk Compare feature.

Table 51: Examples for Comparing Multiple Execution Plans Using Bulk Compare

Example	Command	Description
1	`VEComp -bc 2 -m -l SysC\dbc\dbc -d BulkComQcd1 -rc 1-20 -l SysA\dbc\dbc -d BulkComQcd2 -rc 10-30 -f BulkCompareOut.txt`	Opens plans whose queryID is in the range 1–20 from database qcd 'BulkComQcd1' on server Teradata Database SysC, and whose queryID is in the range 10–30 from database qcd 'BulkComQcd2' on server Teradata Database SysA. It compares the information and saves the Bulk Compare output in multiples files.
2	`VEComp -bc 2 -c SysC\dbc\dbc -d BulkComQcd1 -rc 1-20 -c SysA\dbc\dbc -d BulkComQcd2 -rc 10-30 -f BulkCompareOut.txt`	Opens plans whose queryID is in the range 1–20 from database qcd 'BulkComQcd1' on server Teradata Database SysC, and whose queryid range is 10–30 for database qcd 'BulkComQcd2' on server Teradata Database SysA. It compares the information and saves the Bulk Compare output in 'BulkCompareOut.txt' file.
3	`VEComp -bc 2 -c SysC\dbc\dbc -d qcd -rc 1-20 -c SysA\dbc\dbc -d qcd -rc 10-30 -f BulkCompareOut.txt`	Opens plans whose queryID range is 1–20 from database qcd on server Teradata Database server SysC and whose queryID range is 10–30 from database qcd on server Teradata Database server SysA. It compares the information and saves the Bulk Compare output in 'BulkCompareOut.txt' file.

CHAPTER 12

Working with Other Teradata Applications

Other Teradata applications can be started from Teradata Visual Explain. The menu options and the toolbar buttons used to open these Teradata applications are available only if the applications are installed on the client machine.

See the following topics to learn more about the applications:

- Starting Teradata Index Wizard
- Starting Teradata System Emulation Tool
- Starting Teradata SQL Assistant
- Starting Teradata Statistics Wizard

Starting Teradata Index Wizard

Teradata Index Wizard is a tool designed to increase database performance by analyzing defined workloads and recommending indexes. Teradata Visual Explain provides an interface that starts Teradata Index Wizard.

Note: The menu option and toolbar button used to start Teradata Index Wizard are available only if the application is installed on the client machine.

To start Teradata Index Wizard

- Click .

  The Teradata Index Wizard application opens.

For more information on Teradata Index Wizard, refer to the *Teradata Index Wizard User Guide* B035-2506.

Starting Teradata System Emulation Tool

Teradata System Emulation Tool (Teradata SET) is used to emulate a target or production environment on a test system. Use Teradata SET to import the cost parameters and random AMP samples onto the test system before using the View Execution Plan(s) feature to capture the emulated plan. Teradata Visual Explain provides a facility to start Teradata SET.

Note: The menu option and toolbar button used to start Teradata SET are available only if the application is installed on the client machine.

To start Teradata System Emulation Tool

- Click .

  The Teradata System Emulation Tool application opens.

For more information on the Teradata System Emulation Tool, refer to the *Teradata System Emulation Tool User Guide* (B035-2492).

Starting Teradata SQL Assistant

Teradata SQL Assistant supports several other features related to query execution when compared to the native Execute SQL feature in Teradata Visual Explain. For example, a user can add notes to a query or parallel execution of queries. Teradata Visual Explain provides a facility to start Teradata SQL Assistant.

Note: The menu option and toolbar button used to start Teradata SQL Assistant are available only if the application is installed on the client machine

To start Teradata SQL Assistant

- Click .

  The Teradata SQL Assistant application opens.

For more information on Teradata SQL Assistant, refer to the *Teradata SQL Assistant for Microsoft Windows User Guide* (B035-2430).

Starting Teradata Statistics Wizard

Teradata Statistics Wizard is a graphical tool developed to improve the performance of queries and the entire Teradata Database. Statistics Wizard has been designed to automate the process of collecting statistics for a particular workload or selecting arbitrary indexes or columns for collection or re-collection purposes. Statistics Wizard permits users to validate the proposed statistics on a production system. Teradata Visual Explain provides a facility to start Statistics Wizard.

Note: The menu option and toolbar button used to start Statistics Wizard are available only if the application is installed on the client machine.

To start Teradata Statistics Wizard

- Click .

  The Teradata Statistics Wizard application opens.

For more information on the Teradata Statistics Wizard, refer to the *Teradata Statistics Wizard User Guide* (B035-2503).

Glossary

A

administrator A special user responsible for allocating resources to a community of users.

access right In Teradata SQL, a user's right to perform the Teradata SQL statements granted to the user against a table, database, user, macro, or view. Also known as a privilege.

AMP Access Module Processor. A virtual processor that receives steps from a PE and performs database functions to retrieve or update data. Each AMP is associated with one virtual disk, where the data is stored. An AMP manages only its own virtual disk and not the virtual disk of any other AMP.

B

Basic Teradata Query See BTEQ.

BTEQ Basic Teradata Query. Command-based application that enables users on a workstation to communicate with one or more Teradata Database systems.

C

Call-Level Interface, version 2 See CLIv2

candidate index column A column for which the value or join-range access frequencies stored in the Field table of the specified QCD are greater than 0.

cardinality In set theory, cardinality refers to the number of members in the set. When specifically applied to database theory, the cardinality of a table refers to the number of rows contained in a table.

CLIv2 Call-Level Interface, version 2. A programming interface designed to support SQL access to databases from shrink-wrapped application programs. SQL/CLI provides an international-standard, implementation-independent CLI to access SQL databases. Client-server tools can easily access a database through DLLs. It supports and encourages a rich set of client-server tools.

column In the relational model of Teradata SQL, databases consist of one or more tables. In turn, each table consists of fields, organized into one or more columns by zero or more rows. All of the fields of a given column share the same attributes.

cost The outlay of database resources used by a given query.

D

database A related set of tables that share a common space allocation and owner. A collection of objects that provide a logical grouping for information. The objects include tables, views, macros, triggers, and stored procedures. See also table.

Database Administrator See DBA.

Database Query Log See DBQL.

data definition The statements and facilities that manipulate database structures and the Data Dictionary information kept about these structures. These statements include CREATE, DROP, ALTER, and MODIFY.

Data Definition Language See DDL.

Data Dictionary In Teradata Database, the information automatically maintained about all tables, views, macros, databases, and users known to the Teradata Database system. The dictionary includes information about ownership, space allocation, accounting, and access right relationships between those objects. Data Dictionary information is updated automatically during the processing of Teradata SQL data definition statements. The parser uses the Data Dictionary information to obtain information needed to process all Teradata SQL statements.

Data Manipulation Language See DML.

DBA Database Administrator.

DBQL Database Query Log. A set of measures that can be analyzed over a period of time to verify system usage or to more closely examine a query that uses a large amount of valuable system resources.

DDL Data Definition Language. In Teradata SQL, the statements and facilities that manipulate database structures (such as CREATE, MODIFY, DROP, GRANT, REVOKE, and GIVE) and the dictionary information kept about those structures. In the typical, pre-relational data management system, data definition and data manipulation facilities are separated. The data-definition facilities are less flexible and more difficult to use than in a relational system.

DML Data Manipulation Language. In Teradata SQL, the statements and facilities that manipulate or change the information content of the database. These statements include INSERT, UPDATE, and DELETE.

domain name A group of computers whose hostnames (the unique name by which a computer is known on a network) share a common suffix (the domain name).

F

field The basic unit of information stored in a Teradata Database. A field is either null or has a single numeric or string value.

G

GDO Globally Distributed Object. A data structure that is shared by all of the virtual processors in Teradata Database configuration.

Globally Distributed Object See GDO.

H

hash-ordered secondary index See HOSI.

HOSI hash-ordered secondary index. A secondary index whose rows are ordered by hash.

I

index candidate See candidate index column.

J

join A select operation that combines information from two or more tables to produce a result.

L

log A record of events. A file that records events. Many programs produce log files. Review a log file to assist in determining what is happening when problems occur. Log files use the extension *.log*.

M

macro A set of Teradata SQL statements stored by a Teradata Database and executed by a single EXECUTE statement. Each macro execution is implicitly treated as a transaction.

N

name A user-supplied word that refers to an object, such as column, database, macro, table, user, or view.

non-unique secondary index See NUSI.

null The absence of a value for a field.

NUSI non-unique secondary index. One of two types of secondary indexes. A secondary index can be specified at table creation or at any time during the life of the table. The index can consist of up to 16 columns. To get the benefit of the index, the query has to specify a value for all columns in the secondary index. A NUSI is usually specified in order to prevent full-table scans.

O

object In object-oriented programming, a unique instance of a data structure defined according to the template provided by its class. Each object has its own values for the variables belonging to its class and can respond to the messages, or methods, defined by its class.

object definition The details of the structure and instances of the objects used by a given query. Object definitions are used to create the tables, views, macros, triggers, join indexes, and stored procedures in a database.

ODBC Open Database Connectivity. Under ODBC, drivers are used to connect applications with databases. The ODBC driver processes ODBC calls from an application, but passes SQL requests to Teradata Database for processing.

Open Database Connectivity See ODBC.

P

parameter A variable name in a macro for which an argument value is substituted when the macro is executed.

parser A program executing in a parsing engine that translates Teradata SQL statements entered by a user into the steps that accomplish the user's intensions.

Parsing Engine See PE.

PE Parsing Engine. An instance (virtual processor) of the database management session control, parsing, and dispatching processes and their data context (caches).

privilege See access right.

production system A Teradata Database used in a live environment at a customer site. A system that is actively used for day-to-day business operations. This differs from a test or development system that is used to create new queries or test new features before using them on the production system.

Q

QCD Query Capture Database. A database of relational tables that store the steps of any query plan captured by the QCF.

QCF Query Capture Facility. Provides a method to capture and store the steps from any query plan in a set of predefined relational tables called the QCD.

query A Teradata SQL statement, such as a SELECT statement.

Query Capture Database See QCD.

Query Capture Facility See QCF.

R

random AMP sample An arbitrary sample from an AMP. These are samples of the tables in a query or all of the tables in a given database.

recursive query A named query expression that is allowed to reference itself in its own definition, giving the user a simple way to specify a search of a table using iterative self-join and set operations. Use a recursive query to query hierarchies of data. Hierarchical data could be organizational structures such as department and sub-department, forums of discussions such as posting, response, and response to response, bill of materials, and document hierarchies.

Relational Database Management System A database management system in which complex data structures are represented as simple two-dimensional tables consisting of columns and rows.

request In host software, a message sent from an application program to Teradata Database.

result The information returned to the user to satisfy a request made of Teradata Database.

row The fields that represent one entry under each column in a table. The row is the smallest unit of information operated on by data manipulation statements.

S

session In client software, a logical connection between an application program on a host and Teradata Database. The session permits the application program to send one request to and receive one response from Teradata Database at a time.

SQL Structured Query Language. A standardized query language for requesting information from a database. SQL consists of a set of facilities for defining, manipulating, and controlling data in a relational database.

statement A request for processing by Teradata Database that consists of a keyword verb, optional phrases, and operands. The statement is processed as a single entity.

statistics These are the details of the processes used to collect, analyze, and transform the database objects used by a given query.

stored procedure Combinations of procedural and non-procedural statements run using a single call statement.

Structured Query Language See SQL.

T

table A two-dimensional structure made up of one or more columns with zero or more rows that consist of fields of related information. See also database.

Target Level Emulation See TLE.

TDPid Teradata Director Program ID. Assigned by the system administrator.

Teradata Director Program ID See TDPid.

TLE Target Level Emulation. Emulate a target environment (target system) by capturing system-level information from that environment. The captured information is stored in the relational tables *SystemFE.Opt_Cost_Table* and *SystemFE.Opt_RAS_Table.* The information in these tables can be used on a test system with the appropriate column and indexes to make the Optimizer generate query plans as if it were operating in the target system rather than the test system.

trigger One or more Teradata SQL statements associated with a table and executed when specified conditions are met.

U

user A database associated with a person who uses the Teradata Database. The database stores the person's private information and accesses other Teradata Databases.

Unicode®, UTF-8, and UTF-16 character sets UTF-8 and UTF-16 are two of the standard ways of encoding Unicode® character data. The Teradata Database supports UTF-8 and UTF-16 client character sets. The UTF-8 client character set supports UTF-8 encoding. The Teradata Database supports UTF-8 characters that can consist of from one to three bytes. The UTF-16 client character set supports UTF-16 encoding. The Teradata Database supports the Unicode® 6.0 standard, where each defined character requires exactly 16 bits. There are restrictions imposed by the Teradata Database on using the UTF-8 and UTF-16 character sets. For details, see *International Character Set Support* (B035-1125).

unique secondary index See USI.

USI unique secondary index. One of two types of secondary indexes. A secondary index can be specified at table creation or at any time during the life of the table. It can consist of up to 16 columns. To get the benefit of the index, the query has to specify a value for all columns in the secondary index. A USI has two purposes: to speed up access to a row that otherwise might require a full table scan without having to reply on the primary index, and to enforce uniqueness of a column or set of columns.

V

value-ordered secondary index See VOSI.

view An alternate way of organizing and presenting information in Teradata Database. A view, like a table, has rows and columns. However, the rows and columns of a view are not directly stored by Teradata Database. They are derived from the rows and columns of tables (or other views) whenever the view is referenced.

VOSI value-ordered secondary index. A NUSI can be value ordered which means the NUSI can be sorted on the key values themselves rather than on the corresponding hash codes. This is useful for range queries where only a portion of the index subtable will be accessed. With a value-ordered NUSI, only those blocks in the NUSI subtable that are within the range are scanned. It must be a number value, up to 4 b, versus a longer character column. DATE is the most commonly used data type. The actual data value is stored as part of the NUSI structure.

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